CA NO GE/NORTH- 54/2025-26: PROVN OF RCC FOUNDATION WORK FOR INSTALLATION MACHINERY AT COMPONENT LEVEL REPAIR BLDG. (SHOP NO 18) FOR ESTABLISHING COMPONENT LEVEL REPAIR FOR TANK T-90/SK AT 505 ABW AND PROVN OF EXHAUST FAN IN BARREL TYPES SHED OF TECH AREA AT ESD UNDER GE (NORTH) DELHI CANTT -10

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Signature of Contractor
Dated: _____

AGE (Contracts)
For Accepting Officer

SPEED POST

Tele No: 20892718

Garrison Engineer (North)
Delhi Cantt -110010

8996/54/05/E8

Aug 2025

M/S _____

_____

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CA NO GE/NORTH- 54/2025-26 PROVN OF RCC FOUNDATION WORK FOR INSTALLATION MACHINERY AT COMPONENT LEVEL REPAIR BLDG. (SHOP NO 18) FOR ESTABLISHING COMPONENT LEVEL REPAIR FOR TANK T-90/SK AT 505 ABW AND PROVN OF EXHAUST FAN IN BARREL TYPES SHED OF TECH AREA AT ESD UNDER GE (NORTH) DELHI CANTT -10

Dear Sir(s),

1. Tender documents in respect of above work are uploaded on the site **https://defproc.gov.in.** The tender is on single stage two cover e-tendering system. The contents of Cover I & Cover II are specified in NOTICE OFTENDER.

2. Bids will be received online by ACCEPTING OFFICER upto the date and time mentioned in the NOTICE OF TENDER (NIT). No tender/bid will be received in physical form and any tender/bid received in such manner will be treated as non bonafide tender/bid.

3. Bid will be opened on due date and time fixed for opening in the presence of tenderers / bidders or their authorized representatives, who have uploaded their quotation bid and who wish to be present at the time of opening the bids.

4. Your attention is also drawn to instruction on filling and submission of tender attached herewith. You may forward your points on tender documents and/or depute your technical representative for discussion on tender/drawings and to clarify doubts, if any, on or before date of bid submission start date. You are requested not to write piece meal points and forward your points duly consolidated before above date.

5. Un-enlisted contractors are required to submit the scanned copies (in pdf file) of documents required as per eligibility criteria mentioned in instructions for filling the tender documents and Appendix ‘A’ to NIT alongwith EARNEST MONEY DEPOSIT (EMD) and tender fee on e-procurement portal and submit the physical documents in the office of the Garrison Engineer (North) Delhi Cantt-10, C/o 56 APO within time limit specified in NIT. Inadequacy/deficiency of documents shall make the bid liable for rejection resulting in disqualification for opening of finance bid.

6. Contractor having not executed Standing Security Bond and Standing Security Deposit in any MES formation shall upload scanned copy of EARNEST MONEY DEPOSIT (EMD) mentioned in Notice of Tender and shall ensure receipt of hard copy of EMD in the office of tender issuing authority before date & time fixed for this purpose. In case of failure to abide by any of these two requirements, the finance bid will not be opened.

7. Enlisted contractors of MES shall submit the scanned copies (pdf file) of enlistment letter, tender fee and such other documents as mentioned in Appx 'A' to NIT on e-procurement portal and submit physical documents in the office of the Garrison Engineer Delhi Cantt-10 before dates & time fixed for this purpose.

8. The contractor must ensure that the tender/bid on the proper form is uploaded in time as the Accepting Officer will take no cognizance of any quotations/offer received in any other electronic or physical form like email/fax/by hand/through post from tenderer / bidder even if they are received in time.

9. In view of delays due to system failure or other communication related failures, it is suggested that the tender/bid be uploaded, if necessary, sufficiently in advance of the last due date and time fixed.

10. General Conditions of Contracts (IAFW-2249) (1989 Print) and errata and amendments thereto, Schedule of minimum fair wages and MES SSR (Part-I and Part-II) are not enclosed with these documents. These are available for perusal in the Office of GE (North) Delhi Cantt-10 concerned and this office.

11. Any tenderer, which proposes alterations to any of the conditions, specifications laid down in the tender documents or any new conditions, whatsoever, is liable to be rejected.

Yours faithfully,

Signature of Contractor
Dated: _____

AGE (Contracts)
For Accepting Officer

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

1. EARNEST MONEY DEPOSIT(EMD):-

Contractor(s) who are not enlisted with MES/who are enlisted but have not executed the Standing Security Bond shall submit Earnest Money Deposit as detailed in Notice of Tender in one of the following forms, alongwith their tender/bid:-

(a) Deposit at Call Receipt from a Scheduled Bank in favour of GE (North) Delhi Cantt-10.

(b) Receipted Treasury Challan, the amount being credited to the Revenue Deposit of GE (North) DelhiCantt-10.

It is advisable that Earnest Money is deposited in the form of deposit call receipt from an approved Schedule Bank for easy refund. In case the tenderer /bidder wants to lodge "EARNEST MONEY DEPOSIT' in any other form allowed by MES, a confirmation about its acceptability will be obtained from the Accepting Officer well in advance of the bid submission end date and time. Earnest Money Deposit shall be submitted in the name of GE.

Note:-

Earnest Money Deposit (EMD) in the form of Cheque/Bank Guarantee etc., will not be accepted. NON SUBMISSION OF EARNEST MONEY WILL DEPOSIT (EMD) (scanned copy alongwith technical Bid & hard copy before the date & time fixed for opening of BOQ) WILL RENDER THE BID DISQUALIFIED FOR OPENING OF COVER II (FINANCE BID).

2 PERFORMANCE SECURITY: Within 28 days of receipt of the letter of Acceptance, the successful contractor shall deliver to the Accepting Officer a Performance Security in the form of Bank Guarantee, Govt Security, FDR or any other Govt. instrument for an amount equivalent to 5% of the contract sum. Non compliance of this provision shall constitute sufficient grounds for cancellation of awarded contract and forfeiture of Earnest Money. SSD deposited by Registered contractor shall serve the propose of exemption of Earnest money. If the Performance Security is not deposited by the Registered contractor, the contract shall be cancelled and contractor shall deposit the amount equal to Earnest Money through MRO and issue of tender shall remain suspended to such contractor till the aforesaid amount is deposited in Govt. Treasury

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5. **General**.

5.1 The bids received only in the electronic form will be considered. All bids shall be submitted on web site **"www.eprocuremes.gov.in"** portal. Documents should be scanned and forwarded in 'pdf' form and 'xls' form as indicated.

5.2 Bids shall be uploaded on 'eprocuremes.gov.in' portal on or before the bid closing date mentioned in the tender. No tender/bid in any other electronic or physical form like email/fax/by hand/through post will be considered.

5.3 Bid should be DIGITALLY signed using valid DSC. All pages of tender documents, corrections / alternations shall be signed /initialled by the lowest bidder after acceptance.

5.4 Drawings, if issued in physical form, must be returned duly initialed by the tenderer /bidder in separate envelope indicating his name and address.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

5.5 The tender shall be signed, dated and witnessed at all places provided for in the documents after acceptance. All corrections shall be initialled. The Contractor shall initial every page of tender and shall sign all drawings forming part of the tender. Any tender/bid, which proposes alterations to any of the conditions whatsoever, is liable to be rejected.

5.6 In the technical bid, a scanned copy of Power of Attorney in favour of the person uploading the bid using his/her DSC shall be uploaded. In case the digital signatory himself is the sole proprietor, scanned copy of an affidavit on stamp paper of appropriate value to this effect stating that he has authority to bind the firm in all matters pertaining to contract including the Arbitration Clauses, shall be attached in 'pdf' form. In case of partnership concern or a limited company, digital signatory of the bid/tender shall ensure that he is competent to bind the contractor (through partnership deed, general power of attorney or Memorandum and Articles of Association of the Company) in all the matters pertaining to the contracts with Union of India including arbitration clauses. A scanned copy of the documents confirming of such authority shall be attached with the tender/bid in 'pdf' form, if not submitted earlier. The person uploading the bid on behalf of another partner(s) or on behalf of a firm or company using his DSC shall upload with the tender/bid a scanned copy (in 'pdf' form) of Power of Attorney duly executed in his favour by such other or all of the Partner(s) or in accordance with constitution of the company in case of company, stating that he has authority to bind such other person of the firm or the Company, as the case may be, in all matters pertaining to the contract including the Arbitration Clause.

5.7 Even in case of Firms or Companies which have already given Power of Attorney to an individual authorizing him to sign tender in pursuance of which bids are being uploaded by such person as a routine, fresh Power of Attorney duly executed in his favour stating specifically that the said person has authority to bind such partners of the Firm, or the Company as the case may be, including the condition relating to Arbitration Clause, should be uploaded in 'pdf' form with the tender/bid, unless such authority has already been given to him by the Firm or the Company. It shall be ensured that power of attorney shall be executed in accordance with the constitution of the company as laid down in its Memorandum & Article of Association.

5.8 Hard copies of all above documents should be sent by the contractor to the Tender issuing authority well in advance to be received before the date & time fixed for the same.

5.9 Bid (Cover 1 & 2) shall be uploaded online well in time.

5.10 The contractor shall employ Indian Nationals after verifying their antecedents and loyalty. Attention is also drawn to special condition 3 referred hereinafter and also conditions 24 & 25 of IAFW 2249 (General Conditions of Contracts).

5.11 Tenderers / bidders who uploaded their priced tenders/bids and are desirous of being present at the time of opening of the tenders/bids , may do so at the appointed time.

5.12 The tenderer / bidder shall quote his rates on the BOQ file only. No alteration to the format will be accepted, else the bid will be disqualified and summarily rejected.

INSTRUCTIONS ON FILLING AND SUBMISSION OF TENDER

5.13 In case the tenderer /bidder has to revise/modify the rates quoted in the BOQ (excel sheet) he can do so only in the BOQ, through eprocuremes.gov.in site only before the bid closing time and date.

6 REVOKATION/REVISION OF OFFER UPWARD/ OFFERING VOLUNTARY REDUCTION, AFTER OPENING OF FINANCIAL BIDS BY LOWESTBIDDER

In the event of lowest tenderer / bidder revoking his offer or revising his rates upwards / offering voluntary reduction, after closing of bid submission date & time, his offer will be treated as revoked and the Earnest Money Deposited by him shall be forfeited. In case of MES enlisted Contractors, the amount equal to the Earnest Money stipulated in the Notice of Tender, shall be notified to the tenderer /bidder for depositing the amount through MRO. Bids of such Contractors/bidders shall not be opened till the aforesaid amount equal to the earnest money is deposited by him in Govt Treasury. In addition, bids of such tenderer /bidder and his related firm shall not be opened in second call or subsequent calls. Reduction offered by the tenderer /bidder on the freak high rates referred for review shall not be treated as voluntary reduction.

7. CPM (CRITICAL PATHMETHOD)

7.1 The project planning for work covered in the scope of tender is based on CPM.

7.2 The tenderer/bidder is expected to be fully conversant with the CPM technique and employ technical staff who can use the technique insufficient details. Sufficient books and other literature on the subject are widely available in the market which the tenderer/bidder may make use of.

7.3 The tenderer's/bidder's attention is drawn to special condition of the tender regarding preparation of the detailed network analysis and time schedules for the work and his liability for employing sufficient resources to adhere to this schedule. Any inability on the part of the tenderer/bidder in using the technique will be taken as his technical inefficiency and will affect his class of enlistment and future prospects/invitation to tenders for future works.

7.4 Department may issue amendments/errata in form of CORRIGENDUM to tender/revised BOQ to the tender documents. The tenderer /bidder is requested to read the tender documents in conjunction with all the errata / amendments/ corrigendum, if any, issued by the department.

8. These instructions shall form part of the contract documents.

Signature of Contractor
Dated: _____

AGE (Contracts)
For Accepting Officer

MILITARY ENGINEER SERVICES
NOTICE INVITING TENDER (NIT)

1. A tender is invited for the work as mentioned in Appendix 'A' to this Notice Inviting Tender (NIT).

2. The work is estimated to cost as indicated in the aforesaid Appendix "A". This estimate, however, is not a guarantee and is merely given as rough guide and if the work cost more or less, a tenderer / bidder will have no claim on this account. The tender shall be based on as mentioned in the aforesaid Appendix 'A'.

3. The work is to be completed within the period as indicated in the a fore said Appendix "A" in accordance with the Phasing, if any, indicated in the tender from the date of handing over site, which will be on or about two weeks after the date of Acceptance of the tender.

4. Normally contractors whose names are on the MES approved list for the areas in which the work lies and within whose financial category, the estimated amount would fall may tender/bid, but in case of term contracts, contractors in categories "SS" to "E" may tender/bid. Not more than one tender/bid shall be submitted/uploaded by one contractor or one firm of contractors. Under no circumstance will a father and his son (s) or other closed relations, who have business dealing with one another will be allowed to tender/bid for the same contract as separate competitor. A Breach of this condition will render the tender/bid of both parties liable for rejection.

5. The office of GE (North) Delhi Cantt will be Accepting Officer, here in after refer to as such for the purposes of this contract.

6. The Technical Bid and Financial Bid (Cover-I and Cover-2) shall be uploaded by the tenderer/bidder on or before the date & time mentioned in NIT. A scanned copy of DD with tender/bid enlistment details/documents shall be uploaded as packet 1/cover-1 ('T' bid) of the tender/bid on e-tendering portal. DD is refundable in case T bid is not accepted resulting in non- opening of 'Q bid. The applicant contractor shall bear the cost of bank charges for procuring and encashing the DD and shall not have any claim from the Government whatsoever on this account.

6.1 Tender form and conditions of contract and other necessary document shall be available on eprocuremes.gov.in website for download and shall form part of contract agreement in case the tender/bid is accepted.

6.2 In the case of contractor who has not executed the Standing Security Bond, the Cover-I shallbeaccompaniedwithbyEarnestMoneyofamountasmentionedinaforesaidAppx"A"inthe form of deposit at call receipt in favour of GE (North) Delhi Cantt-10 by a Schedule Bank or in Receipt Treasury Challan, the amount being credited to the revenue deposit of the GE (North) DelhiCantt-10.

6.3 A contractor who is not enlisted for the area in which the work lies but whose name is in the MES approved list of any MES formation and who has deposited standing security and executed standing security Bond may bid without depositing earnest money alongwith the tender; but if the Accepting Officer accepts the tender/bid, the contractor will be required to lodge with the Controller of Defence Accounts concerned the amount of "Individual Security deposit" within thirty days of the receipt by him of notification of acceptance of his tender/bid failing which this sum will be recovered from 1st RAR payment of from the first final bill. In the case of term/running contracts, remaining sum shall be recovered from subsequent bill(s) of the contractor.

6.4 ----BLANK----

6.5 ----BLANK----

6.6 The GE will either return the Earnest Money to the successful tenderer / bidder by endorsing an authority on the deposit call receipt for its refund on receipt of an appropriate amount of security deposit or will retain the same in part of full on account of security deposit if such a transaction is feasible.

6.7 Copies of the drawings and other documents pertaining to the work signed for the purposes of identification by the Accepting Officer or his accredited representatives, samples of materials & stores to be supplied by the contractor will also be available for inspection by the tenderer/bidder at the office of Accepting Officer and concern GE during working hours.

7. The tenderers / bidders are advised to visit the site of work by making prior appointment with the GE who is also the executive agency of the work. A tenderer /bidder is deemed to have full knowledge of all relevant documents, samples, site etc. whether he has inspected site or not.

8. Any tender/bid which proposes any alterations to any of the conditions laid down or which proposes any other conditions or prescriptions whatsoever is liable to be rejected.

9. The uploading of bid implies that bidder has read this notice and the Conditions of Contract & has made himself aware of the scope and specifications of work to be done & the conditions & rates at which stores, tools and plants etc, will be issued to him and local conditions & other factors having bearing on execution of the work.

10. Tenderers/bidders must be in possession of a copy of MES Standard Schedule of (Rates see appendix ‘A’) including amendments and errata there to.

11. Invitation for e-tender does not constitutes any guarantee for consideration validation of “T” bid and subsequent opening of finance bid of any applicant/bidder, even of enlisted contractors of appropriate class, merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the “T” bid and not open the financial bid of any applicant /bidder.”T” bid validation shall be decided by the Accepting Officer based on inter alia, capability of the firm as per criteria given in Appx “A” to this NIT. The applicant/contractor/bidder will be informed regarding non- validation of his ‘T’ bid assigning reasons thereof through the “defprocurement” website. The applicant/contractor if he so desires, may appeal to the next higher Engineer Authority i.e HQ CWE Delhi Cantt on email id of HQ CWE Delhi Cantt with copy to the Accepting Officer on email before the schedule date of opening of finance bid. The decision of the next higher authority (NHEA) shall be final and binding. No applicant/ contractor shall be entitled for any compensation whatsoever for rejection of his bid.

12. The Accepting Officer reserves the right to accept a tender submitted by a Public Undertaking giving a price preference over other tenders(s)/bids which may be lower, as are admissible under the Govt. Policy. No claim for any compensation or otherwise shall be admissible from such tenderer/bidder whose tender/bid is rejected.

13. Accepting Officer does not bind himself to accept the lowest or any tender/bid or to give any reason for not doing so.

14. This **Notice Inviting Tender (NIT)** Including Appendix ‘A’ shall form part of the contract.

Signature of Contractor
Dated: _____

AGE (Contracts)
for Accepting Officer

APPENDIX 'A' TO NOTICE INVITING

1.	Name of work	:	**PROVN OF RCC FOUNDATION WORK FOR INSTALLATION MACHINERY AT COMPONENT LEVEL REPAIR BLDG. (SHOP NO 18) FOR ESTABLISHING COMPONENT LEVEL REPAIR FOR TANK T-90/SK AT 505 ABW AND PROVN OF EXHAUST FAN IN BARREL TYPES SHED OF TECH AREA AT ESD UNDER GE (NORTH) DELHI CANTT -10**
2.	Estimated Cost of Work	:	Rs 48.07 Lakhs
3.	Period of completion	:	**06 (Six) Months**
4.	Cost of tender documents	:	Rs 500/- in the form of DD / Bankers cheque from any Scheduled/ Nationalised bank in favour of **GE (North) Delhi Cantt-10** and payable at **Delhi Cantt-10**
5.	Website/portal address	:	https://defproc.gov.in
6.	Type of Contract	:	The tender shall be based on specification and IAFW-1779A and GCC IAFW-2249 with list of works to be priced by tenderer. The tenderers are required to quote their rates against items of list of works.
7.	Information & Details:		

	(a) Publishing date and time of tender documents	Refer critical dates on the website.
	(b) Starting date and time of downloading of tender documents	
	(c) Bid submission start date	
	(d) Last date of bid submission	
	(e) Date of bid opening	

8 **Eligibility Criteria**

	(A) **For MES enlisted contractors**	:	Contractor shall be enlisted with MES in **Class 'D'** and above and category **a(i)** subject to satisfactory remarks wrt performance in respect of works in hand as reflected in Work Load return (WLR) or any other report circulated by competent Engineer Authority.
	(B) **For other Contractors**		(a) (i) (Contractor not enlisted with MES shall meet the enlistment criteria of **'D' class MES Contractors & category a(i)** contractor with regard to satisfactory completion of requisite value of works with Central /State Government /Central/State PSUs/AWHO/AFNHB /CGEWHO/DGMAP and shall submit documents mentioned in Para '5' under notes. The other requirement given in Para 1.4 & 1.5 of section I of MES Manual on Contracts 2020 as available in all MES formations as well as MES website (www.mes.gov.in) are also required to be uploaded. However, these other documents/requirements will not be criteria for qualification/disqualification in technical evaluation and mandatorily to be completed before award of contract in case non-enlisted Contractor / Firm emerges as L-1. (ii) Not carrying adverse remarks in Work Load Report (WLR) or any other similar report circulated by any competent authority, if already working in MES (ii) Not suspended/debarred/blacklisted (either permanently or temporarily) from participating in any bid or for business dealings by any central/state Government department or any Central /State Government PSU or any Autonomous Body under Central/State Government or any local body as on the bid submission end date.
	(c) Technical PQC criteria		Not Applicable
9.	Tender issuing and Accepting Officer	:	**GE (North) Delhi Cantt** **Nic Mail id : ndcntdz4-mes@nic.in**
10.	Executive Agency	:	**GE (North) Delhi Cantt** **Nic Mail id : ndcntdz4-mes@nic.in**
11.	Earnest Money	:	**Rs 96,140.00** in form of Deposit Call Receipt or Receipted Treasury Challan from any Scheduled/Nationalized bank in favour of **GE (North) Delhi Cantt-10**.

APPENDIX 'A' TO NOTICE INVITING TENDER (NIT)

NOTES :-

1. After opening of Cover I if the number of MES enlisted contractors of eligible class as well as un-enlisted contractors, if any, fulfilling the other eligibility criteria given in NIT are less than 7 (Seven), applications in respect of contractors one class or two classes (in case of remote and difficult areas to be decided as per list circulated by CE command/ADG) below the eligible class shall also be considered subject to fulfillment of their eligibility criteria given in the NIT. Therefore MES contractors one class below (two classes below in case of remote and difficult areas) may also bid for this tender, However contractors of one/two classes below the eligible class shall not be considered in case their present residual work in hand is more than five times their present tendering limit, However in case such contractors fulfill the criteria of upgradation to the stipulated eligible class based on past experience of completed works (individual work experience and/or average annual turnover, as applicable) and financial soundness (solvency/financial soundness and working capital), the celling of present residual work will not apply and they will be considered for issue of tender. Such bidders shall upload in their Cover-I details of works in hand showing names of work, Names of Accepting Officers. Contract amounts, dates of commencement and completion (stipulated) and progress as on bid submission date. These details shall be verified by the Tender Issuing Authority from concerned formations in case bids of such contractors are considered for evaluation.

2. In case after opening of Cover I, the number of MES enlisted contractors of eligible class as well as un–enlisted contractors, if any fulfilling the other eligibility criteria given in NIT are 7 (Seven, or more applications of only those one class below the eligible class bidders shall be considered, who have previously completed similar works satisfactorily and are meeting the criteria of up gradation in respect of past experience of competed works (individual work experience and/or average annual turnover as applicable and financial soundness (solvency /financial soundness and working capital) as per details given in Manual on Contracts. Therefore such contractors shall upload the requisite information /documents in the Cover-I.

3. Un-enlisted contractor shall be considered provided he meets the criteria. Foreign firms shall not be eligible for this tender. However Indian Firms having foreign nation/Indian national staying abroad/Indian national having taken foreign citizenship, as director(s) shall be considered subject to security clearance from the concerned authorities.

4. Contractors enlisted with MES will upload following documents in Cover I for checking eligibility:-

(a) Application for tender on Firm's letter head.

(b) Enlistment letter issued by the Registering Authority duly renewed for the cycle period in vogue.

(c) Scanned copy of DD/Bankers Cheque toward cost of tender and EMD instrument in case SSD bond is not signed at the time of registration.

(d) Scanned copy of Provident Fund Code Number.

(e) Scanned copy of GST registration number.

5. Contractor not enlisted with MES will be required to upload following documents in Cover I for checking eligibility.

(a) Application for tender on Firm's letter head.

(b) Scanned copy of DD/Bankers cheque toward cost of tender and Earnest Money Deposit (EMD) instrument.

(c) Copy of police verification report/Police Clearance certificate/Character certificate from the Police authority of the area where te registered office of the firm is located/notarized copy of valid passport of Proprietor/each Partner/each Director.

(d) Details of works being executed in MES, if any.

(e) Scanned copy of Provident Fund Code Number.

(f) Scanned copy of GST registration number

(g) All documents required for enlistment in MES for the class mentioned in Para 8 (b) above as per Para 1.4 & 1.5 of section I of MES Manual on Contracts 2020. However, these documents/requirements will not be criteria for qualification/disqualification in technical evaluation and mandatorily to be completed before award of contract in case non-enlisted/Firm emerges as L-1.

6. Tenders not accompanied by scanned copies of requisite DD/ Banker Cheque towards cost of tender and earnest money (as applicable) in Cover I shall not be considered for validation of "T" bid and their Financial bids will not be opened.

7. Contractors should ensure that their original physical DDs and Earnest Money Deposit (EMD) instruments (as applicable) reach the office of Accepting Officer within 5 days of bid submission end date failing which following action shall be taken.

(a) In case of tenders from an enlisted contractor of MES, where scanned copies of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover I but physical copies are not received within the stipulated period, their financial bids (Cover 2) will be opened. However non-submission of physical copies of cost of tender shall be considered as willful negligence of the tenderer with ulterior motives and such tenderer shall be banned from bidding for a period of six months commencing form the date of opening of Financial Bid (Cover 2).

(b) In case of tenders from un-enlisted contractor, scanned copes of requisite DD/Bankers Cheque towards cost of tender have been uploaded in Cover I but physical copies are not received within the stipulated period, their financial bids (Cover 2) will not be opened, Name of such contracts along with complete address shall be circulated for not opening of their bids for a period of six months commencing form the date of opening of Financial bid (Cover 2).

(c) In case of tenders from enlisted and un-enlisted contractors, where scanned copies of instruments for Earnest Money Deposit (as applicable) have been uploaded in Cover I but the same are not received in physical from within stipulated period, such tenders shall not qualify for opening of financial bid (Cover

8. Contractor will not be allowed to execute the work by subletting or through power of attorney to a third party/another firm on his behalf. However a contractor can execute the work through power of attorney to sons/daughters/spouse of Proprietor/Partner/ Director and firm's own employees, director, project manager **provided they are not having a separate enlist firm in MES in their name as Proprietor/Partner/Director**.

9. After opening of Cover I and during its technical evaluation, in case any deficiency is noticed in the documents required to be uploaded by the tenderer as per NIT, a communication in the form of e-mail/SMS/Speed Post etc shall be sent to the contractor to rectify the deficiency within a period of seven days from date of communication failing which their financial bid (Cover2) shall not be opened and contractor shall not have any claim on the same.

10. Invitation for e-tender does not constitute any guarantee for validation of technical bid and subsequent opening of financial bid of any applicant /bidder merely by virtue of enclosing DD. Accepting Officer reserves the right to reject the Technical bid and not to open the financial bid of any applicant/bidder. Technical bid validation shall be decided by the Accepting Officer based on eligibility of the firm as per criteria given in this Appendix. Tenderer/bidder will be informed regarding non-validation of his Technical bid assigning reasons therefore through tender evaluation report which shall be uploaded on the website. Such tenderer, if desires, may appeal to the Next Higher Engineer Authority (NHEA) viz HQ CWE Delhi Cantt one mail id dcntdz3-mes@nic.in with copy to the accepting officer on email before the scheduled date of opening of Cover 2. NHEA shall decide the matter within a period of seven working days from the date of receipt of appeal. The decision of the NHEA shall be final and binding. The tenderer/bidder shall not be entitled for any compensation whatsoever for rejection of the bid.

12. In case an un-enlisted contractor is already executing works in MES he shall not be considered eligible for the subject tender if the total value of such work is more than twice the tendering limit of the MES class of contractor for which it is eligible. For this purpose, details of the works being executed by such a contractor shall be uploaded in the Cover-Iof the bid and shall be checked /verified by the Accepting Officer.

13. In case the BOQ is revised through the corrigendum and the bidder has failed to quoted on revised BOQ (ie he has quoted on pre revised BOQ), such bid shall be treated as willful negligence by the bidder and his quotation shall be considered non – bonafide. In such cases the lowest tender shall be determined from amongst the valid/ bonafide bids only. Accepting Officer may decide whether to re-tender or consider the lowest bonafide tender for acceptance

14. Revoking the offer or revising the rates upward or offering voluntary reduction by the lowest tenderer after opening of Cover-2 shall be considered as a wilful default, For this default a penalty of an amount equal to Earnest Money shall be levied. In case of an un-enlisted tenderer Earnest Money deposited by him shall be forfeited, in case MES enlisted tenderer having deposited the standing security bound and amount equal to the earnest money stipulated in the NIT Shall be notified to the tenderer for depositing through MRO and consideration of such tenderer in tender evaluation for future works shall remain suspended till the afore mentioned amount is deposited in the Government Treasury. No other disciplinary/administrative action shall be taken against such. In such a situation, the next lowest offer shall not be considered for acceptance. Instead, retendering shall be resorted to in a transparent and fair manner and the defaulting tenderer and his related firm if any, shall not be eligible for this tender in second call or subsequent calls.

15. Tender to related firms shall not be issued simultaneously. Firms shall be termed as related if Proprietor/one or more Partners/Directors are common. Decision of Accepting Officer on issue/deny the tender to any one of the related firms shall be final and binding.

Signature of Contractor
Dated: _____

AGE (Contracts)
For Accepting Officer

MILITARY ENGINEER SERVICES
ITEM RATE TENDER AND CONTRACT IAFW-1779A

(To be used in conjunction with General Conditions of Contracts (IAFW-2249-1989-Print) (Print 1989).

REGISTERED/AD

Tele: 20892718

Garrison Engineer (North)
Military Engineer Services
Delhi Cantt-110010

8996/54/05/E8

Aug 2025

CA NO GE/NORTH- 54/2025-26 PROVN OF RCC FOUNDATION WORK FOR INSTALLATION MACHINERY AT COMPONENT LEVEL REPAIR BLDG. (SHOP NO 18) FOR ESTABLISHING COMPONENT LEVEL REPAIR FOR TANK T-90/SK AT 505 ABW AND PROVN OF EXHAUST FAN IN BARREL TYPES SHED OF TECH AREA AT ESD UNDER GE (NORTH) DELHI CANTT -10

1. M/s …….. is/are hereby authorised to submit quoted bid/e-tender for the above work. The e-tender value of works is Rs **48.07 lakhs**. The tender is to be submitted through eprocuremes.gov.in portal as pfer NIT to **GARRISON ENGINEER (NORTH) DELHI CANTT-10** marked on the outside **"Contract Number GE/NORTH-54/2025-26.**

2. All documents must be returned whether or not a tender has been submitted.

3. Any correspondence concerning this tender shall be addressed as indicated at the top of this sheet quoting the reference as given, above.

THE PRESIDENT OF INDIA DOES NOT BIND HIMSELF TO ACCEPT THE LOWEST OR ANY TENDER

Signature of Contractor
Dated: _____

AGE (Contracts)
for Accepting Officer

NOTES: -

1. The contract covers name of work.: **PROVN OF RCC FOUNDATION WORK FOR INSTALLATION MACHINERY AT COMPONENT LEVEL REPAIR BLDG. (SHOP NO 18) FOR ESTABLISHING COMPONENT LEVEL REPAIR FOR TANK T-90/SK AT 505 ABW AND PROVN OF EXHAUST FAN IN BARREL TYPES SHED OF TECH AREA AT ESD UNDER GE (NORTH) DELHI CANTT -10**

NOTES: - 1. The entire works covered under this contract shall be completed within **06 (Six) months** from the date of handing over of site given in W.O. No.1 to be issued by GE after acceptance of the tender.

2. Unit rates quoted by the tenderer(s) under Col 5 of Sch "A" shall include for all necessary works required for the entire completion of the items as described and shall cover supply of all materials, labour, fixing, jointing, connecting up and testing etc. complete all as specified except where specifically mentioned otherwise in the description of the respective item(s). Tenderer's special attention is invited to visit site area and make apprise themselves with working condition, accessibility of site, nature of buildings, height, safety of workman etc before quoting their rates. Contractor,s quoted rate shall have deemed to be included the cost of all the aspects and no extra payment is to be made on this account. Contractor's attention is invited to the method of measurements and preambles given in MES Schedule 2009 (Part –I) & SSR 2020 (Part-II) including errata & amendment there to.The quoted rates by the tenderer are deemed to be inclusive of all the taxes, e.g sales tax, excise octroi, GST, work contract Tax VAT labourers etc as applicable and no claim whatsoever on account of these taxes or increase in such taxes shall be admissible. Makes of the materials/fittings to be incorporated in the work shall be as indicated in Appx to particular specifications which is attached herewith and shall form part of the contract.

3. Tenderer(s) are required to quote their unit rates under Col 5 of Sch. "A" in "figures" Only" the amount shall be extended automatically under column 6 & 7 of Sch 'A' as figure and words respectively.

4. The items and quantities given in Sch "A" are provisional and shall be measured in accordance with the stipulated method of measurements **through E-MB & payment also made through E-MB** and paid for as actually carried out as per written order(s) of GE, however, shall not exceed beyond the limits laid down in condition 7 of IAFW-2249 (General conditions of contracts). Tenderers are requested to register their selves in CCEP portal of MES. If any Contractor has not got themselves registered in CCEP portal of MES, their work executed on ground shall not be able to recorded in E-MB and they shall able to claim payment through RAR/Final Bill. For non-registration in CCEP portal leading to non-payment of RAR/Final Bill, the contractor only shall be held responsible and no claim whatsoever shall arise against the govt for non-payment of RAR/Final Bill.

5. The rates quoted by the contractor in Sch "A" shall be deemed to allow for all "minor extras" and "constructional details" which are not specifically given in particular specifications but are essential for the execution of work services in a sound & workman like manner. In case of difference of opinion between the contractor & Engineer-in -Charge as to whether or not certain items(s) of work constitute "minor extras" and "constructional details" included in the contractor's prices, the decision of the Accepting Officer shall be final, conclusive and binding.

6. Unit rates under col 5 of Sch 'A' shall be deemed to include for all the provisions given in the preambles in MES Schedule as applicable to respective items(s) unless otherwise stated in the tender documents.

7. Dismantling/demolition work/taking down shall be executed carefully with trained labours in order to ensure that the work to the extent ordered is only dismantled/demolished/taken down. Any excess dismantling/demolition/taking down shall be made good by the contractor to the entire satisfaction of the Engineer -in-Charge to match with the adjoining work without any extra cost to the Government. The all old material retrieved from above work except items which shall be credit to contractor through " Sch of credit " shall become Government property and to be deposited in MES store yard by contractor without any extra cost of govt.

8. Unless otherwise stated, cost of any cutting/forming holes & making chases etc. and making good the same in cement and sand mortar 1:3 required for execution of work shall be deemed to be included in the respective items of Sch. "A" . However consequent to variation in quantities of respective items during execution, no price adjustment on account of the above shall be made.

SCHEDULE 'A'(Contd/----------)

9. Before actual commencement of the work, the site shall be jointly inspected by the Engineer-in-Charge and the contractor. During inspection, necessary tests/site checks shall be carried out by the contractor as per directions of the Engineer -in-Charge to ascertain the actual quantum and repairs required for the various items of schedule "A". Thereafter the site shall be inspected by GE for verification of the above requirements ascertained by Engineer-in-Charge and the contractor and the repairs and replacements to the tune of actual requirements shall only be executed as per orders of GE. The cost of all the above aspects shall be deemed to be included in the unit rates quoted by the contractor. Also completion period of the contract given in schedule "A" note 1 here-in-before shall be deemed to include for time required for inspection and testing etc.

10. Wherever indicated in Schedule as "S&F" and "M&L" the same may please be read as "Supply and Fix" and "Material and Labour" respectively.

11. The description of items is complete for material and labour unless otherwise mentioned in description of items.

12. Tenderers are required to get fully acquainted with the scope of work as catered for in the tender documents and also the site of work. No claim shall be entertained for any misunderstanding or any in-accuracy with regard to the site condition and provision made in the tender documents.

13. The rate quoted by the tenderer shall be deemed to be inclusive of all the taxes i.e. excise, otroi, GST, construction labour welfare cess, all other taxes which will include by the Govt in time to time as applicable and no claim what so ever on account of increase in such taxes shall be admissible.

14. The work covered under this contract is to be carried out in buildings under occupation. Therefore the contractor shall plan and progress the work with least disturbance to the occupants. The above aspect shall be deemed to be taken into consideration by the tenderers while quoting the rates and no compensation whatsoever shall be admissible on this account.

15. The rate quoted of the Sch 'A' Item by the tenderer are deemed to be amplified and read in conjunction with the purchase voucher of Items as provision in CA. In absence of Purchase voucher of Item (other than items which envolve CAT Part Number), 10% of prevailing market cost of material shall be recovered from Contractor's dues. Engineer-in-Charge shall confirm or otherwise shall recover from Contractor's due. No in any circumstances the Items involve CAT part number shall be measured and paid to contractor in absence of purchase voucher (CAT Part Number shall be shown in purchase voucher). Equivalency of CAT part shall be verified and justified by the Contractor as well as Engineer-in-Charge. It shall deemed that tenderer is well aware about Purchase Voucher and their recovery while quoting his rates.

16. Please ensure that at the time of uploading of e-tending, a copy of provident fund code number in addition to other documents required shall be mandatory. If the contractor not in possession of this number shall be disqualified in 'T' bid evaluation and his financial bid shall not be opened.

17. EPF Code & UAN: All contractors shall mandatorily be in possession of the EPF Code Number and UAN of employees working under their establishment under EPF & MP Act 1952. So far this tender is concern, before release of work order the contractor shall submit EPF Code Number to GE. The contractor shall also ensure that Sub Contractor if engaged by him shall also comply with provisions of EPF & MP Act 1952.

18. **Payment to Labour during execution work**: The payment to labour engaged by the contractor shall be made as per Wages Act. No payments to the contractor /firms will be made unless proof of payment of wages is submitted by the contractor/firm. In case payments are made through current coins or currency notes, certificate will be submitted by the contractor alongwith undertaking by the concern workmen asking for payments other than by cheques or by crediting in the bank account of the employee/workmen. Adherence to these instructions will be checked during various inspection. Being Principal Employer, the GE shall ensure that the contractor fully comply with above provisions. The documents held by the contractor shall be subject to scrutiny by law enforcement agencies. The contractor shall also provide proof of monthly payment details under EPF & ESI with account numbers alongwith details of regular payments which shall be verified by GE.

19. **Electrical License**: **In terms of IE Rule 1956, Rule No 45, the contractor shall have Electrical License holder issued by State Govt. under his establishment. The GE shall ensure that the work is executed through authorised person only.**

SCHEDULE 'A'(Contd/----------)

20. **Electrical License**: **In terms of IE Rule 1956, Rule No 45, the contractor shall have Electrical License holder issued by State Govt. under his establishment. The GE shall ensure that the work is executed through authorised person only.**

21. **Performance Security Deposit**: As per condition 19 of IAFW-2249 within 28 days of receipt of the letter of Acceptance, the successful contractor shall deliver to the Accepting Officer a Performance Security in the form of Bank Guarantee (Issued by Nationalised /Scheduled Bank as per Appx 2.1 to Manual on Contract 2020 with modifications with respect to Performance Security in place of Security Deposit), Govt Security, FDR for an amount equivalent to 5% of the contract sum. Non compliance of this provision shall constitute sufficient grounds for cancellation of awarded contract and forfeiture of Earnest Money. SSD deposited by Registered contractor shall serve the propose of exemption of Earnest money. If the Performance Security is not deposited by the Registered contractor, the contract shall be cancelled and contractor shall deposit the amount equal to Earnest Money through MRO and issue of tender shall remain suspended to such contractor till the aforesaid amount is deposited in Govt Treasury.

22. **Defect Liability Period:** In terms of Condition 46 of IAFW-2249 (As amended vide amendment No 48), the Defect Liability Period in respect of this tender shall be **24 (twenty four) months**.

23. **Refund of Performance Security**: The Performance Security Deposit mentioned above (As amended vide Amendment No 45 to Condition 68 of IAFW-2249) shall be refunded to contractor after expiration of Defect Liability Period mention above. Recovery if outstanding against the contractor shall be effected from the Performance Security as stipulated under Condition 67 of IAFW-2294 (as amended)

24. **COVID 19**:-

(a) Contactor ensure location identified for labour camp shall be disinfected completely using user friendly disinfectant mediums.

(b) Contractor ensure proper supervision of labour Camp for hygiene and Sanitation. Personnel in the camp are to be screened periodically for fever, coughs or any other symptoms of COVID-19.

(c) Contractor to ensure provision of Mask / Sanitizer / Saop water for all personnel at sites carrying out activity. Wearing of face cover is compulsory in all places.

(d) There should be strict ban of gutka, tobacco etc. and spitting should be strictly be prohibited.

(e) Proper disposal of waste generated at the labour camp to avoid flight safety hazard by attacting insects, mosquitoes, flies and birds.

(f) The workers are to be employed on shfit basis as far as possible and stagger the lunch breaks of labour to ensure social distancing.

(g) Contractors to be ensure Medical insurance of labours.

(h) At least 6 feet distance is to be maintained by the labours.

SCHEDULE 'A'(Contd/----------)

25. The subject work consists of the following work:-

SRL No	Name of work	Authority and amount	TS with date	Amount (in Rs.)
(a)	PROVN OF RCC FOUNDATION WORK FOR INSTALLATION MACHINERY AT COMPONENT LEVEL REPAIR BLDG. (SHOP NO 18) FOR ESTABLISHING COMPONENT LEVEL REPAIR FOR TANK T-90/SK AT 505 ABW DELHI CANTT- 10	HQ Delhi Area letter No 3700/13/CW/Q3W dt 05 Jan 2024	2040/09/E2 dt 17 May 2025	43,12,000.00
	PROVN OF EXHAUST FAN IN BARREL TYPES SHED OF TECH AREA AT ESD UNDER GE (NORTH) DELHI CANTT - 10	HQ Delhi Area letter No 3702/13/Misc/Q3W dt 24 Jan 2025	2068/04/E2 dt 11 Mar 2025	4,95,000.00

Signature of Contractor
Dated :_____

AGE (Contracts)
for Accepting Officer

SCHEDULE 'B'
ISSUE OF STORES ETC TO THE CONTRACTOR (SEE CONDITION 10 OF IAFW-2249)

Srl. No.	Particular	Rate at which the stores etc will be issued to the contractor		Place of issue by Name	Remarks
		Unit	Rate		

.....NIL.....

SCHEDULE 'C'
LIST OF TOOLS & PLANT (OTHER THAN TRANSPOT) WHICH WILL BE HIRED TO THE CONTRACTOR (SEE CONDITION 14,15,34 AND 35 OF IAFW-2249)

Srl. No.	Particular	Rate at which the stores etc will be issued to the contractor		Place of issue by Name	Remarks
		Unit	Rate		

.....NIL.....

SCHEDULE 'D'
TRANSPORT TO BE HIRED TO THE CONTRACTOR (SEE CONDITION 16 & 35 OF IAFW-2249)

Srl. No.	Particular	Rate at which the stores etc will be issued to the contractor		Place of issue by Name	Remarks
		Unit	Rate		

.....NIL.....

Signature of Contractor
Dated: _____

AGE (Contracts)
for Accepting Officer

TENDER

To,

THE PRESIDENT OF INDIA

Having examined and perused the following documents: -

1. Specifications signed by, AGE (Contracts) c/o Garrison Engineer (North) Delhi Cantt-10

2. Drawing(s) detailed in the list of drawing(s).

3. Schedule 'A', 'B', 'C' and 'D' attached hereto.

4. MES Standard Schedule of Rates 2009 (Part-I specifications) and Standard Schedule of Rates-2020 Part-II: (rates) (here-in-after and in IAFW-2249 referred to as the MES Schedule) together with errata/amendments as under :-

SSR -2009 Part-I - 1 to 3

SSR -2020 Part-II 1 to 122

5. General conditions of contracts, IAFW-2249 (1989 Print) together with erratas 1 to 20 and amendment Nos.1 to 49.

6. Water: Refer condition 31 of IAFW-2249 (General conditions of contracts): water will be supplied by the MES.

7. Should this tender be accepted ** I/we agree

*(a) That the sum of Rs _____ forwarded as Earnest money shall either be retained on account of security deposit or be refunded by the Government on receipt of an equal amount as security deposit within the time specified vide condition 22 of IAFW-2249.

(b) To execute all the works referred to in the said documents upon the terms and conditions contained or referred to therein and as detailed in General Summary on the subsequent pages and to carryout such deviations as may be ordered vide condition 7 of IAFW-2249 upto a maximum 10% (Ten percent) and further agree to refer all disputes as required vide condition 70 of IAFW-2249 to the sole arbitration of a serving Officer having degree in Engineering or equivalent or having passed final/direct final examination of Sub Division II of Institution of Surveyors (India) recognized by the Govt of India to be appointed by CE Delhi Zone, Delhi Cantt or in his absence the officer officiating as CE Delhi Zone, Delhi Cantt, whose decision shall be final, conclusive and binding.
* To be deleted where not applicable.

Contd…/

Signature ……. …. In capacity of …… duly authorised to sign the tender for and on behalf of M/S …… ……(IN BLOCK LETTERS).

Dated……..

Witness …… Postal address: _____

Address…… Telephonic Address: _____

……. Telephone Number: _____

ACCEPTANCE

1. …… alterations have been made in these documents and as evidence that these alterations were made before the execution of the contract agreement, they have been initialled by the contractor and AGE (Contracts).

2. The said officer is hereby authorised to sign and initial on my behalf the documents forming part of this contract.

3. The above tender was accepted by me on behalf of the President of India for the contract sum of Rs……..…….. (Rupees …….....

…….....)

at the item rates contained in schedule 'A' on the ……day of ……

Signature. ……Date this ……. Day of ……)

Appointment: Garrison Engineer (North) Delhi Cantt-10
(For and on Behalf of The President of India)

SPECIAL CONDITIONS

1. **General**:-

The following special conditions shall be read in conjunction with General Conditions of contracts IAFW-2249 (1989 Print), including errata/amendments thereto. If there are any provisions in these special conditions, which are at variance with the provisions in the above-mentioned documents; the provisions in these special conditions shall be deemed to take precedence there over

1.1 The work under this contract shall be carried out in accordance with Schedule 'A' the particular specifications, drawings, general specifications and other provisions in MES Schedule.

1.2 The term General specifications referred to hereinbefore, as well as referred to in IAFW-2249 (General conditions of contracts), shall mean the specifications contained in the MES schedule.

1.3 General Rules, preambles, special conditions, method of measurements etc. specified in MES Standard Schedule of Rates shall be applicable unless specific provisions to the contrary are made in these tender documents.

2. INSPECTION OF SITE BY THE TENDERER: The tenderers are advised to visit the site(s) of work with prior permission of GE and make themselves familiar with the local working conditions, accessibility to sites(s), availability of material and other relevant conditions which effect their pricing, workman like execution and entire completion of work. The tenderer shall be deemed to have visited the site(s) before tendering whether they actually visit or not.

3. CONDITION FOR WORKING IN UNRESTRICTEDAREA:

3.1 The work lies in unrestricted area. However, the contractor, his agents, servants, work people and vehicles may pass through the unit lines, in which case the Engineer-in-Charge shall, at his discretion has the right to issue passes, control their admission to the site of work or any part thereof. The contractor shall on demand by the Engineer-in-Charge submit a list of personnel; etc concerned and shall satisfy the Engineer-in-Charge as to the bonafides of such people. Passes shall be returned at any time on demand by the Engineer-in-Charge and in any case, on completion of work.

3.2 The contractor and his work people shall observe all the rules promulgated by the authority controlling the area in which the work is to be carried out e.g. prohibition of smoking, lighting and fire precaution, restricted hours of working etc. Any person found violating the security rules laid down by the authority, shall be immediately expelled from the area without assigning any reasons whatsoever and the contractor shall have no claim on this account. Nothing shall be admissible for any man hours lost on this account.

3.3 IDENTITY CARD OF WORKMEN**:** Every workman shall be in possession of an identity card. The identity card will be issued after thorough investigation of the antecedents of the labourers by the contractor and attested by the Officer in Charge of the unit concerned in accordance with the standing rules and regulations of the unit. Contractor shall be responsible for the conduct and action of his workmen, agent or representative.

3.4 SEARCH: Thorough search of all persons and transport shall be carried out at each gate and for as many times a gate is used for entry or exit. Search may also be carried out any time or any number of times at the work site within the restricted area.

3.5 FEMALE SEARCHER**:** If the contractor desires to employ female labour on works to be carried out inside the Area of a factory, Depot, park etc and a female searcher is not borne on the authorized strength of the factory, Depot, Park etc at the time of submission of the tender he shall be deemed to have allowed in his tender for pay and allowances etc, for a female searcher (class IV servant /Gp'D' servant) calculated for the period, female labour is employed by him inside that area. If more than one contractor employs female labour during any month and female searcher(s) has/have to be employed in addition to the authorised strength of the factory, Depot park etc, the salary and allowances paid to the additional female searcher(s) shall be distributed on an equitable basis between the contractors employing female labour taking into consideration the value and period of completion of their contracts. The GE's decision in regardto the amount recoverable on this account from any contractor shall be final and binding.

4. CONTRACTOR'S REPRESENTATIVES/AGENTS AND WORKERS: The contractor shall employ only Indian Nationals and verify their antecedents and loyalty before employing them on the work. He shall ensure that no person of doubtful antecedents and nationality is in any way associated with the work. As a proof that he has employed only Indian Nationals, the contractor shall render a certificate to this effect to GE within one month from the date of commencement of work. The GE shall have full power, and without giving any reasons to order the contractor immediately to cease to employ in connection with this contract any representatives, agent, servants and workmen or employees whose continued employment in his (GE's) opinion is undesirable. The contractor shall not be allowed any compensation on this account. Tenderer's attention is also drawn to condition 25 of IAFW-2249 in this connection. Verification of antecedents of the Contractor's representatives/ labour deployed at site in connection with execution of work under the Contract, as per security requirement of user unit/ installation shall be the responsibility of the Contractor and all expenses in connection with verification of antecedents by police authority/ security agency shall be born by the Contractor.

SPECIAL CONDITIONS

5. LAND FOR STORAGE OF MATERIALS, WORKSHOP, and OFFICE AND ACCCOMMODTION FOR LABOURERS: The contractor shall apply to Station Headquarters on prescribed Proforma available in GE's office for storage purpose only. Site/ land measuring 20mX10m for storage shed shall be allotted for the duration of work but not exceeding one year, subject to approval of the same by the Station Headquarters. Contractor is required to give undertaking that he is not occupying any other temporary site for any other contract in Delhi Cantt and site will be vacated immediately on termination of allotment period. Final payment to the contractor will be made after vacation of allotted site/ land. Same site/land once allotted will not be subjected to extension after expiry of allotment period but the contractor may apply for an alternative site, if required.

6. CONTRACTOR'SPLANT/EQUIPMENTATSITE: The contractor shall furnish the Engineer- in-Charge every morning a distribution return of the plant and/or equipment on the site of work required for incorporation in the work. The Engineer-in-charge shall record the particulars of plant/equipment supplied by the contractor in the works diary and send the return to the GE for record in his office.

7. USE OF EXISTING BUILDINGS ETC: The contractor shall not use the existing latrines, baths and other existing buildings if any for use by his work people or for storage of his materials or for any other purpose unless permitted to do so in writing by Garrison Engineer.

8. WATER: Refer condition 31 of General Conditions of contracts IAFW-2249. Water will be supplied by MES at the point(s) decided by the GE. As the water supply of MES is likely to be intermittent the contractor shall make his own arrangements for storing the water required for the works labour and workmen etc at his own expenses. Water charges will be recovered at the rate of Rs 3.75 per Rs 1000/- worth of work done priced at contract rates. No claim/ whatsoever shall be admissible on account of compensation for intermittent /insufficient water supply by the MES. However, any subsequent change in rate of recovery of water charges shall be deemed to be included in CA. If the supply of water may not be continuous and adequate. The contractor shall be deemed to have ascertained for the hours of availability quantity of water before submitting his tender. The MES do not guarantee the continuity of water supply and no compensation shall be allowed for intermitted or inadequate water supply and break down in the system. If the supply is not sufficient the contractor shall make his own arrangement to supplement the water supply at his own cost for this purpose the contractor shall be allowed to install hand pumps, tubewells at the site of work at places as directed by the Engineer-in-Charge or GE and in any case on completion of work no compensation whatsoever shall be admissible to the contractor if the GE requires him to remove the hand pumps, tube wells before completion of work.

9. ELECTRICITY: Electric supply required for the work upto a maximum of 20KVA single phase 50 Hz shall be made available by the MES at points decided by GE. KWH meters to register the electric energy supplied and main switch shall be provided and installed by the MES. Contractor shall provide all necessary cables, fittings etc from the tapping point in order to ensure a proper and suitable supply of electricity for execution of work. The contractor will be charged for the electric energy consumed for execution of works at the rates of Rs 9.00 (Rupees Nine only) per unit for lighting and power. MES do not guarantee continuity of supply and no compensation whatsoever shall be allowed for supply becoming intermittent or for breakdown in the system. In case GE or his representative is not satisfied with device or electric lines provided by the contractor, shall be disconnected from the supply if so directed by him.

10. DAMAGE TO EXISTING BUILDINGS, ROADS AND DRAINS ETC: The contractor shall be responsible to reinstate and make good at his own expense, the damage if any caused by his workmen etc. to existing buildings, roads, drains, cables etc., while carrying out the work under this contract. Rectification, reinstatement, replacement, making good and touching up, etc, shall conform to the standard of workmanship and materials, originally used and the same shall be done to the entire satisfaction of Engineer-in-charge.

SECIAL CONDITIONS

11. OCTROI AND OTHER DUTIES: The price quoted by the contractor shall be deemed to include all duties, taxes Octroi, GST, surcharge, cess etc. The contractor shall not include any condition for fluctuation in price of labour and/or materials in his tender shall be binding on contractor tothe full extent the regulations mentioned therein and Appendix C to IAFW 2249. Statutory Labour Laws contained under, Contract Labour (regulation & Abolition) Act 1970, EPF & MP Act 1952, Payment of Wages Act 1936 shall be in force and binding as per law. The payment to labour engaged by the contractor shall be made as per Wages Act. No payments to the contractor /firms will be made unless proof of payment of wages is submitted by the contractor/firm. In case payments are made through current coins or currency notes, certificate will be submitted by the contractor alongwith undertaking by the concern workmen asking for payments other than by cheques or by crediting in the bank account of the employee/workmen. Adherence to these instructions will be checked during various inspection. Being Principal Employer, the GE shall ensure that the contractor fully comply with above provisions. The documents held by the contractor shall be subject to scrutiny by law enforcement agencies. The contractor shall also provide proof of monthly payment details under EPF & ESI with account numbers alongwith details of regular payments which shall be verified by GE

12. CO-OPERATION WITH OTHER AGENCIES: The contractor shall permit free access and afford reasonable facilities to other Agencies or departmental workmen engaged by Govt. to carry out other work services (whether or not in connection with the work under the scope of this contract) on the site, simultaneously. The contractor shall not be allowed any extra payment on this account.

13. ACCEPTABLE QUALITY OF WORK AND FINISHES: To determine the acceptable standard of workmanship, the Garrison Engineer may order the contractor to execute certain typical portion of works and services such as some length of wall, portion of different type of floors, plaster, painting and other finished/items, sanitary fittings, plumbing, electrical fittings etc under strict and constant supervision of MES staff and label these works as guiding sample so that further works shall be executed to confirm these samples.

14. TIME AND PROGRESS CHART: The time and Progress Chart to be prepared as per condition 11 of General Conditions of Contracts (IAFW-2249) immediately after handing over the site shall consist of detailed network analysis and a time schedule. The critical path network shall be drawn jointly by the GE and the contractor soon after acceptance of the tender to complete the as per schedule. The Contractor shall mobilize and employ sufficient resources to achieve the detailed schedule within the broad frame work of the accepted methods of working and safety. drawn to condition 2A & 3 of General Conditions of Contracts(IAFW-2249) for compliance. The contractor shall not communicate any classified information regarding the work either to sub-contractor(s) or to any other person(s) without prior written approval of the Engineer-in-Charge.

15. RECORD OF CONSUMPTION OF CEMENT: The contractor shall maintain a properly bound register in the form approved by the Engineer-in-Charge showing the daily receipt of cement, quantity used in works and balance at the end of each day. This register shall be signed daily by the contractor's representative in token of correctness of the entries made and it shall be signed by JE every day and AGE once in week. The register shall be kept at site of work in the safe custody of the contractor. The register shall handed over to the Engineer-in-Charge for record with MES on completion of work.

16. CLEANING DOWN: The contractor shall clean all floors, joinery, glass panes etc. and remove drops of cement/lime paint etc. and touch up all painter's work and carry out other items of work as are necessary to make and leave the premises clean and tidy before handing over the buildings and/or works. The contractor shall not be allowed any extra payment on this account.

17. VALIDITY OF TENDER: The tender shall remain open for acceptance for a period of 60 days (Sixty days) commence from the next date subsequent to last date of bid submission.

SPECIAL CONDITIONS

18. PROPRIETARY MATERIALS/ARTICLES AND RECORD OF MATERIALS: The proprietary materials/articles such as paint, bitumen, water proofing admixture, etc. when brought to site shall be inspected and approved by the Engineer-in-Charge. The quantity of all proprietary articles brought to site shall be recorded in measurement book (IAFW-2261) as 'NOT TO BE ABSTRACTED' and signed by the contractor and the Engineer-in-Charge. The contractor shall ensure that the materials are brought to site, in original sealed containers/ packing, bearing manufacturer's marking except in the case of requirement of material (s) being less than smallest packing.
Original purchase vouchers/bills including manufacturer's test certificate (where applicable) in respect of the following item shall be provided by contractor to the Engineer-in-Charge alongwith RAR against claim for material lying at site raised in the RAR:-

Water proofing compound	GI pipes and fittings	Steel windows /ventilators
Paint, distemper & cement base paints	Sanitary fittings and fixtures	PVC door
Galvalume Sheet.	Tiles, paver blocks	Aluminium doors/windows/ventilators
All grades of bitumen	False ceiling	Factory made panelled door shutter
Electrical fittings, fixtures, cables, conductors, equipments, LT panels, Transformer, DG Set, APFC Panel, Servo Stabilizer	Steel (TMT and structural)	GI/DI pipes and fittings, RCC pipes
Cement	MCB/ELCB/RCCB/MCCB, exhaust/ceiling fans, DB, Street light fittings, LED	lamps/fittings, Aviation light, Tube light fittings
High mast light, Steel tubular poles, Lightening arrestor, Insulators	Sealing compound, Water proofing membrane	CHS equipments and hot water piping

19. The contractor shall, on demand, produce to the GE, and original receipted vouchers/invoices in respect of the materials other than as stated in condition 14 above. Vouchers/invoices so produced and verified shall be stamped by Engineer- in-Charge indicating contract number. The contractor shall ensure that the materials are brought to site, in original sealed containers/packing, bearing manufacturer's marking except in the case of the requirement of materials(s) being less than smallest packing.

20. After vacation of site/area of land/building allotted to the contractor. Payment of final bill shall be made to the contractor only after such areas of land/building are vacated by the contractor

21. OFFICIAL SECRET ACT: The contractor shall be bound by the Indian official Secret Act 1923 particularly Section 5 thereof, all as per condition 2A ofIAFW-2249.

22. ADVANCE ON ACCOUNT OF MATERIALS (APPLICABLE TO CONTRACTS OF VALUE RS. 50 LAKHS AND ABOVE):

(a) For the purpose of such contracts, the following shall be deemed to be added in continuation of para 8 of condition 64 of IAFW-2249: "Provided further the contractor may be paid advance on account to the full value of under mentioned materials (as applicable in this contract) brought on the site, on his furnishing guarantee bond on the prescribed form from a scheduled bank for the amount of the retention money which should otherwise be recoverable from him under the contract.

(i) Electric cables/fittings/fixtures and electric equipments

(ii) Any other non-perishable and manufactured materials which do not lose their identity after incorporation in the work, as decided by GE. Materials like bricks, aggregate, pre-cast concrete and similar items are not to be taken in the list.

(b) The bank guarantee bond(s) shall be executed for a period of six months and on a form as directed by the Accepting Officer. The contractor shall further arrange to extend the period of guarantee bond(s)as and when necessary, and as directed by the Accepting Officer or shall furnish fresh guarantee bond(s)if and when necessary and as directed by the Accepting Officer of similar value in lieu.

SPECIAL CONDITIONS

23. **SPECIAL CONDITION FOR CONCILIATION:** The scope of conciliation shall be as per condition 71 of IAFW 2249.

(a) **APPOINTMENT OF CONCILIATOR:** Disputes (other than those for which the decision of the CWE or any other person is by the contract expressed to be fixed and binding) shall be referred to the sole conciliator viz serving Officer not below the rank of superintending Engineer/ Superintending Engineer (QS&C) appointed by Chief Engineer Delhi Zone, New Delhi or in his absence the Officer officiating as Chief engineer.

(b) **STATUS & EFFECT OF SETTLEMENT AGREEMENT** The settlement agreement signed by the parties as a result of conciliation proceedings shall be the same status and effect as it is an arbitral award on agreed terms.

24. **QUALIFIED TRADESMEN (Applicable for estimated cost one crore and above work as per NIT):** In compliance with the condition 26 of IAFW-2249 (General Conditions of Contracts) the contractor shall employ skilled/semi-skilled tradesmen who are qualified and possessing certificate in particular trade from Industrial Training Institute (ITI)/Similar reputed and recognized Institutes by State/Central government to execute the works of their respective trade. The number of such qualified trades men shall not be less than 25% of total skilled/ semi-skilled tradesmen required in each trade.

25. The rates quoted by the Contractor shall be deemed to be inclusive of all taxes and cessesviz GST, labour welfare cess etc, duties, Royalties, Octroi & other levies payable under the respective Statutes. No reimbursement/refund for variation in rates of taxes, cesses, duties, Royalties, Octroi& other levies, and/or imposition/abolition of any new/existing taxes, duties, Royalties, Octroi & other levies shall be made except as provided in sub para (b) here-in-below. These taxes shall be levied on gross amount of work:-

(a) The taxes which are levied by State/Central Govt at certain percentage rates of Contract Sum/Amount shall be termed as “taxes directly related to Contract value” such as ,GST, Labour Welfare Cess/tax and like but excluding Income Tax. The tendered rates shall be deemed to be inclusive of all “taxes directly related to Contract Value” with existing percentage rates as prevailing on last due date of submission of bid. Any increase in percentage rates of “taxes directly related to Contract value” with reference to prevailing rates on last due date for submission of bid shall be reimbursed to the Contractor and any decease in percentage rates of “taxes directly related to Contract value” with reference to prevailing rates on last due date for submission of bid shall be refunded by the Contractor to the Govt/deducted by the Govt from any payments due to the Contractor. Similarly imposition of any new “taxes directly related to Contract value” after the last due date for receipt of tenders shall be reimbursed to the Contractor and abolition of any “taxes directly related to Contract value” prevailing on last due date for receipt of tenders shall be refunded by the Contractor to the Govt/deducted by the Govt from the payments due to the Contractor.

(i) The contractor shall, within a reasonable time of his becoming aware of variation in percentage rates and/or imposition of any further “taxes directly related to Contract value”, give written notice thereof to the GE stating that the same is given pursuant to this Special Condition, together with all information relating thereto which he may be in a position to supply. The Contractors shall submit the other documentary proof/information as the GE may require.

(ii) The Contractor shall, for the purpose of this condition keep such books of account and other documents as are necessary and shall allow inspection of the same by a duly authorized representative of Govt, and shall further, at the request of the GE furnish, verified in such a manner as the GE may require, any documents so kept and such other information as the GE may require.

(iii) Reimbursement for increase in percentage rates/imposition of “taxes directly related to Contract value” shall be made only if the Contractor necessarily & properly pays additional “taxes directly related to Contract value” to the Govt, without getting the same adjusted against any other tax liability or without getting the same refunded from the concerned Govt Authority and submits documentary proof for the same as the GE may require.

SPECIAL CONDITIONS

26. **EMPLOYESS PROVIDENT FUND SCHEME**: Under the GoI Employees Provident Fund Act/Scheme 1952, the contractor shall have a EPF Code No and all labours employed by him or sub-contractor shall be enrolled as member of Provident Fund and should be given Universal Account Number (UAN). A certificate to effect that all working labours employed for execution of work are registered for EPF and due distribution have been credited into their account, shall be furnished by contractor along with running advances or final bill.

27. **DAMAGE TO EXISTING WORK**: Any damage done to the existing structure during the execution of work shall be made good by the contractor at his own cost and the site of work left clean and tidy on completion. Rectification/reinstatement etc. shall conform to the standard of materials originally used in the work and finished work shall match with existing work in all respects to the entire satisfaction of the GE. In case of any dispute on this account the matter shall be referred to the CWE, whose decision in writing shall be final and binding.

28. **SITE LABORATORY AND TESTING OFMATERIALS**

(a) **A LEVEL TESTS FOR WORKS COSTING Rs100 LAKHS AND ABOVE:-** The contractor shall set up site labor at or of minimum size 3mx 4m with proper ventilation and lighting for testing of materials for A level tests as tabulated below. The contractor shall arrange all equipments/ machines at his own cost with prior approval of GE. The cost shall be included in the costs quoted by the contractor. The contractor shall employ a competent technical representative as approved by the GE for the purpose of testing and all such tests shall be carried out in the presence of Engineer-in-Charge. The successful test results thereof shall be recorded and signed jointly by the contractor and Engineer-in-Charge. The charges of these tests i.e. A level tests carried out in site laboratory of the contractor shall not be recovered. However in case a test which was specified to be carried out in site laboratory and which could not be carried out in site laboratory due to some reason, such test shall be carried out in Zonal/Command lab/ any other laboratory approved by GE and the recovery will be made at the rates given in Appendix for tests carried out in Zonal/Command lab or otherwise at the actual rates in case the tests are carried out in other labs. Testing charges will be deducted from the RAR's irrespective of the tests results for such tests.

(b) **A LEVEL TESTS FOR WORKS COSTING UP TO Rs100 LAKHS:-**The contractor may set up site laboratory at his option for works costing up to 100 lacks. The other stipulations will be same as specified in preceding Para (a). However in case the contractor has not set up the site laboratory and tests are carried out in Zonal or any other laboratory approved/ set up by GE, the recovery shall be made at the rates given in table below for tests carried out in Zonal lab or otherwise as per actual rates in case the tests are carried out in other labs. Testing charges will be deducted from the RAR's irrespective of the test results for such tests.

(c) The recoveries on account of testing charges wherever applicable shall be effected from the running account payments due to the contractor payable after completion of the respective test or whenever the test is due whichever is earlier.

29. **POST-CONSTRUCTION ANTI-TERMITE TREATMENT**

29.1 Anti-termite chemical treatment shall be carried out to the buildings as specified under BOQ and in particular specifications. The work of Anti-termite treatment shall be carried out either by a specialist firm or by any other firm/agency enlisted for carrying out anti-termite treatment work in MES and who is holding a valid license under Insecticides Act to carryout the anti-termite pest control operation, with the prior written approval of CWE.

29.2 The contractor shall stand a guarantee to the Government for a period of ten years from the certified date of completion of the work for the effectiveness of the treatment carried out. The contractor shall furnish a written guarantee for the effectiveness of the treatment for a period of ten years in favour of GE.

29.3 Chemical used for Anti-termite treatment shall be treated as proprietary item and the quantity procured shall be recorded in measurement book duly signed by Engineer-in-Charge and the contractor.

29.4 Should GE at any time during construction or reconstruction or prior to the expiry of the guarantee period, find that the buildings have been infected with termites, the contractor shall on demand in writing from the GE specifying the building complained of notwithstanding that the same may have been inadvertently passed, certified and paid for, undertake to carryout such treatment as may be necessary forthwith to render the building(s) free from termite infection at his own expense, till expiry of the guarantee period. In the event of his failure to do so, within the specified period to be specified by the GE in his aforesaid demand, the GE may undertake such treatment at the risk and expense in all respects of the contractor. The liability of the contractor under this condition shall not extend beyond the period of ten years from the certified date of completion, unless the GE had previously given notice to the contractor to rectify the defects.

SPECIAL CONDITIONS

29.5 The contractor shall provide a plate/board of dimensions 45cm x 30cm x2cm thick in CM (1:4) on all the buildings as directed by the Engineer-in-Charge and engrave there on the contract No, Name of the Agency executing the termite treatment and the date of expiry of the guarantee period.

29.6 The amount calculated as per the scales given for individual security deposit on the total cost of Anti-termite treatment at contract rates adjusted to nearest hundred rupees shall be retained by Govt from the contractor's dues against the guarantee period for Anti-termite treatment. This amount shall be released only after satisfactory expiry of guarantee period. The contractor may however furnish a fixed deposit receipt in lieu, from a scheduled Bank in favour of GE.

29.7 The Security Deposit referred to be held from the contractor against the guarantee period for anti-termite treatment shall be calculated as per the scale laid down for calculating Earnest Money deposit enhanced by 25% on the amount equal to 10% of the cost of estimated cost of anti-termite treatment at the contract rates to for the buildings for which treatment is to be done and may be refunded to the contractor after the expiration of the period off ten years from the certified date of completion by the GE provided always that the contractor shall first have been paid the final bill and have rendered a No Demand Certificate (IAFA-451). The facility of furnishing fixed deposit receipt in lieu of the sum to be retained as security may be accepted.

29.8 Condition 10, 46 & 68 of the General Conditions of the Contract (IAFW-2249) shall be deemed to be amended to the extent mentioned above.

29.9 In addition to the above special conditions, following points shall also be ensured/ complied:-

(a) The chemicals shall be procured only from manufactures or their authorised agents/dealers.

(b) Chemicals brought to site in sealed containers bearing ISI specification marks shall only be permitted to be used.

(c) Chemicals shall be stored carefully at site, seals of the containers shall be broken only in the presence of the Engineer-in-Charge (EIC). Empty containers should be got removed from the site promptly. If on any particular day the contents of full containers could not be used in the work, the containers should be got sealed at the end of the day in the presence of EIC and opened when required also in the presence of the EIC.

(d) The Engineer-in-Charge (EIC) should ensure that paid vouchers are produced by the contractor or the agency executing the work, for the full quantity of chemical required and brought to site and a record of such vouchers should be kept by the EIC.

(e) Tests may be carried out in a National Test House Alipore, Calcutta or IIT or any Govt approved Lab at the discretion of the GE, the chemical brought by the contractor/agency executing the work, to satisfy that spurious materials are not being used. Expenditure incurred on account of getting the chemical tested shall be borne by the contractor.

(f) A copy of ISI 6313 should be made available at site at all times during execution of the work.

Signature of Contractor
Dated: _____

AGE (Contracts)
for Accepting Officer

PARTICULAR SPECIFICATION

1. GENERAL

(a) Work under this contract shall be carried out in accordance with Schedule `A' (BOQ), Particular Specifications, drawings if any, General Specifications and other provisions contained in MES Schedule Part-I (2009) and Part-II (2020) read in conjunction with each other. Where specifications for any item of work are not given in MES Schedule or in these particular specifications, specifications as given in relevant Indian Standard Code of Practice shall be followed. In addition to makes indicated in the BOQ, a list of manufacturer is enclosed at Appx to this PS. The contractor shall procure the material from the manufacturer indicated there in.

(b) **MATERIALS:** Unless specific makes/manufacturers are specified in the tender documents all the materials to be procured by the contractor for incorporation in the work under this shall be with ISI certifications mark. For materials of specific make/manufacturer's names refer Appendix enclosed with tender here- in -after. ISI marking shall not be mandatory for such items where makes/ manufacturers have been specified unless the items of makes/manufacturers specified are available both with IS marking and without ISI marking. If any material is not manufactured with IS certifications mark in the country, it shall be confirming to relevant IS and other specifications specified elsewhere and shall also conform to samples displayed in CE's/CWE's/GE's office as applicable. The makes specified in Schedule 'A' (BOQ) shall take precedence over the makes specified in Appendix to these particular specifications.

(c) **LOCAL MATERIALS:** Irrespective of actual distance involved, local materials such as stones, aggregates, sand, road metals etc shall conform to or superior to the specifications given here in after and to the samples kept in the office of CEs/CWEs/GEs. The tenderers are advised to inspect these samples. The tenderers shall be deemed to have inspected the samples and have full knowledge thereof whether they inspect them or not before quoting their tender.

(d) **SAMPLES OF MATERIALS:** Refer condition 10 of IAFW-2249. Specific requirements regarding dimensions, strength, weight and finishes, as per IS, MES SSR and the particular specifications given hereinafter vis-à-vis actual properties check, tests carried out, reference to test certificates and markings, etc based on which samples of each materials are approved as Conforming to relevant specification shall be recorded in the sample approval register.

(e) **MATERIAL PROCUREMENT**: The contractor shall not procure materials unless the samplesare first got approved from the Garrison Engineer. All material costing more than Rs 1.00 Lakh shall be procured directly from the manufacturer. The contractor shall ensure that the materials incorporated in the work are identical with the approved samples. If, the same is not identical with the approved samples, the same shall be replaced by the contractor to the entire satisfaction of GE without any extra cost of the Govt. All material incorporated shall comply with relevant Indian Standard.

(f) **RECORD OF MATERIALS**: Refer Special condition 27. The quantity of all the proprietary materials shall be recorded in measurement books and signed by the contractor and the Engineer-in-Charge. The contractor shall procure all the materials (where specific makes/manufacturer's specified) directly from their manufacturer's or from their authorised dealers only. The contractor shall produce to the GE original printed and machine numbered purchase vouchers/invoices including manufacturer's test certificate for all the materials mentioned here-in-before in Special Conditions.

(g) **STANDARD OF WORKMANSHIP:** The workmanship shall be as per standard engineering practices and as specifies and directed. All relevant Indian Standard shall be applicable to entire CA items and complied with.

PARTICULAR SPECIFICATION (Contd….)

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2. EXCAVATION AND EARTH WORK:

(a) **EXCAVATION**: Unit rates for items of Schedule 'A' Part-I shall include for excavation and earth work in soft soil. If rock (soft/hard/disintegrated) is met at site, contractor shall immediately notify the fact to the GE in writing, who shall after due verification, regularize the change through a proper deviation order.

(b) **FILLING IN TRENCHES/UNDER FLOORS:** The approved earth obtained from excavation in foundations shall be used for filling in trenches, under floors and any other situation. Filling around pipes after the pipes are laid and tested shall be earth-free from pebbles/stones. Filling under floors/sides of trenches shall be in layers not exceeding 250mm thick, and each layer shall be watered and well rammed. Surplus soil shall be removed and spread at places as directed by the Engineer-in-Charge, to a distance exceeding 250 metres but not exceeding 500 metres, and leveled as directed by Engineer-in-Charge.

(c) **TRENCHES FOR FOUNDATIONS AND PIPES:** The excavation shall be restricted to dimensions shown on the drawings. Excavation made, if any, in excess of required depth or or width same shall be made good by the contractor with cement concrete mix (1:5:10) without extra cost to the Govt.

(d) **BAILING/PUMPING OF WATER:** All water that may accumulate in trenches of foundation, column footings, rafts etc, during the progress of work from rains, sub soil water, spring etc shall be bailed/pumped out or other wisere moved all as per clause 3.17 of MES SSR Part-I and nothing extra.

(e) **HARD CORE:** Unless otherwise specified, the hard core shall be of hard broken stone or boulders broken to gauge not exceeding 63mm. The material of hard core shall be well graded for providing dense and compact sub grade. Hard core shall be deposited, spread and leveled in layers not exceeding 15cm thick and watered and well rammed to a true surface and compacted with sufficient fine material. The thickness of hard core specified or indicated in the drawing, is the thickness after consolidation. Where thickness has not been mentioned it shall be 150mmconsolidated.

3. CEMENT: - Cement to be used in all types of works shall be ordinary Portland cement Grade 43 conforming to IS: 8112-1989(Latest Revision) or Portland Pozzollana cement confirming to IS 1489-1991. The Contractor shall procure cement for all sections of Schedule ‘A’ required for this contract as the work precedes at site.

Cement shall be procured by the contractor from the main producers of cement enumerated as under:

Srl No.	Company Name	Address	Type of Cement
1.	The Associated Cement Companies Ltd Brand “**ACC**”	414-421, Splendor Forum (4thFloor), 3, District Centre, Jasola, New Delhi-110044	All
2	M/s Ultra tech Cement Ltd Brand:” **ULTRATECH**	‘B‘ Wing 2nd Floor Mahakali Caves Road, Andheri (East), Mumbai-400093 Ph- 022-66917800	All
3.	The India Cement	Dhun Building, 827, Anna Salai, Chennai-600002	All
4.	M/sDalmia Cement Bharat Ltd)Brand	Dalmiapuram Distt-Truchirappalli, Tamil Nadu-621651	All
5.	Century Cements Brand “**Century**”	Industry House, 159 Chruch gate Reclamation, Mumbai-400020 Ph-022-22023936	All

PARTICULAR SPECIFICATION (Contd….)

6.	Saurashtra Cement Brand " **SAURASHTRA**"	Gala No. A-1 Ground Floor, UdhyogSadan No 3 MIDC, Central Road, Andherei (East), Mumbai- 400093 Ph- 022 32955557/67	All
7.	The Ramco Cements Ltd (formerly Madras Cement)	Auras Corporate Centre, 98-A, Dr Radha Krishnan Salai, Mylapore, Chennai-600004, Ph-044- 28478666	All
8.	Mangalam Cement Ltd. Brand "**MANGLAM**"	PO Adityanagar, Morak, Distt-Kota, Rajasthan-326520 Mob-9351468076	All
9.	Birla Corporation Ltd Brand "**BIRLA**"	Birla Building (3rd & 4th Floor) 9/1 R.N Mukherjee Road Kolkata-700001 Ph-033-30573700	All
10	Orient Cement Brand "**ORIENT**"	5-9-22/57/D, 2nd and 3rd Floor, GP Birla Centre, Adarsh Nagar, Hyderabad-500063 Ph- 04423688600	All
11.	NuvocoVistas Corporation Ltd **(**FormerlyLafarge Cement) Brand **"NUVOCO"**	Equniox Business park, Tower-3, East Wing, 4thFloor, LBS Marg, Kurla (west), Kurla, Mumbai,Maharashtra-400070	All
12.	Shree Cement Brand "**SHREE**"	Bangur Nagar, Beawar, Distt-Ajmer, Rajasthan- 305901 Ph-01462228101/06	All
13.	JK Cement Brand "**JK**"	Kamla Tower, Kanpur-208001	All
14.	JK Lakshmi Cement Ltd Brand "**JK LAKSHMI**"	Jaykaypuram, Dist-Sirohi, Rajashthan Ph-02971 244409/10	All
15.	Jaypee Rewa Cement Brand "**JAYPEE**"	Jaypeenagar, P.O Jaypee Nagar, Rewa-486450	All
16.	Ambuja Cement Ltd. Brand "**AMBUJA**"	Kodinar, PO-Ambujanagar, Taluka-Kodinar, Distt-Junagadh, Gujrat-362715 Ph-02795237000	All
17.	M/s Shree Guru Kripa Cement Pvt. Ltd. Brnad "**SARTAJ**"	4/4 Trikuta Nagar Jammu Ph-0191-2472043.	(a) OPC43 (b) PPC

PARTICULAR SPECIFICATION (Contd….)

18.	M/S Parasakti CementsLtd. Brand "**PARASAKTICEMENT**"	123/3RT, Plot # 8-3-214/21 Srinivasa Nagar Colony (west) Hyderabad-500038 Tel-040-44119100/200 Fax-040-23747562	(a) OPC43 (b) PPC
19.	M/s My Home Industries Ltd. Brand "**MAHA CEMENT**"	9th Floor, My Home Hub, Madhapur, Hyderabad- 500081 Ph- 040-66929696 Fax-040-66929797/98	(a) OPC43 (b) PPC (c) PSC
20.	M/s Chettinad Cement CorporationLtd. Brand "**CHETTINAD CEMENT**"	4th Floor, Rani Seethai hall Building, 603, Anna Salai Hall, Chennai-600006 Tele No. 044- 42951800, Fax No.044-28291558 Email: info@chettinalcement.com Website : www.chettinadcement.com	(a) OPC43 (b) PPC
21.	M/s Sanghi Industries Ltd Brand "**SANGHI**"	10thFloor, Kataria Arcade, Off SG Highway, PO- Makarba, Dist-Ahmedabad, Pin-380051, Mo- 09825803690, Tel-079-26838000, Fax-079- 26838111, Website :- www.sanghicement.com	(a) OPC53 (b) PPC
22.	M/s Wonder Cement Ltd Brand : **"WONDER CEMENT"**	17, Old Fatehpura, SevaMandir Road, Udaipur- 313004, Rajasthan (India), Tele:- 91-294-33991133, Fax-91-294-3006333, Email Corp.office@wondercement.com	(a) OPC43 (b) OPC53 (c) PPC
23.	M/s Kesoram Industries Ltd Brand : **"BIRLA SHAKTI"**	Cement Division, office No 613 to 616, white House, Block-III, VI Floor, 6-3-1192/1/1, Kundanbagh, Begumpet, Hyderabad-500016, Telengana, Tele : 91-40-4334-4555, Fax-91-40- 4334-4534 Email : corp.office@wondercement.com	(a) OPC43 (b) OPC53 (c) PPC
24	M/s J&K Cement Corp, Kathua **(Only for J&K region)**	Industrial Extn Area, Kathua-184102, J&K	OPC
25	M/s Shiva Industries, Kathua **(Only for J&K region)**	Industrial Extn Area, Kathua-184102, J&K	OPC
26	M/s Star Cement Meghalaya **(Only for CE Shilong Zone & CE (AF) ShilongArea)**	Lumshnog, PO-Khaliehriat, Distt-JantiaHills,Meghalaya, Ph-03655 278215/16/18	(a) OPC (b) PPC
27	M/s Meghalaya Cements Ltd Brand "**TOP CEM**" **(Only for Eastern region)**	Lohia House, MG Road, Factory Bazar, Guwahati-781001, Tel: 0361-260367/671	(a) OPC (b) PPC
28	M/s Green Valliey Industries Ltd Brand "**MAX CEMENT**" **(Only for NE region)**	Vill – Nongsning, P.O- Chiehruphi, P.S- Khliehriat, Dist-Jaintia Hills, Meghalaya-793200, Tele- 91-361-2465482/84 Fax : 91-361-2465481 Email : info@greenvalliey.com	(a) OPC43 (b) OPC53 (c) PPC

PARTICULAR SPECIFICATION (Contd….)

4. The particulars of the manufacturer / supplier of cement alongwith the date of manufacture shall be produced by the contractor for every lot of cement separately. The documents in support of the purchases of cement shall be produced before the Engr-in-Charge for verification by the GE.

4.1 The following conditions shall have to be satisfied before PPC can be allowed to be used in the work:-

(a) PPC should meet the strength criteria as perIS-8112-1989.

(b) The minimum period before striking off formwork while using PPC shall be as given in PS Clause 3.8hereinafter.

(c) In cold climate regions where temperature is lower than 15 ^{o} C only OPC shall be used.

(d) Mixing of OPC & PPC shall not be allowed in a work. However with the strict control of ground executives different buildings can have different types of cement. hile procuring PPC, the following requirements are to be ensured and certificate to the effect is to be obtained by the contractor for each batch from the manufacturer and the same shall be submitted to the GE for approval to procure PPC.

(i) The quality of fly ash is strictly as per IS-1489 (Part-I) –2002

(ii) Fly ash is inter-ground with clinker and not mixed with cement.

(iii) Dry fly ash is transported in closed containers and stored in Silos. Only pneumatic pumping should be used.

(iv) The fly ash received from thermal power plants using high temperature combustion above 1000°C should be used.

4.2 **TESTING**:-

The contractor shall submit the manufacturer's test certificate in original alongwith the Test Sheet giving the result of each physical test as applicable and the chemical composition of the cement or authenticated copy thereof, duly signed by the manufacturer with each consignment clearly bringing out lot No. The Engineer-in-Charge shall record these details in the Cement Supply & Acceptance register as given in Appx 'E' here-in-after after due verification. The GE shall also organize in dependent testing of random samples of cement (both physical and chemical properties) drawn from various lots for each consignment to cement brought out by the contractor before incorporation in the work from the National Test House, SEMT, Regional Research Laboratories, Government approved laboratories, as per IS: 3535 (Method of sampling Hydraulic cement), IS:4031(Method of physical test for Hydraulic Cement) and IS:4032(Method of chemical an alysis of Hydraulic cement.)

4.2.1 Following mandatory tests shall be carried out for cement procured by the Contractor:-

(a) Initial and final setting time (b) Soundness test

(c) Compressive strength test at 3, 7 & 28 days as specified in relevant IS code.

4.2.2 The cement shall conform to chemical requirements and physical requirements as specified in respective IS. The tests carried out as per provisions of IS codes specified herein before shall be the criteria for acceptance of cement by Engineer-in- Charge. If samples from a lot/lots are not within the acceptance limits of Indian standard the lot/lots shall be rejected without any claims or compensation to the contractor for the lot/lots purchased. The contractor shall replace the lot/lots with the fresh one, which shall be tested again for acceptance. The cost of all tests carried out on cement before acceptance for incorporation in the work shall be borne by the contractor whether the results are acceptable or not.

4.2.3 **STORAGE**:-

Cement shall be stored over dry platform at least 20cm high in such a manner as to prevent deterioration due to moisture or intrusion of foreign matter. In case of storerooms, the stock should be at least 20cm above from floors and 60 cm away from walls in addition to precautions specified in clause 4.3.1 on Srl Page 51 & 52 of SSR (Part-I). Inspections shall be carried out once a day by the Engineer-in-Charge. It shall be ensured by the Engineer-in-Charge that tested and untested cement are segregated and stored separately with distinct identification. The cement godown shall be provided with two locks on each door. The key of one lock at each door shall remain with the Engineer-in-Charge or his representative and that of the other lock with the contractor's authorized representative at site of works so that cement is removed from the godown only according to daily requirement with the knowledge of both the parties.

PARTICULAR SPECIFICATION (Contd….)

4.3 DOCUMENTATION:-
The contractor shall submit original vouchers from the supplier for the total quantity of cement supplied under each consignment to be incorporated in the work. All consignments received at the work site shall be inspected by GE alongwith the relevant documents before acceptance. The original vouchers and the Test Certificates shall be defaced by the Engineer-in Charge and kept on record in the office of GE duly authenticated and with cross-reference to the control number in the cement Supply & Acceptance Register. The cement supply & Acceptance Register will be signed by Supdt B/R Gde-I/ JE (Civil), Engineer-in-Charge, GE and the Contractor. The Accepting Officer may order a Board of Officers for random check of cement and verification of connected documents. The entire quantity of all types cement shall also be suitably recorded in the Measurement Book for record purpose not to be abstracted before incorporation in the work and shall be signed by the Engineer-in-Charge and the contractor.

4.5 SCHEDULING OFSUPPLY:-
Schedule of procurement of cement shall be finalized by the contractor with GE and shall be in corporate in the CPM chart so that procurement is in accordance with the progress con template in the CPM prepared. The Complete requirement of cement shall be worked out before making any RAR payment and procurement to cement by the contractor shall be completed sufficiently in advance of the execution of work.

4.6 AGGREGATES: Aggregates for concrete work shall conform to specifications all as specified in clauses **4.4.1 to 4.4.7** of MES Schedule Part-I.

(i) **FINE AGGREGATES (SAND):**Fire aggregates for all concrete work shall be naturally occurring river sand/crushed sand confirming to clause 4.4 of SSR Part-I. Sand confirming to grading zone IV of IS- 383 shall not be used for RCC work.

(ii) **COARSE AGGREGATES:** Coarse aggregates for all concrete work in all situations shall be graded crushed stone aggregate (crushed in mechanically operated crusher) of approved quality, irrespective of whether indicated specifically in the description of items or otherwise or SSR provisions. Nallah shingle will not be used even for plain cement concrete work in any situation. Coarse aggregate shall be obtained from queries.

(iii) **GRADING OF COARSE AGGREGATE**: Grading of coarse aggregate (Broken/crushed stone) unless otherwise specified shall be as follows:-
(aa) For all reinforced/plain cement concrete of thickness up :12.50mm graded to 25 mm or under.
(ab) For all reinforced cement concrete work over 25mmthick.:20mm graded (ac) For plain cement concrete over 25 mm and under 75 mm thick. : 20mm graded (ad) For plain cement concrete 75 mm thick and over.:40mmgraded

4.7 WATER PROOFING ADMIXTURE: - Irrespective of specifications given elsewhere water proofing for all cements concrete/mortar where indicated shall be integral (liquid) water proofing admixture of make as specified in respective appendix attached to these particular specifications and shall be mixed in the proportion recommended by the manufacturer. Incase of deviations, water proofing admixture shall be considered @1% by weight of cement.

PARTICULAR SPECIFICATION (Contd….)

4.8 **WATER:** Water shall conform to the requirements stipulated in IS-456-2000.

4.9 **MIX OF CEMENT CONCRETE**

4.9.1 Cement concrete wherever shown specified on drawings shall be of the following mix unless otherwise mentioned elsewhere: -

(a) PCC lean concrete in foundations of raft slab, column Footing, brick walls, brick pillars under brick steps, Dwarf wall and lean concrete not mentioned elsewhere	PCC (1:4:8) type D-2 (Nominal Mix)
(b) Sub base of floors except as mentioned in the above (a)	PCC (1:4:8) type D-2 (Nominal Mix)
(c) Lean concrete under floors	-do-
(d) Plinth protection including toe	PCC (1:3:6) type C-2 (Nominal Mix)
(e) PCC bed blocks, PCC bedplates and concrete around pipes, PCC block (Other than for hold fast).	PCC (1:3:6) type C-1 (Nominal Mix)
(f) PCC blocks for hold fasts, PCC cills, PCC kerbs, coping, benching and any other PCC item not specifically mentioned elsewhere	PCC (1:2:4) type B-1 (Nominal Mix)
(g) All RCC works 40 DUs	M-30 Design Mix

4.9.2 Minimum cement content and max water cement ratio for PCC nominal mix of various grades shall be as per Table 5 and 6 of IS-456-2000.

4.10 **DESIGN MIX CONCRETE**

4.10.1 Where the concrete is specified by grade i.e. M-25, M-30, M-35 and M-40 the same shall be design mix. The design concrete mix shall be as per IS 456: 2000, IS-10262-1982, SP-23 (S &T) 1982 and as specified hereinafter. Design Mix M-25 & M-30 shall be carried out on recommended guide lines for concrete mix design with following stipulations:-

(a) Grade of Concrete : M-25/M-30

(b) Characteristic compressive strength Compliance requirement : As per IS-456: 2000 Table 11

(c) Type of cement Port land pozzolona cement : Ordinary Portland Cement 43 grade or

(d) Maximum size of aggregate : 20mm (Crushed aggregate) for sections more than 75 mm and 10mm (Crushed aggregate) for sections less than or equal to 75 mm

PARTICULAR SPECIFICATION (Contd….)

(e) Type of exposure	: Moderate: as per IS-456: 2000 Table 3
(f) Maximum water cement ratio	: 0.45
(g) Degree of quality Control	: Very Good(Refer table 40 of SP23- 1982)
(h) Degree of workability	: As per IS-456: 2000 with slump for pumped and non-pumped concrete
(i) Minimum quantity of cement	: As per IS 456: 2000 Table-5.
(j) Admixture	: As per manufacturer's instructions

4.10.2 At the time of tendering, the contractor after taking into account the type of aggregate, plant and method of laying, he intends to use, shall allow for the water cement ratio which he consider will achieve the strength requirement specified and will produce workability which will enable concrete to be properly compacted for its full depth.

4.10.3 Actual proportion of cement, aggregate and water to obtain the required strength shall be determined by preliminary tests as specified hereinafter. The water cement ratio may be reduced in the trial mix design but not increased.

4.10.4. ----------BLANk----------

4.10.5 As soon as possible, after receiving the order to commence the work, the Contractor shall procure sufficient quantities of the aggregate and cement samples for testing and mix design. The optimum mix to achieve the target mean strength shall be determined in the laboratory Conditions and the Concrete shall conform to relevant IS Specification. Design mix shall got done from National Test house, IIT or any other Govt approved laboratory approved by GE. Cost of materials and transportation and testing charges are to be borne by the Contractor and same shall deemed to be included in the rates quoted by the Contractor. Details/results of trial mixes from the engineering establishment along with their recommendations regarding quantities of cement, water, sand and coarse aggregates to be used in the work to achieve requisite strength in the field at 28 days shall be submitted by the contractor to the GE for final approval.

4.10.6 Govt approved lab as approved by MES or NABL accredited lab.

4.10.7 The approval of the GE will not absolve the contractor of his responsibilities to obtain the required minimum strength of concrete at any stage.

4.10.8 Contractors unit rates shall include for the above provisions and as brought out here-in-after. Further, in case there is any change in source of local materials viz sand/aggregates or grade of cement etc, the procedure as in Para 3.6.5 & 3.6.6 here-in-before shall be repeated.

4.10.9 Quality Assurance :- Measure as per Clause 10 of IS-456.

4.10.10 In proportioning design mix concrete, the quantity of both cement and aggregate shall be determined by mass and all in accordance with clause 9.2 and sub clauses thereon of IS-456-**2000** as applicable. All the materials shall be tested at a frequency as laid out in relevant IS. Use of weigh batcher is compulsory for all grades of concrete.

4.10.11 MIXING OF CONCRETE **OTHER THAN DESIGN MIX**

The contractor shall use power driven mechanical concrete mixer of adequate capacity with hopper of approved type of mixing. The coarse and fine aggregate for mixing of cement concrete shall be put in the hopper after measuring the same by weigh batching, the drum shall be loaded with about the half of the coarse aggregate, then with the fine aggregate, then with the cement and finally with remaining coarse aggregate at the top. Water shall be added immediately before the rotation of the drum is started. The period of mixing shall be not less than 2 minutes after adding all the materials in drum and shall continue till the resulting concrete is uniform in appearance.

4.10.12 WEIGH BATCHING:

(i) The quantity of cement and aggregate shall be determined by weight. In no case volume batching of aggregates based on weight- volume conversion in the mix design shall be allowed. Water shall be either measured by volume in calibrated tanks or weight. Due care shall be given to adhere to maximum free water cement ratio indicated here in before. All measuring equipment shall be maintained in clean serviceable condition and their accuracy periodically checked. In this connection also refer clause 4.11.3.1 to 4.11.3.5 of SSR Part-I.

(ii) Quantity of concrete mixed in any one batch shall not exceed the rated capacity of the mixer. The whole of the mixed batch shall be removed before adding new batch. Concrete mixed above shall not be modified by addition of water or otherwise in order to facilitate handling or for any other purpose.

(iii) On cessation of work and other stoppages exceeding 20 minutes, the mixer and other plant used for the wet mix shall be thoroughly washed with clean water. Pick up and throw over blades in the drums of the mixer, which are worn down 20mm or more in depths shall be replaced with new blades.

4.11 **Ready Mix Concrete (RMC)**

4.11.1 All Design Mix Reinforced Cement Concrete work covered under the subject work e.g. foundations, walls, columns, beams, slabs, etc shall be with **Ready Mix Concrete (RMC)** of design Mix as specified here in clause 4.11.2 from **approved manufacturers having fully automatic batching plants installed of capacity over 30 cum/hr** with tested and calibrated water meter, control panel and provision for computer generated batch report all as per IS-4925 (amended up to date). However, for smaller quantity i.e. for concreting to lintels, chajjas, shelves, etc mechanical mixer may be used with prior approval of the GE and mixing of concrete shall be as per clause 10.3 of IS:456-2000.

4.11.2 The relevant details of Design Mix Concrete as specified earlier shall be applicable along with the following specific details for Ready Mix Concrete**.** The contractor shall use ready mix concrete (RMC) of grades as specified. RMC shall conform to the requirements as per IS-4926 (amended up to date) and IS 456-2000.

4.11.3 RMC shall be procured by the contractor from any one of RMC manufacturer given below as per IS-4925 (amended up to date), having fully automatic computerized batching plant of capacity over 30 cum/hr with tested and calibrated water meter, control panel and provision for computer generated batch report all as per IS-4925 (amended up to date), as approved by GE after verification of the plant and various infrastructure, facilities testing and quality control of materials and concrete at the plant lab and their method of transportation and placing of concrete at site, etc:-

(a) M/S Larsen & Toubro Ltd: ECC Division (L&T Concrete) (Now M/S Lafarge)
(b) M/s RDC Conc. India Ltd.
(c) M/S Grasim Industries Ltd (UltraTech Concrete)
(d) M/S Associated Cement Co. Ltd
(e) M/s Shiala Enterprises

4.11.4 Alternatively, contractor shall install at site one or more fully automatic computerized batching plant of capacity 12 to 20 Cum/Hr with tested and calibrated water meter, control panel and provision for computer generated batch report all as per IS-4925 (amended up to date). Qualified civil engineer and operator will be employed by the contractor & same shall be approved by GE.

4.11.5 In proportioning design mix concrete, the quantity of both cement and aggregate shall be determined by mass and all in accordance with clause 9.2 and sub clauses thereon of IS-456-**2000** as applicable. The mix design shall be revised with change in properties of cement and aggregate. All the materials shall be tested at a frequency as laid out in relevant IS. Use of weigh batcher is compulsory for all grades of concrete. The Contractor shall ensure that concrete is manufactured under computer-controlled operations and pumped and placed at site using sophisticated equipment's and methods. For that the various operations of production of concrete like weigh batching of all ingredients, adjustment for moisture, water cement ratio control, admixture dose if required, auto timer for mixing, delivery slips etc shall be fully computerized. This record of quality assurance shall be submitted to the GE and shall be kept on record. Fully automatic batching plant at site shall conform to relevant IS and shall be of standard make as approved by GE. The concrete produced from the batching plant shall be pumped using pumps of suitable capacity. Admixtures/plasticizers of make as given below can be added (as per mix design) to make concrete plastic for facilitating pumping.

PARTICULAR SPECIFICATION (Contd….)

4.11.6. Specification for all materials / ingredients, etc as given earlier shall hold good for RMC.

4.11.7 ADMIXTURE / plasticizers : Admixtures / plasticizers @ not more than 1% of weight of cement at no extra cost to the Department may be used conforming to IS 9103-1999 (Revised) and shall be from any one of the following firms:-

(a) MC Bauchemic (India) Pvt Ltd

(b) Fosroc Chemicals (India) Ltd

(c) STP Speciality Chemicals Ltd

(d) CICO Technologies Ltd

(e) Fairmate Chemicals Pvt.Ltd.

(f) M/s Sika India Pvt Ltd

(g) M/s Thermax Ltd

(h) M/s MAPEI Const Product India Pvt Ltd

4.11.8 Initial mix design from RMC manufacture shall be submitted by the contractor in quadruplicate with complete data adopted for mix design along with test results of all materials and concrete. The initial mix design should take into account the aspects such as loss of workability and strength during transportation, till placement of concrete.

4.11.9 RMC manufactured and supplied shall be in conformity with the initial mix design approved the GE. Any change shall be made only with approval of GE after necessary revision in design mix.

4.11.10 The contractor shall ensure that all facilities are made available by the manufacturer of RMC at the plant for the departmental officials to inspect the materials incorporated, test carried out for all materials, concrete etc. Copies of all tests carried out for materials used/concrete shall also be made available to the department.

4.11.11 The concrete shall be transported in concrete transit agitators conforming to IS-5892.

4.11.12 The concrete shall be delivered and placed by pumping using BOOMER. The decision as to whether the RMC can be placed by pumping or not shall be decided by GE whose decision shall be final and binding. The tenderer will have no claim any extra on this account.

4.11.13 The pump and pumping arrangement shall be inspected and approved by GE, while approving the RMC manufacturer.

4.11.14 No water / admixture shall be allowed after initial mixing of concrete at the plant. The use of PPC / Fly ash / mineral based admixture in RMC shall not be permitted.

4.11.15 Slump test shall be carried out at site by the department in the presence of contractor and representative of RMC manufacturer. The concrete shall be placed in position within the designed initial setting time and at the end of initial setting time, the remaining concrete shall be rejected.

4.11.16 In addition to the tests carried out by RMC manufacturer at the plant site, sampling and testing of concrete shall be carried out at the site before delivery as per IS-456-2000 by the department along with the representative of the contractor and RMC manufacturer. However, the cost of testing shall be borne by the contractor. The concrete, which does not meet the requirement of acceptance criteria as per IS-456-2000 shall be rejected and the contractor shall make good the same at his risk and cost.

PARTICULAR SPECIFICATION (Contd….)

4.11.17 The records indicating following details duly signed by the competent representative of RMC manufacturer / contractor and Engineer-in-Charge shall be maintained.

(a) Date of delivery

(b) Time of mixing

(c) Time of delivery at site.

(d) Quantity and Grade of RMC.

4.11.18 The contractor shall enter into an agreement with the approved RMC manufacturer to ensure compliance of above aspects. The overall liability to the department, for the RMC supplied shall rest with the contractor only. The department shall not be liable for any breakdown or problems with the RMC manufacturer.

4.11.19 The department will recommend to the traffic Police to issue entry permits for the entry of vehicles into the working area. However, absence of such permits will not be treated as reason for delay in completion of work.

4.11.20 SAMPLING AND ACCEPTANCE CRITERIA FOR CONCRETE : The criteria for sampling and acceptance of the concrete shall be as per Para 15.1 to 16.1 of IS-456:2000, (Code of practice for plain and reinforced concrete). The minimum frequency for the work tests shall be as specified therein and as specified here in after.

4.11.21 PACKING AND TRANSPORTATION OF SAMPLES The Contractor must, at his own expenses, properly pack the samples. The Contractor shall bear the cost of transportation of the samples required to be tested from site of work to the laboratory of Chief Engineer Delhi Zone or any other laboratory.

4.11.22 TESTS**:** The following tests shall be carried out during the execution of work. Tests shall be carried out in accordance with IS-516 and IS-1199. The contractor shall provide all facilities and equipment for casting and curing of test cubes.

(A) SLUMP TEST:
The frequency of the test shall be decided by the GE. The slump for the vibrated concrete shall be as specified in IS-456. The GE, however, reserves the right to vary the limit which will be ascertained at the time of deciding the mix design for each grade. Any batch from which a slump test is being made shall not be transferred to the places of laying until the slump in excess of the required for the particular batch is not removed from the site.

(B) COMPRESSIVE STRENGTH TEST :
(a)Three cubes shall be tested at 7 days and three at 28 days for crushing and the remaining three cubes shall be kept as reserved exclusively for Govt use for subsequent testing (if so desired by Govt) and preserved for one year from certified date of completion of work under the contract. The contractor cannot object testing of other cubes for the purpose as a matter of right.

(C) For the purpose of subsequent identification of the work test cubes, the concrete to which these pertains shall be cross referred and record of this maintained and signed by the Engineer-in-Charge and contractor or his authorized representative.

4.11.23 TRANSPORTING, PLACING, COMPACTION and CURING :

(i) It should be as per clause 13 of IS 456:2000. Concrete shall be transported without delay and incorporated in works at the position of laying within 20 minutes from the time of discharge from the mixer.

PARTICULAR SPECIFICATION (Contd….)

(ii) Mixed concrete shall be deposited in final position and solidly packed around reinforcement, carefully poured and consolidated by means of portable vibrators or mechanically operated and of the kind as suitable for a particular situation as directed by the GE. Care shall be exercised that no voids or honey comb pockets are formed. The concrete shall not be laid in position for more than 1 metre in height in one concrete operation. The compaction of concrete in RCC roof and floor slabs shall be done with plate vibrators.

(iii) Exposed surfaces of concrete shall be kept continuously in a damp or wet Condition by pounding or by covering with a layer of sack, canvas, Lenin and kept constantly wet for at least 14 days. The concrete shall be protected from premature drying for at least 8 days after pouring and shall be cured as directed by the Engineer-in-Charge for a period of not less than 14 days.

4.12 CURING AND MINIMUM PERIOD FOR STRIKING OF FORM WORK WITH POZZOLANA CEMENT USED

4.12.1 Where pozzolana cement is used for RCC works in various locations, the minimum period for curing and striking off form work shall be as under in lieu of that specified in clause 4.11.6.3 of MES Schedule Part-I :-

(a)	Curing: - Structural RCC work, PCC work, Plastering etc.	:	14 days
(b)	Striking off Form Work Walls, columns and vertical sides of beams	:	3 to 4 days
(c)	Slabs (Props left under)	:	8 days
(d)	Beams soffits (Props left under)	:	9 days
(e)	Removal of props to slab/beam		
	(i) Spanning upto 6 M span	:	16 days
	(ii) Spanning beyond 6 M span	:	22 days

(f) For cantilever portion of slab form work with support shall be retained until the completion of the casting of the entire frame work of the building.

4.12.2 In case of bad weather, periods mentioned above may be revised at the discretion of the Engineer-in-charge. The contractor shall be deemed to have considered the above provision before tendering and quote lump sum accordingly. No claim will be entertained if longer periods required for striking off form work and curing and all such effected matters is looked into upon the use of pozzolana cement as a result being issued to the contractor in full or in part or for bad weather.

PARTICULAR SPECIFICATION (Contd….)

4.12.3 Curing shall be carried out using pump of suitable capacity at no extra cost to the Govt. all as directed by Engineer-in-Charge.

4.12.4 **FORM WORK:** Form work shall comply with requirements of paras 4.11.6.1 to 4.11.6.5 and 7.15.1 to 7.15.12 of MES Schedule (Part-I). All form work for columns, beams/bands, slabs, walls etc shall be provided of steel. However the form work for the items other than mentioned above may be of steel, plywood or timber of adequate strength. In case of any deviation involving form work, the pricing shall be done on the basis of timber form work.

4.12.5 **FINISHING TO EXPOSED SURFACES OF CONCRETE:** Unless otherwise indicated elsewhere or shown in drawings, exposed (RCC/PCC)/Cement concrete surfaces, shall be plastered with cement and sand mortar (1:3), minimum 5mm thick finished even and smooth after removal of form work. Exposed surfaces of lintels, beams, columns etc, which are continuous with plastered surfaces of walls shall be plastered in the same manner as specified for the walls.

5 **PRECAST CONCRETE ARTICLES:** Cement concrete lintels with or without integrally cast Chhajjas upto 1.5 metre clear span, shelves, bed blocks/plates, cover slabs, fins and the like may either be precast or castin situ at the contractor's option, unless otherwise specified elsewhere. If precast, these shall be set in cement mortar (1:3). In case of deviation involving these items, pricing shall be done on the basis of cast-in-situ work.

6. **BRICK WORK:**

MATERIALS : (i) **Bricks** :- Bricks shall be kiln burnt, available locally or in the neighboring region, best quality conforming to the samples kept in CWE/GE's office with sub class 'B' old size bricks having minimum crushing strength of 75 Kg/Sqcm. Water absorption of bricks shall not exceed 20% when tested in accordance with IS-3495. Sampling and testing of bricks shall be carried out as per IS-5454 and IS- 3495. The nominal size of brick shall be 230mm x 115mm x 75mm. The tolerance in the dimensions shall be $\pm$ 8 percent.

PARTICULAR SPECIFICATION (Contd….)

6.1. **Material**: **CEMENT**: Refer clause under 'Concrete' hereinbefore.

6.2. **Material**: **SAND**: Sand for mortar shall be as specified in clause 5.4 of MES Schedule Part-I.

6.3. **WORKMANSHIP**: Workmanship shall be all specified under relevant clause of SSR Part-I.

7. **WOOD WORK AND JOINERY:** Timber for all joinery and woodwork shall conform to specifications given in clause 7.3 of the MES Schedule Part-I and shall be within the permissible limits of defects in clauses 7.4 and 7.5 of the MES Schedule Part-I. Timber shall be well seasoned whether air or kiln dried at the discretion of the contractor (except factory made door shutters which shall be kiln seasoned only) but without any price adjustments. The moisture contents of timber shall not exceed the limits laid down in clause 7.7 of MES Schedule Part-I.

8. **PARTICLEBOARD:** Particle board shall conform to the requirement of IS-3097(duly ISI marked) exterior grade (grade-1), specifications for veneered particle board (medium density). Particle board shall be of uniform thickness and density throughout and bonded with phenol formaldehyde synthetic resin (BWR type). Both the faces of particle board shall have sanded smooth finish. Where thickness of particle board has been shown as 20mm in the drawings / else-where, the same shall be amended to read as 18 /19mm.

9. **PLYWOOD:** All plywood where indicated shall be BWR exterior grade conforming to IS-303. Decorative plywood where indicated shall conform toIS-1328.

10. **PRELAMINATED PARTICLE BOARD:** Pre-laminated particle board where indicated shall be factory impregnated exterior grade One or both side decorative type lamination as approved by GE. In case of one side decorative lamination, other side shall be balancing white pre lamination; shall confirm to IS-12823 Gde-I Type-Thickness of the board shall be 18/19 mm if not indicated in drawings.

11. **WORKMANSHIPFORJOINERYINGENERAL:** Joinery shall be wrought all over. Timber exposed to view shall be wrought and timber not exposed to view shall be clean sawn. The workmanship and fixing of joinery shall be as per clause 8.15 to 8.26 of MES Schedule Part I. The dimensions of the various components of joinery (other than block board shutters) shown on drawings, wherever at variance, shall supersede the standard dimensions mentioned in clause 8.24 of the MES Schedule Part I. However, for pricing deviations involving any joinery work, the rates in MES Schedule (Part-II) for the corresponding joinery items shall be applicable. Timber members upto 3.00 metre in length shall be in one piece.

12. **FACTORYMADEPANELED/GLAZED/WIREGAUGESHUTTERS:**Paneled/glazed/Wiregauzed shutters of doors shall be factory made with 12mm thick plywood exterior gradepanels, manufactured in accordance with IS-1003, made of well selected kiln seasoned, chemically pressure treated, 2nd class hardwood 'Hollock'. Styles and rails shall be as specified. A Tolerance of (plus/minus) 3mm on width and height only shall be allowed, provided the shutter snugly fits into the frame. The thickness of shutters given in drawing shall be the finished thickness and no tolerance on this shall be permissible. The GE should ascertain whether the testing facilities are available for seasoning, preservative treatment of timber at approved factory and shutters are manufactured as per IS-1003specifications.

13. **PVC DOORS AND DOOR FRAMES**:

PVC doors and door frames shall be provided of dimension as per BOQ. PVC door frame(Chowkhatsshall be of size 50 mm x 47 mm with a wall thickness of 5 mm made out of cotruded5 mm rigid PVC foam shocked mitre cut at two corners and jointed with 2 Nos 150 mm long brackets of 15mm x 15 mmMS square tube. The two vertical door profiles shall be reinforced with 19 mm x 19 mm square tube of 19 gauge Rubber gasket weather seal shall be provided through out the frame. The door frames shall be fixed with wall using 65/100 mm long MS screws all as per manufacture's instructions and as directed by Engineer-in-Charge. Factory made solid panel PVC door shutters 30 mm thick consisting of frame made out of MS tubes of 19 gauge thickness of size of 19 mm x 19 mm for stiles and 15mm x 15 mm for top and bottom rails MS frame shall be covered with 5 mm thick heat moulded PVC channel of size 30 mm x 50 mm forming stiles and 5 mm thick 75 mm wide PVC sheet for top rail, lock rail and bottom rail oneither side. 10 mm thick (2 x 5 mm). 20mm wide cross PVC sheet shall be provided as gap insert for top bottom rails . Panel insert of 5mm thick PVC sheets shall be fitted in the MS frame welded/scaled to the stiles and rails with 30mm wide x 5 mm thick PVC sheet beading on either side and jointed together with solvent cement adhesive. The size of stiles and rails shall be 75 mm and 100 mm respectively. The door shutter shall be fixed to PVC doorframe as per manufacturer's instructions.

PARTICULAR SPECIFICATION (Contd….)

14. **BUILDERSHARDWARE**:

GENERAL: All articles of builder's hardware shall bear ISI marking. In case ISI marked articles are not manufactured, those shall conform to the relevant IS specifications given in the MES schedule for the relevant item. Finish of articles shall be as specified in clause 9.2.4 of the MES Schedule Part-I. Screws used for fixing the articles of builders hardware shall be as specified in clause 9.2.6 of the MES Schedule (Part-I). Article shall be as under:-

15. **BUTTHINGES**: But things shall be of MS steel medium weight all as specified in clause9.7.1& 9.7.2 of MES Schedule Part-I.

16.1 **CONTINUOUS (PIANO) HINGES:** Continuous (Piano) Hinges shall be finish as specified in clause 9.7.6 of MES Schedule Part-I. The width of the flap shall be to suit the thickness of the shutter.

16.2 **BARREL TOWER BOLTS**: Aluminum an odised barrel tower bolts of extruded section shall be as per clause 9.3 of SSR Part I. Size of barrel tower bolts shall be pae BOQ. Where size of barrel tower bolts not indicated, the same shall be 150 mm for doors and 100 mm for windows.

16.3 **HANDLES:** Extruded aluminium alloy handles fabricated type, anodised shall be as per clause 9.11 of SSR Part-I. Size of handles shall be as per BOQ. Where size of handles not indicated, the same shall be 150 mm for doors and 125mm for windows.

16.4 **HASP AND STAPLES**: Hasp and staples, safety type, aluminium anodised with mild steel hinge pin shall be as per clause 9.10 of SSR part I.Size of Hasps and staples shall be as per BOQ. Where size of Hasps and staples not indicated, the same shall be 75 mm.

16.5 **DOORS SPRINGS:** Ordinary rat tail rod pattern MS spring including roller and roller plate shall be as per clause 9.7.8 of SSR part I. Size of door spring shall be as per BOQ . Where size of door spring not indicated, the same shall be 150mm.

16.6 **HYDRAULIC DOOR CLOSER**: Hydraulic door closer shall be hydraulically regulated designation No 2, Universal type aluminium body suitable for door shutter of weight between 36 to 60 Kg shall be provided as specified in clause 9.16 of SSR Part-I. One number hydraulic door closer shall be provided to each shutter in all Aluminium doors.

16.7 **TOWEL RAILS**: Towel rails shall be tubular, mild steel, chromium plated, 22 gauge, min 18/19 mm bore, 'D' shaped withflangedendsforfixing.Incaselengthisnotmentionedindrawings,itshallbe60cmlong.
Wherever aluminium towel rails are indicated in the drawings, these shall be deemed to read as chromium plated mild steel rails as specified in this clause.

16.8 **WIRE CLOTH:** Wire cloth to be used for wire gauzed shutters shall be stainless steel with average width of aperture 1.18mm and nominal dia of wire 0.457 mm. In case of deviation, rates given in MES schedule part II for wire cloth of nominal dia of wire of 0.56 mm shall be applicable. Where SS wire mesh is specified in the BOQ, it shall be 0.36 mm thick and 1.40mmaperture.

16.9 **KNOBS:** Shutters and drawers of small size built in cabinets, switch boxes and built in furniture shall be provided with aluminium anodized knobs of not less than 38mm dia.

16.10 **ALUMINIUM DOOR STOPPER/WOODEN CLEATS**: Aluminium wooden door stoppers/wooden cleats of suitable size shall be provided to each open able shutter of window, ventilators and doors.

16.11 **HELICAL DOOR SPRING:** Where shown on drawings, wire gauged door shutters shall be provided with mild steel helical door spring 150mm long. MAKE: RITZ or Equivalent.

PARTICULAR SPECIFICATION (Contd….)

16.12 chromium plated pipe should be more than1mm.

16.13 **HANGINGROD**: Hanging rod of 19 mm (±1 mm) dia G.I. pipe light grade with appropriate socket for hanging rod in cupboards shall be provided as shown on the drawings.

16.14 **LOCKING BOLTS:** Aluminium alloy (anodized) locking bolts (10 mm dia, 100 mm size) shall be ISI marked (IS 2681) and be provided to each shutter(one each)

16.15 **HOOK & EYES:** Hook and eye shall be of aluminium anodized and matching with window and of size as shown on drawings. If the size is not specified in drgs, the same shall be of 125mm size.

16.16 **MAGIC EYE:** Magic eye shall be provided to each entrance door (one per DU) as per locations shown on drawings and shall be as per sample approved by the GE. It shall be of brass body, with both sides lenses, and of approved make.

16.17 **BALL CATCH:** Aluminum magnetic catcher of heavy duty of standard make shall be provided in all type of Dwelling Units.

16.28 **ALDROP BOLT:** Aldrop bolt shall be aluminium anodized aldrop bolt (sliding door bolt) type 3, bolt of extruded section 16mm dia, ISI marked(IS-2681).

16.19 **LOCK FOR CUPBOARD:** Locks for cupboard shall be of Grade -2, size 50 mm 4 levers unless otherwise specifiedandwhereversshownindrawingsshallbeofbrassallasspecifiedinclause
9.9.4 of MES SSR Part-I. Make of locks : Godrej/Harrison/link. In case of Aluminium windows/CSWs, the edges of wire gauge shall be bent over Aluminium beads and the beads pressed well into the angle of Aluminium section to hold the gauge on two faces as shown on drawings.

16. **STEEL AND IRONWORK**

17.01 The entire steel required for in corporation in the work shall be arranged by the contractor and shall comply with the requirement of following specifications:-

(a) **REINFORCEMENT STEEL**: High strength deformed steel bars produced by Thermo Mechanical Treatment process (TMT) steel bars of Grade Fe 500 (D) meeting all other requirements of IS-1786-2008. The procurement shall be done as per procedure givenbelow.

(b) **STRUCTURAL STEEL**:- Standard quality : Fe-410-S Conforming to IS- 2062, Structural Steel Tube shall be YST-310grade.

(c) **HOLD FASTS ,GRILLS ETC & OTHER THAN STRUCTURAL:** Fe-310-0 conforming to IS – 1977

(d) **GALVANISED SHEETS PLAIN OR CORRUGATED**: Conforming to IS-227.

(e) **FABRIC REINFORCEMENT FOR CONCRETE**: Conforming to IS- 1566 . Wire fabric for General use, window Grills, etc shall be conforming to IS-4948 and all as per MES Sch Part I clause10.33

PARTICULAR SPECIFICATION (Contd….)

18 PROCUREMENT

18.01 REINFORCEMENT STEEL:- TMT Reinforcement steel shall be procured bythe contractor directly from main producers enumerated as under:-

Ser No	Firm Name & Address	Type of Steel
1.	RashtriyaIspat Nigam Ltd. (RINL) Brand " RINL" Visakhapatnam Steel Plant, Visakhapatnam-530031, India Tel : (91891)518226, 518376, Fax : (91891) 518316 Email: cmdvsp@itpvis.ap.nic.in	All
2	Tata Iron & Steel company (TISCO, or Tata Steel) Brand : " TATA" Bombay House, 2,4 HomiModi Street, Mumbai-400001, India Tel : (9122) 2049131, Fax : (9122) 2049522, 287,840 Email : corpcomm@isr.tatasteel.com (Br office for North: Jeevan Tara Building , Patel Chowk New Delhi	All
3	Steel Authority of India Ltd. Brand : " SAIL" Central Marketing Organisation, Northern Region, 17thFloor, scope Minar, Laxmi Nagar Distt Centre, Delhi-110092	All
4	M/S Jai Balaji Industries Ltd .Brand : " JAI BALAJI" 5 Bentek Street, Kolkata-700001, Delhi Office : 510, Block-b Navraung House, 21 Kasturba Gandhi Marg, Nerw Delhi-110001 Ph- 011-43620219, 43620220, Mob : 783827772/99589367103 E-mail-info@jiabalajigroup.com	TMT bars of Grade Fe 500 & Fe 500D
5	M/S Shyam Steel Industries Ltd. Brand " SHYAM TMT" Shyam Towers EN-32, Sector-V, Salt Lake, Kolkata-700091 Tel : 033- 40074007, Fax-033-40074010, E-mail-marketing@shyamsteel.com	TMT bars of Grade Fe 500 & Fe 500D CRS
6	M/S Steel Exchange India Ltd. Brand : SIMHADRI TMT" My home Laxminivas, Apartments, Ameerpet, Hyderabad-500016, A.P., Tel : 040-23403725, Fax : 040-23413267, E-mail-info@seil.co.in	TMT bars of Gde Fe 500 & Fe 500D HSCRM
7	M/S Jindal Steels and Power Ltd. Brand : "JINDAL" OP Jindal Road, Hissar, Haryana Pin-125005 Tel : +91 1662 222471-84, Fax : +91 1662 220476	TMT bars of Gde Fe 500, Fe 500D, Fe 550 & Fe 550D and CRS with sizes (8-40mm)
8	M/S SRMB Srijan Ltd. Brand : "SRMB"	TMT bars of Gde
	Srmb House, ;7, khetra das lane, Kolkata-700012 Tel : 033-66006600, Fax : 033-2211 0483	Fe 500, Fe 500 D, Fe 550 & Fe 550 D (8mm to 32mm)

PARTICULAR SPECIFICATION (Contd….)

9	M/S Concast Steel & Power Ltd, Kolkata. Brand : " CONCAST MAXX" 21, HemantBasuSarani Suit Nos-511 & 512, 5thFloor, Kolkata- 700001 Phone : 91-33-2213 0481-87, Email-info@concastgroup.com	TMT bars of Grade Fe 500 & 500D (8mm to 32mm)
10	M/S ShriBajrang Power &Ispat Ltd, Brand : " GOEL TMT" Raipur, Chhatisgarh, Vill : Borjhara, Urla Industrial Area, Raipur- 493221, Chhatisgarh, Tele : 0771 4288019/29/39	TMT bars of Grade Fe 500 & Fe500D (8mm to 36mm)
11	M/S JSW Steel Ltd, Karnataka, Jindal Mansion, 5A Dr G Deshmukh MargMumbai-400026 Tel : 022-2351-3000 & 022-235206400 Brand :"NEOSTEEL"	TMT Bars of Grade Fe 500, 500D CRS (8mm to 40mm)
12	M/s. ShyamMetalics& Energy Ltd. , Viswakarma, 1stFloor, 86 C, Topsia Road, Kolkata-700046 Tel : 33-2285-2212 Email www.shyamgroup.com, Brand : " SEL"	TMT Bars of Gde Fe 500 , Fe 500D (size 8-32mm)
13	M/s. Kamachi Sponge & Power Corporation Ltd. ABC Trade Centre, 3rdFloor (Inside Devi Theatre Complex) Old No. 50, New No. 39, Anna Salai, Chennai-600002, India : Tel 044- 42961100 : Fax 044-42961122 Email sales@kamachitmt.com Brand : " KAMACHI"	TMT Bars of Gde Fe 500, Fe 500D, Fe550, Fe 550D HCRM (size 8-40mm)
14	M/S BDG Metal & Power Ltd., Brand: "BDG 6", HMP House, 4 fairlie Place, 5thfloor Kolkata-700001, India, Tel: 91- 33- 4005-9005, Fax : 9133-4005-9095, E-mail: info@goyalgroup.in, Website: www.goyalgroup.in	TMT bars of Grade Fe 500 , Fe 500D, Fe 550, Fe 550D HCRM (8mm to 40mm)
15	M/S Gallant Metal Ltd, Brand: "GALANTT TMX", Ward 10BC, Plot No. 123, Ground Floor, GanddhiDham Kutch, Gujarat-370201, Tel: 91- 2836-228164, Fax: 91-2836-235787, Email: gml@gallant.com , Webite: www.gallant.com	TMT Bars of Grade Fe 500, Fe500D, CRS (8mm to 32mm)
16	M/S RashmiMetaliks Ltd., Brand: "RASHMI TMT", Premlata Building, 39, Shakespeare Sarani, 6thfloor, Kolkata-700017, Tel: 033-2894255/56, Fax: 033-22894254, Email : domesticdip@rashmigroup.com, Website:www.rashmigroup.com	TMT Bars of Gde Fe 500 size (8- 32mm), Fe 500D, Fe 550D (size 8-25mm)
17	M/S Real Ispat& power Ltd, Brand: "GK TMT", Vrindavan, Near IDBI Bank Civil Lines, Raipur-492001, C.G., Tel: 91-771-4224000, Fax: 91-771-4224010, E-mail: real@realispat.com, Website: www.realispat.com	TMT Bars of Gde Fe 500, Fe 500D (size 8-36mm)
18	M/s Super Smelters Ltd, Kolkata, Brand: "SUPER SHAKTI", Premlata Building, 39, Shakespeare Sarani, 3rd floor, Kolkata-700017, Tel / Fax: 91-33-2289-2734/36, Email: info@supershakti.in, Website: www.supershakti.in	TMT Bars of Gde Fe 500, Fe 500D, Fe 550 (size 8-32mm)

PARTICULAR SPECIFICATION (Contd….)

19	M/s Electrotherm (India) Ltd, Brand: "ET TMT", Survey No.72, Palodia, Via-Thaltej, Ahmedabad Gujarat-382115, Tel: 91-2717- 660649/660550, Fax: 91-2717-234866,Website: www.electrotherm.com	TMT Bars of Gde Fe 500, Fe 500D & CRS (size 8-32mm)
20	RashtriyaIspat Nigam Ltd. (RINL) Brand " RINL" Visakhapatnam Steel Plant, Visakhapatnam-530031, India Tel : (91891)518226, 518376, Fax : (91891) 518316 Email: cmdvsp@itpvis.ap.nic.in	Structural Steel (angle, Beam, Colum, Channel, plate)
21	Tata Iron & Steel company (TISCO, or Tata Steel) Brand : " TATA" Bombay House, 2,4 HomiModi Street, Mumbai-400001, India Tel : (9122) 2049131, Fax : (9122) 2049522, 287,840 Email : corpcomm@isr.tatasteel.com (Br office for North: Jeevan Tara Building , Patel Chowk New Delhi	-do-
22	Steel Authority of India Ltd. Brand : " SAIL" Central Marketing Organisation, Northern Region, 17th Floor, scope Minar, Laxmi Nagar Distt Centre, Delhi-110092	-do-
23	M/S Jindal Steels and Power Ltd. Brand : "JINDAL"	-do-
24	OP Jindal Road, Hissar, Haryana Pin-125005 Tel : +91 1662 222471-84, Fax : +91 1662 220476	

(b) Reinforcement steel for water retaining structure shall be as per table

below:- SAIL : HCRRebar
TISCO : TISCONCRS
RINL : High Strength Corrosion Resistant Rebar(HSCRM)

18.01.1 **STRUCTURAL STEEL:**

Structural steel sections of all sizes shall be procured directly from main producers SAIL or TISCO or RINL only. In case of non-availability of Structural Steel sections of any size with any of the main producers ie SAIL, TISCO or RINL (non-availability produced duly signed by auth reps of each main producers i.e SAIL, TISCO and RINL) the same can be procured from approved secondary producers who manufacture structural steel out of ISI marked billets and are having BIS certification / ISI Marking on their products at the option of contractor with minus price adjustment of 5% (five Percent) of "Supply only" SSR rate of Structural Steel adjusted by quoted percentage by Contractor for Schedule / percentage inserted by MES for Deviation purpose as per Contract Agreement. In case of item rate quoted by the Contractor the minus adjustment shall be 5% (five percent) of the quoted rate by the Contractor of Structural Steel. In case the desired section of Structural Steel is not rolled/manufactured by main producers i.e. SAIL, TISCO & RINL, there shall be no price adjustment in use of Structural Steel procured from approved Secondary producers.

18.01.2 GALVANISED STEEL SHEETS & FABRIC REINFORCEMENT OF CONCRETE**:** These shall be procured directly from main producers. Testing provisions are given here in after in these particular specifications.

18.02 Contractor shall have to check from GE as to whether the proposed secondary producer is approved by the Deptt. or not. No claim whatsoever shall be entertained in case proposed secondary producer is rejected due to non-approval of Deptt.

18.03 Structural steel, Galvanised steel sheets and Fabric reinforcement for concrete may be procured from authorized dealers of main producers in case total requirement of structural steel / Galvanised steel sheets / Fabric reinforcement for concrete is less than 05Tonnes.

18.04 Steel sections for railings, gates, fencing. Guard bars, grills , steel chowkhats , hold fasts etc. which do not constitute structural members, may be procured from main producers / secondary producers / BIS marked manufacturers or their authorized dealers at the option of contractor without any price adjustment . Testing of these articles is optional at the discretion of GE.

PARTICULAR SPECIFICATION (Contd….)

18.05 The contractor may procure TMT steel from the authorized conversion agents of main producers provided that the main producers have given such direction in a specific sale / purchase order and they are producing TMT steel using Thermax / Tempcore / EVCON Turbo Technology. In such case the purchase Voucher or cash memo shall be from main producers.

18.06 All finished steel shall be well and clearly rolled to the dimensions, sections and weight specified. The finished material shall reasonably free from cracks, surface flaws, laminations rough, jagged and imperfect edges and other harmful defects and shall be finished in a work man like manner.

18.07 Tolerance in size and weight of reinforcement bars shall be as specified in clause.
10.17.14 and clause10.17.5 of MES Sch. (Part I).

18.08 **TESTING OF STEEL**:-

18.08.1 The contractor shall submit the manufacturer's test certificate in original along with test sheet giving the results of each chemical test as applicable and the chemical composition of steel or authenticated copy thereof, duly signed by the manufacturer with each consignment. The Engineer-in- Charge shall record these details in Steel Supply & Acceptance Register, as given at Appendix,, 'after due verification and send a certified true copy of test sheet to GE for record. The CWE/GE shall also organize independent testing of random samples of steel drawn from various lots from National Test House, SEMT Wing CME, Regional Research Labs, Govt. approved Labs, Zonal Labs etc. as per recommended minimum frequency shown in Table below. Samples from each lot should be tested for quality and elongation. The elongation shall not be less than 18%.

Sl No	Nominal size of steel for concrete Reinforcement	Quantity
1.	Bars of size less than 10mm	1 sample (3 specimen) for each test for every 25 tonne of steel or part thereof
2.	Bars size 10mm to 16mm	1 sample (3 specimen) for each test for every 35 tonne of steel or part thereof
3.	Bars size over 16mm	1 sample (3 specimen) for each test for every 45 tonne of steel or part thereof
	STRUCTURAL STEEL	
4.	Tensile Test	1 Test for every 25 tonne of steel or part thereof
5.	Bend Test	1 Test for every 10 tonne of steel or part thereof

18.08.2 Independent testing of reinforcement steel by GE is optional at the discretion of GE in case of procurement of steel from main producers and in case of testing, testing charges shall be borne in accordance with condition 10 A of IAFW-2249. i.e. testing charges shall be borne by the Deptt if test results are found in order otherwise these shall be borne by the contractor.

18.08.3 Independent testing of reinforcement steel by the GE is mandatory in case of procurement of steel from conversion Agents, authorized dealers and testing charges shall be borne by contractor irrespective of outcome of test results.

18.08.4 Independent testing of structural steel by GE is optional at the discretion of GE in case of procurement of steel from main producers and in case of testing, testing charges shall be borne in accordance with condition 10 A of IAFW-2249. i.e. testing charges shall be borne by the Deptt if test results are found in order otherwise these shall be born by the contractor.

18.08.5 The Steel Supply & Acceptance register shall be signed by JE, Engineer-in-Charge, GE and contractor. The entire quantity of steel be suitably recorded in the measurement book "not to be abstracted" before incorporation in the work and shall be signed by Engineer-in-charge and contractor.

PARTICULAR SPECIFICATION (Contd….)

18.09 LIST OF APPROVED SECONDARY PRODUCERS OF STRUCTURALSTEEL

Sr. No.	Name of Secondary Producers
1	M/S ShriBadrinarain Alloy and Steel Ltd. ,95 Stephan House 4 BBD Bag, Kolkata
2	M/S KL Steel Pvt Ltd. , Post Box No 61 , LalKuan , Bulandshahr Road, Gaziabad UP
3	M/S Pushpak Steel Industries Pvt Ltd, Gate No 119 , Alandi Market Road, Dhanore , Tahkhed Pune
4	M/S Shree Parashnath Re-rolling mills Ltd, 4C, MaharshiDevendra Road, 3rdFloor, Kolkatta-700007
5	M/S KL ConcastPvt Ltd, Z-18, Naraina, New Delhi
6	M/S SRMB Udhyog Ltd , 46 BB Galguly street Kolkata
7	M/S Karam Steel Corporation, Nasrali Road, PO Box No.56, Mandi Gobindgarh-147301 (PB)

18.10 **STEEL IN COILS ETC**:-

Any bar of any dia for reinforcement may be procured in round bundles or coils and the cost of straightening the same shall be borne by the contractor. When bars are procured in bundles, the length of each bundle shall be worked out on the basis of unit weight predetermined by the GE by getting suitable length (not less than 3 m) out of each consignment received, getting it straightened, length measured and weighed in presence of contractors accredited representative. The said length and the weight shall be recorded from which unit weight (weight per unit length) shall be calculated. The length of bars worked out on the basis of unit weight determined as above shall form the basis for the purpose of calculating quantity of steel used/to be used in work and making payment of materials lying at site. However if the unit weight works out more than the unit weight given in SSR, then unit weight given in SSR shall be followed for computing weight of steel for the purpose of making payment of steel lying atsite.

18.11 **STORAGE**:-

Steel of different sizes shall be stacked separately. For each classification of steel, separate areas shall be earmarked. Steel shall be marked with distinct painting marks for easy identification. All steel shall be so stored that it is always at least 15cm above the GL Steel shall be stored in a manner so as to prevent distortion and corrosion. Any section that has deteriorated and corroded or if considered defective by Engineer-in-Chargeshall not be used in the work and shall be removed by the contractor without any extra cost. It will be the responsibility of the contractor to make sure that all possible arrangements are made for the safe custody of the steel. In case of any loss of steel only contractor will be responsible and the loss will/shall be made good without any delay or claim what soever.

18.12 **SCHEDULING AND SUPPLY**:-

Schedule of supply of steel shall be finalised by contractor with GE and shall be incorporated in CPM chart so that supply of steel is monitored in a way to avoid any delay in completion of the work. The complete requirement of steel of various sizes will be worked out before making any RAR payment and procurement of steel by the contractor will be completed sufficiently in advance of the date of completion, No extension of time will be considered for non availability of any section of steel.

18.13 Assembly of reinforcement shall be as per clause 12 of IS456.

PARTICULAR SPECIFICATION (Contd….)

18.14 **NOMINAL COVER TO REINFORCEMENT**:-
If nominal cover is not shown on drawing, it shall be as under :-

Exposure	Nominal cover in mm and less than **	Slab	Wall	Beams	Colums
Mild	20mm	20m	20mm	20mm	40mm *
Moderate	30mm	30m	30mm	30mm	40mm *

* In case of columns of minimumdimensionof200mmor under, whose in forcing barsd on ot exceed 12mm, a nominal cover may be of25mm.
** For further details refer clause 26.4 of IS 456 of 2000.

19. **WELDING**: Welding wherever shown on drawings shall be by metal arc process in accordance with IS-816 andIS-823.

20. **GRILL/GUARD BARS ONLY FOR WINDOWS &VENTS OTHER THAN ALUMINIUM WINDOW/VENTS:** Mild steel Grill/Guard bars of respective type as indicated shall be provided to all external steel windows and ventilators of buildings covered under Sch 'A' Part I as per detail shown in the drawings. In case type of grill/guard bars is not mentioned in any drawings, Guard bars type-F shall be provided.

21. **HOLD FASTS/LUGS**: Flat iron hold fasts/lugs shall be provided by welding except those, to be provided to wooden chowkhats which shall be fixed with bolts/nuts as per details. Holes in wooden chowkhats shall be plugged with hard wood plugs.

22. **STEEL DOOR:-**Steel door shall be fabricated as per details. The size of door unless otherwise not shown on drawing shall be 1200mm x 1200mm and thickness of MS sheet shall be 2 mm . All iron and steel work shall be given two coats of synthetic enamel paint over a coat of red oxide primer.

23. **GALVALUME SHEET ROOFING:** Galvalume sheet shall be0.50 mm thick (total coated thickness) of profile and colour as specified of 550Mpa minimum, yield strength. It shall be coated with hot dip alloy of 55% Aluminium 43.5% Zinc 1.5% Silicon and finished with resin coat on both surfaces @ 150 gm/Sqmof coating (total both surfaces) and shall be fixed using hot dip galvanized, self drilling and self tapping, screws neoprene and EPDM washers. Penetrations and laps in sheet shall be sealed by using proper sealant profile. HDPE fillers shall be provided wherever required to close voids between sheets, sheet & fasteners etc. The sheeting shall be fixed with side lap and end lap as recommended by manufacturer. Ridges and hips shall be covered with Galvalume sheets of 0.5mm thickness. These shall be fixed to roof as per manufacturer instructions and standard Engineering practice. The entire fixing process of Galvalume sheets shall be executed through as pecialist agency indentified by the supplier and as approved by GE. The contractor shall submit the original manufacturer's test certificate in connection with chemical composition of the steel, tensile/ yield strength of steel sheet, corrosion resistane (saltspray) test, dimentional (thickness of both basemetal and coating), coating adhesion (180degreebendtest)etc and also submit manufacturer's guarantee certificate of product confirmation of standard. GE shall also carry out independ enttest as per the test mentioned above from random samples throught Govt approved laboratories/command/zonal laboratories. The cost of testing, transportation of materials etc shall be borne by the contractor irrespective of the results of testing and no extra claim whatsoever shall be admissible. The sheets shall be protected with guard film. The material shall be procured directly from manufacturer.

PARTICULAR SPECIFICATION (Contd….)

24. FLOORING:

GENERAL: Ordinary portl and cement as specified in clause 3.2.1 here-in-before shall be used for all flooring. Provisions contained in clauses 13.25, 13.27, 13.32 & 13.38 of MES Schedule (Part-I) are to be adopted for laying floors and pavements. Floors shall be laid to levels or to falls as shown on drgs and as directed by the Engineer-in- Charge. Floor finish shall be extended over dwarf walls, doors and other openings. Floor topping of cast-in-situ cement concrete shall be finished even and smooth without using extra cement as specified in clause 13.32.8 of MES Schedule Part-I. Cast-in-situ terrazzo shall be in grey cement (OPC) without pigment unless otherwise indicated. In location where terrazzo with white cement has been indicated, it shall be casted with white cement and pigment. White cement shall be Birla/L&T/JK Cement. Marble aggregate (Chips) shall be of size 6mm to 12mm for terrazzo floors. The colour of marble chips shall be as shown on drawings or as approved by GE, when not indicated in drawings. Surfaces finished in terrazzo shall be grinded and wax polished as specified in clause 13.38.6 to 13.38.8 of MES Schedule Part-I. The grinding by hand may be allowed in WC/Bath, if grinding by machine is not feasible in such situations. These situations are in addition to those as specified inclause 13.38.6 of MES Schedule Part-I. Sub base shall not be laid in panels.

24.1 PCC: 40/25 mm thick PCC (1:2:4) type B-1/B0 (using 20/12.5 mm graded crushed stone aggregates) finished even and smooth without using extra cement laid in alternate bays. 4mm thick glass dividing strips shall be provided only where mentioned in BOQ.

24.2 NON SKID CERAMIC TILES FLOORING: Non skid ceramic tiles flooring shall be Ist quality provided with mat superior finish ceramic tiles of size specified in the BOQ and Coloured/Pattern, Make/Brand as specified conforming to IS 15622-2006/ISO-3006 (Grade B-II (a)) Shall be laid over screed as mentioned in the BOQ using cement slurry @ 3 Kg/Sqm over PCC 1:2:4 type B-0 (using 12.5 mm graded crushed stone aggregate, over 75 mm thick PCC.

25. PLASTERING, POINTING & SKIRTING/DADO: Plaster and skirting/dado shall be returned in jambs, soffits of lintels and windows etc. All plastered surfaces shall be trowelled to even and smooth surfaces without using extra cement. Thickness of cement plaster mentioned is exclusive of dubbing. Dubbing may, however, be done in one operation with plaster and no separate payment for dubbing shall be admissible. All corners, angles, junctions shall be truly vertical or horizontal as the case may be and shall be carefully finished. Corners around joints openings and junction of walls shall be rounded to minimum radius of 5mm. For specification of refer clause on concrete herein before. The sand shall comply with clause 14.5.2 of SSRPart-1.

26. GLAZED VITRIFIED, CERA MICTILES DADO/SKIRTING: It shall match with the floor finish and all specified in MES SSR Part-I.

26.1 KOTA STONE FLOORING: 18 to 20mm thick Kota stone slabs (Machine cut) of best quality as approved by GE, jointed with grey cements lurry, with pigment to match the shade of the slab shall be provided over 15-18mm thick bedding layer of cement mortar (1:3) using cement slurry @ 4 Kg/Sqm. The floor shall be grinded and polished to mirror finish all as specified in MES SSR Part-I.

26.2 GRANITE FLOORING:- Granite flooring shall be 18 to2 0mm thick polished granite of best quality as approved by GE, jointed and pointed in white/colour cement using cement slurry @ 4 Kg/Sqm. Wherever it isused in the steps, jambs, plat forms etc, the stone shall be in single piece. The floor shall be provided as specified in MES SSR Part-I.

26.3 MARBLE FLOORING: Marble flooring shall be white polished marble of size not less than 600mmx600mmx18mmto20mm thick of best quality as approved by GE, jointed and pointed in white cement using cements lurry@4Kg/Sqm. White marble shall be finished (mirror polished) shall be provided. It shall be 20mm thick pre mirror polished hand cut in best quality. All exposed edges shall be round/bull nosed to a minimum radius 5mm. Wherever it is used in the steps, jambs, platforms etc, the stone shall be in single piece. The floor shall be provided as specified in MES SSR Part-I.

26.4 PRE-POLISHED KOTA STONE: Pre-polished Kota stone shall be of 18 to 20mm thick pre mirror polished hand cut of best quality set and jointed with grey cement slurry with pigment to match the shade of slab using cement slurry @ 4 Kg/Sqm all as directed. Pota stone floor and skirting / dido shall be laid as specified in MES SSR Part-I.

PARTICULAR SPECIFICATION

27. **INTERLOCKING PAVING TILES/ BLOCK:** The thickness and grade of factory made cement concrete inter locking paver block shall be as mentioned in BOQ. The paver blocks shall be made by block making machine with strong vibratory compaction of approved size and design/ shape in required colour and pattern. The joints shall be filled with the material as specified, complete all as per the direction of Engineer-in-charge/ GE. Precast concrete Paver blocks shall conform to IS 15658:2006. Paver blocks shall be sound and free from cracks or other visual defects. The tolerance on length or breadth of paver blocks shall be +2mm and tolerance on thickness of tiles shall be +3mm. Water absorption shall not be more than 6 percent by mass. Shapes shall be triangular, Zigzag, Haxagon or other shape as indicated.

28. **CHAIN LINK FENCING:** Chain link fencing both Galvansied Iron and PVC coated shell be provided as per specified in BOQ. Make /brand as specified confirming to IS and approved by GE having the required specification's as per clause No 10.34 of MES SSR Part –I.

29. **BARBED WIRE:-** Galvansied steel barbed wire for fencing shall confirm to specification for galvanized steel barbed wire for fencing. The galva barbed wire shall be manufactured from for fencing. The galvanized barbed wire shall be manufactured from galvanized mild steel wire conforming to IS280-2006. Complted work as per clause No 10.3.27 MES SSR Part –I.

30. **INTERGRAL WATER PROOFING COMPOUND:-** Material and workman ship of water proofing compound shall be as specified in clause No 4.8 of MES Schedul 2009 (Part –I). Internal water proofing compound shall be one of the brands / manufacturers listed in the list of approved makes / manufaterure and as approved by Engr-in- Charge Charge and shall be brought in sealed tins or closed packets.

31. **CEMENT BASE PAINT:-**

(a) First quality brand of the makes listed in 'List of approved makes/manufactures' shall be used.
(b) Cement paint shall comply with the requirement mentioned in IS : 5410-1969 be as per para 15.7 and 15.15 of SSR Part-I (2009)

32. **WHITE WASHING / COLOUR WASHING:** Whitewash / colour wash shall be done in 3 coats. The colour wash shall consist of 2 coats of colour wash over one coat of white lime wash. However, soffit of rooms shall be provided with 3 coats of white (lime) wash. Lime wash shall be carried out all as specified in 15.2 & 15.11& 15.12 of SSR Part-I. To achieve better finish of white wash zinc oxide shall be mixed with lime wash at the rate of 2 percent of slacked lime. In final coat of white wash fevicol DDL shall be mixed as per manufacturer's instructions. No ultramarine blue shall be mixed in any coat. However, sodium chloride, as specified in clause 15.12.3 of SSR part I shall be mixed in first and second coats.

33. **OIL BOUND DISTEMPER (WASHABLE DISTEMPER):** Oil bound distemper, shall be applied in two coats of oil bound distemper over one coats of cement primer conforming to IS: 428 -2000 & shall be provided over prepared surface. All cracks, voids and minor damages shall be patched/repaired with white cement putty or with plaster of peris and surface shall be prepared with white cement based putty all as specified in clause15.19 of SSR Part-I.

33. **GLAZING:** All glazing shall be with sheet glass of ordinary quality and shall conform to IS -2835. Glass shall be free from all defects viz bubbles, waviness etc. and shall be of uniform thickness. In case thickness not mentioned elsewhere glazing to windows/doors shall be with 5mm thick sheet glass. Glass used shall be plain sheet glass except in baths/WC/Toilets/Lav/WC as applicable where it shall be pinhead glass. The glazing to steel frames, shall be fixed with special spring glazing clips with putty and glazing to wooden joinery shall be fixed with wooden beading with putty all as specified in clauses 16.5 to 16.10.2 of MES Schedule Part-I. Glass shall be fixed to aluminium windows /vents with aluminium snap beading along with PVC/neoprene gasket.

PARTICULAR SPECIFICATION (Contd….)

34. **PAINTING:** All synthetic enamel paints and emulsion paints shall be of 1st quality manufactured by the standard firm. The contractor shall inform the GE, within three weeks of the acceptance of the tender, the brand/ names of the manufacturers of paint proposed to be used in the work and submit samples thereof to GE before their use in work and obtain prior approval of GE. Paints for priming coat, under coat and finishing coat shall be of the same brand. Tint of paint, if not mentioned in drgs/schedule of finishes, shall be as approved by the GE. Pink primer shall be used on wooden surface and red oxide primer shall be used on steel surface. Concrete surface shall be applied with cement primer. The workmanship shall be as under:-

35. All surfaces required to be painted shall be smoothened size and knotted and then applied with priming coat, stopping and filling [filler coat] shall be done after priming coat and surface rubbed down to a level and smooth surface and thereafter under coat and finishing coat shall be applied all as specified inclause 17.6 of MES Schedule Part-I. Steel and iron work shall be painted in the manner as specified in clause 17.8 of MES Schedule Part-I. All steel and wooden surfaces including wooden based surfaces such as plywood, particle board etc shall be given two coats [under coat and finishing coat] of synthetic enamel paint. Aluminium painted shall be given two coats of aluminium paint conforming toIS-2339 over coat of appropriate primer.

36. **WATER PROOFING TREATMENT TO RCC ROOF SLABS:** Water proofing shall be provided with APP modified water proofing membrane 3 mm thick (weighing not less than 3.00 Kg/Sqm) and shall be under strict supervision of manufacturer's technical representative . The treatment over roof slab shall be carried out by providing 10 mm thick plaster in cement mortar 1:4 mixed with water proofing compound as per manufacturer' instructions. While the concrete is green. The surface shall be then be painted with one coat of bitumen primer conforming to IS : 3384 - 1965@ 0.3 Kg/Sqm. A layer of APP based polymeric membrane 3 mm thick (minimum), weighing not less than 3.00 Kg/Sqm reinforced with polyester non woven mat 160 GSM thick shall be laid by torch application with side over lap of 75mm and end lap of100mm.The joints /overlap shall be sealed with torch application. For parapet, the membrane shall be taken up to 30 cm and then inserted inside the parapet in a groove. 40 mm thick brick tile grouted in cement mortar 1:3 mixed with 5% crude oil by weight of cement and flush pointed in the same mortar work shall be provided as protection. Covered fillet in PCC 1:2:4 type B 0 of radius 75 mm to be provided at junction of roof and parapet.

37. **GUARANTEE CLAUSE**: The contractor shall stand written guarantee to the Government for a period of 10 years for the effectiveness of the treatment from the certified date of completion of the work. Should the GE at any time, prior to the expiry of the side guarantee of ten years, find defects in the water proofing of roof, the contractor on demand in writing from the GE, specifying the building/structures complained of shall forth with undertake to carry out such treatment as may be necessary to render the said building /structures water proof at this own expense, till expiry of the guarantee 10 years from the certified date of completion. In the event of contractor's failure to comply with GE's directions (within the period to be specified by the GE) the GE may under take rectification work at the risk and cost of the contractor. The liability of the contractor under this condition shall not however extend beyond the period of 10 years from the certified date of completion unless the notice was served on the contractor previously to rectify the defects. The amount of security deposit against the guarantee for water proofing treatment calculated as per the scales laid down for calculating "Performance Security" on the amount of water proofing treatment at contract rates, rounded off next Rupees.

37. **GUNITING**: Guniting shall be done as indicated in Sch'A' BOQ. The steps to be taken while carrying out the gunting are: Taking down cement plaster, raking out joints, hacking for key. The parts of brick work which is intact shall be roughened to needful guniting. Removal of dust from joint and keys by sand blasting and wire brushing as directed. Applying polycrete for concrete and mixture. Providing Guniting with pressure 2.2 to 2.50 kg/Sq cm with mixture of 1 cement and 2 of sand (1:2) of thickness as indicated including stripping, floating with wooden stripper for proper dimensional control including necessary form work, scaffolding complete. Dismantling of concrete member by chiseling from RCC member (Column, Beams, Slab, Chajjas, faciaetc). Exposing the existing reinforcement without disturbing the core of RCC member, cleaning the surfaces with wire brushes, providing additional reinforcement as directed at site, where required, binding with 75x75x2mm thick (dia) welded mesh with binding wire and tack welding using adhesive, plaster create for concrete admixture. The unit rate quoted by tenderer shall be deemed to be inclusive of the above provision. Gunited surfaces shall be cured by the contractor for seven days at his own cost and arrangement. The contractor shall make arrangement at his own cost for removing the rubbish etc before and after the Guniting work.

PARTICULAR SPECIFICATION (Contd….)

38. **SANITARY APPLIANCES AND TOILET FITTINGS:** All Sanitary ware (except flushing cistern) shall be of white vitreous china best quality ISI marked of makes as specified. The samples of sanitary fittings, fixing accessories shall be approved by the GE wherever occurs in tender documents before bulk procurement for incorporation in the work. ISI marked fittings of these brands shall be provided instead of those without ISI marked. General requirement and specific requirements mentioned in Para 18.32 to 18.39 as applicable of MES Schedule Part- I shall be complied with. The contractor shall employ qualified and licensed plumber for supervision of installation and testing of the sanitary fitting and plumbing. The sanitary items shall comply with following specification also. Sanitary fittings issued under Schedule 'B' shall be fixed as per SSR Part-I relevant clause.

38.1 **WATER CLOSET (SQUATT PATTERN- ORISSA PAN)**: WC shall be **Orissa pan** type conform to IS- 2556 Part-III and shall be as per Cat No. 20004 of long pattern of Hindware or other equivalent approved make as specified of size 580mm x 440 mm shall be provided with 'P' or 'S' trap. The pan shall be set in cement concrete (1:2:4) using 20mm graded crushed stone aggregate at least 25cms around and finished just below the rim to receive the specified thickness of floor finish.

38.2 **WATER CLOSET PADESTAL PATTERN (EWC):** Water closet (EWC) shall be as per Cat No. **as specified and** standard with 'P' trap or 'S' trap of all as per IS-2556 Part-II. Plastic seat and cover with flat under side, solid moulding closed front pattern, with cover. Seat cover shall be white in colour with chromium plated hinges and nuts, of make as specified.

38.3 **CISTERN**: 10 litre discharge capacity polythene low level flushing cistern, solid moulded with valve less 7ymphonic fitting, manually operated with one PVC ball valve horizontally plugged type and polythene float valve shall be provided alongwitha pair of standard brackets/clamps as supplied by manufacturers. The accessories includes ; 15mm dia PVC connection pipe, 32mm dia flush pipe of chromium plated brass tube bent to required shape including chromium plated brass coupling.

39. **WASH HAND BASIN**: Wash hand basins shall consist of ; Wash hand basin of size 530x425mm vitreous china glazed ware, flat back pattern as per Cat No CASINO and along with pedestal Cat No-1117 of Make CERA or any other equivalent model of approved make as specified with waste union and perforated grating (both fittings of brass chromium plated). Colourof WHB shall be white in all locations, 15mm bore central hole basin mixture Cat Part 167B of Jaquar or any other equivalent Model of approved Make as specified (the cost of this provision shall also be considered included quoted under Sch'A' BOQ, apair of cast iron brackets, 32mm diagal 7anished steel medium grade waste pipe fitted with brass chromium plated waste coupling outlet complete, length of waste pipe shall be as indicated on drgs/ as directed by Engineer-in-Charge, Chromium plated steel chain and rubber plug.

39.1 **MIRROR**: Mirror shall be polished sheet glass 5.5mm thick on 5.5mm thick AC board and provided with wooden/polythene (PVC) moulding frame. Mirror shall be hung by key slots on screws fixedto plug embedded in walls unless otherwise shown on drawings. Size of mirror where not shown on drawings shall be 60x45cm. Colour of PVC moulding frame shall be same as that of wash handbasin.

39.2 **WASTE PIPE AND COUPLING**: CP coupling with 32mm dia, PVC flexible, waste pipe of required length shallbeprovidedwithbrasschromiumplatedwastecouplingwithperforatedgrating32mmdia.

39.3 **URINALS:** Urinal shall be provided of white vitreous china full stall urinals of approved makes as specified conforming to IS-2556 Part-VI (Section-1) with grating and union including 5 litres discharge capacity PVC automatic flushing cistern. (For set of four urinals or part thereon only one flushing cistern is to be provided as per main drgs, 20mm bore medium grade 7alvanized iron flush pipe, 32mm bore medium grade 7alvanized iron waste pipe up to outlet drain, Chromium plated brass coupling to connect urinal pan with waste pipe and chromium plated adopter to connect pan with flush pipe.

39.4 **SOAP NICHE:** Soap niche shall be of size 30cmx20cmx10cm deep. The location of soap niche shall be as directed by Engineer-in-Charge and all the surfaces of niche (cill, sofit, back and sides) shall be provided with glazed ceramic tiles (coloured or white depending upon the colour of tiles to be provided in the adjoining dado), set and jointed in neat cement slurry, pointed neatly in white cement.

39.5 **PEG SETS:** Pegset where ever shown/ indicated shall be of aluminium anodized. These shall be screwed with steel screw on Deodar planks (French Polished) 10cm wide x 20mm thick and of such length that the end peg is at a minimum distance of 5cm from end edge and the distance in between the respective pegs is not less than 15cm, The planks shall be fixed to wall with wooden gutties/plugs and screws. Where peg sets are indicated to be provided on the back of the door, peg sets as indicated above shall be provided on the top rail of back of the door with screw without providing wooden planks.

PARTICULAR SPECIFICATION (Contd….)

39.6 STAINLESS STEEL SINK WITH/WITHOUT DRAINING BOARD: Kitchen sink without draining board shall be made out from non magnetic stainless steel sheet 1.00mm thick of specified make of size as per BOQ. The surface of the sink shall be plastic coated to make it safe and scratch free installation. If size of sink and draining board as specified are not available in the market then next (higher) nearest size available shall be provided with accessories/fittings viz; Brass chromium plated chain rubber plug arrangements, Waste coupling brass chromium plated, screwed with fly nut and washer suitable for 50mm bore pipe, Waste coupling shall be of the same make as that of sink, 50 mm bore GI light grade waste pipe from basin to trap and a pair of cast iron brackets. Kitchen sink with draining board shall be made out from single piece on magnetic stainless steel sheet1.00mm thick.

39.7 HDPE OVER HEAD WATER STORAGE TANKS & OTHER ALLIED WORKS: HDPE Water storage tanks shall be 3 layers type rotational moulded, cylindrical vertical type. HDPE Water storage tanks shall be ISI marked (engraved). In case the tanks of capacity as shown/indicated are not manufactured, the contractor shall provide tanks of next higher capacity without any extra cost to the department. The shall consist of Mild steel galvanized medium grade 25mm bore over flow and wash out pipe and float ball valve with brass rod and polythene ball shall be provided. Wash out pipe shall be provided with stop valve and anti-mosquito rose. Over flow pipe shall be connected to wash outlet pipe beyond the stop valve at an approximate distance of 50cm from take off point of wash out pipe and beyond this the combined wash out and over flow pipe shall be provided. In case RCC floor slab for placement of O.H tanks it shall be placed on PCC (1:2:4) Type B-1, 40mm thick shall be provided after applying neat cement slurry @3 kg of cement per sqm over RCC slab. The soffit of tank slab shall be plastered in CM (1:3), 5mm thick using liquid type water proofing compound before hoisting and placing of tanks in position.

39.8 PLUMBING WORK (UPVC): Plumbing work shall be carried outas specified inclause18.13to18.23, 18.40 and 18.62 to 18.67 of MES Schedule part-I. In Case the soil/waste/vent pipes are of UPVC pipes these shall be conforming to IS-13592-1992 & fitting conforming to IS-14735-1999. All the pipes and fittings shall have ISI certification mark. UPVC pipes and fittings shall be jointed with solvent welded routs all joints as specified in clause18.52 of MES Schedule Part-1 & IS code. UPVC pipes and fittings shall be fixed to wall with mild steel clamps as specified in of MES Schedule Part-1.

40. INTERNAL WATER SUPPLY:

40.1 MATERIAL: Polypropylene Random copolymer (PP-R) pipes and fittings shall be all as specified in clause 18.114.1 & 18.114.2 of MES Schedule Part-I, bib taps, stop valves and pillar taps shall be all as specified in clause 18.14, 18.15 and 18.16 of MES Schedule Part-I. Shower rose shall be of approved make as per sample kept in GE's office. Mild steel galvanised tubes (pipes) and fittings shall be all as specified in clause 18.4 of MES Schedule part-1, bib taps and stop valves shall be all as specified in clause 18.14 and 18.15 of MES Schedule part-1. Shower rose shall be of approved make as per sample kept in GE's office.

40.2 LAYING AND FIXING PIPES : The pipes for supply of water to all fittings in the buildings shall run on the internal walls as approved by the Engineer-in-Charge at site. Pipes for internal water supply shall run concealed in walls/floors. The cost of cutting chases, forming holes etc. and making good for concealing the pipe shall be deemed to be included in lump sum cost of building of Sch'A' Part-I. Where pipes are laid underground, the trenches shall be excavated as directed by the Engineer-in-charge. The pipes running along face of the walls shall be clamped in the walls as specified in SSR. Where pipe is passing through a wall, a mild steel tube sleeve shall be fixed all as specified in clause 18.51.1 of SSR part-1. Testing of pipes shall be carried out all as per clause 18.114.5 of MES Schedule Part- i

40.3 PVC PIPE CONNECTION: Readymade PVC connection pipes of standard make (ISI marked) shall be 450mm long suitable for 15mm bore GI pipe complete with brass chromium plated coupling nuts and rubber washer. The weight of brass coupling nuts shall not be less than 40 grams. The weight of PVC pipe 450mm long shall not be less than 40 grams.

40.4 MILD STEEL GAVANISED TUBES AND FITTINGS:-Refer clause 18.4 of MES Schedule, Part-I. Laying & jointing of pipes shall be done general y in accordance with MES Schedule Part-I Clause 18.46, 18.47, 18.48, 18.50 and 18.51 unless otherwise directed by GE.

40.5 UNPLASTICISED PVC PIPES: UPVC pipes shall conform to IS 4985-2000. specification for Unplasticised PVC pipes for potable water supply. The pipes shall be reasonably round. Internal and external surfaces of the pipes shall be smooth and clean, reasonably free from grooving and other defects. The ends shall be cleanly cut square with the axis of the pipes, UPVC pipes shall be of pressure ratings (working pressure) as 2.5, 4.5 and 10 Kg/sqcm as indicated. Fittings for Unplasticised PVC pipes shall be made of PVC or shall be as recommended by the manufacturers of the pipe issued.

PARTICULAR SPECIFICATION

40.6 **HIGH DENSITY POLYETHYLENE (HDPE) PIPE**: HDPE pipe shall conform to IS 4984-1995. specification for high density polyethylene pipes for potable water supply. The Pipes shall be smooth internal and external surfaces. Slight shallow longitudinal grooves or irregularities in the wall thickness shall be permissible provided that the wall thickness remains within the permissible limits. HDPE pipe shall be of pressure ratings (working pressure) as indicated. The pipes shall carry colour bands to indicate class of pipes. Fittings to use with HDPE Pipes Shall be injection moulded or fabricated type confirming to IS 8008 (Parts I to 6)-2003, Specification for injection moulded high density polyethylene (HDPE) fittings for potable water supplies or to IS 8360 (Parts I to III)-1977, Specification for fabricated high density polyethylene fittings for potable water supplies or as recommended by the manufacturers of the pipes used.

40.7 **UPVC SOIL,WASTE AND RAIN WATER (SWR) DRAINAGE PIPES**:UPVC(SWR) pipes shall conform to IS 13592-1992, Type 'A' Pipe for use in ventilation pipe work and rain water application. Type 'B' Pipe shall be for use in soil and waste discharge system. The size of pipes shall be indicated. Rubber rings for pipes and fittings shall conform to IS 5382-1985. UPVC (SWR) fittings shall be used as per recommendations of the manufacturer of the pipes. In case there is no IS code for the fittings, these shall generally conform to the requirements of BS 4515DIN 19531 and DIN 19534. UPVC (SWR) pipes and fittings shall be strong, dimensionally stable and shall be free defects.

40.8 **PT MT BATH FITTINGS:** The fitting shall be as per manufacturer's instruction and as approved by GE.

41. **PURCHASE VOUCHERS**

In addition to this mandatory submission of purchase vouchers for any material specified elsewhere in tender documents, contractor shall submit the original purchase vouchers to the Engr-in-Charge for his verification for the following items. The said original purchase vouchers shall be released to the contractor after defacing the same as having verified with respect to this contract. No payment shall be released in absence of purchase Vouchers:-

(a) GI pipe.
(b) Gate valve.
(c) Foot valve.
(d) Ball valve
(e) Star delta Starter
(f) CI Pipe.
(g) PVC Tank.
(h) Sluice valve. LT panel. Tank nipple. NRB.

42. **INTERNAL ELECTRIFICATION:**

(a) **GENERAL:** The materials, workmanship and the installation as a whole shall generally be governed by specifications, and rules and the principles laid down in SSR 2009, Part-I Section 19, Specifications given here under are in amplification and modifications, the specifications given here under shall take precedence. Items of electrical work shall be deemed to include interlayer minor items to make the installations in the building complete in all respects, duly tested and certified as fit for switching electric supply on the taking in side the buildings. The installation shall strictly comply with the provisions contained in the latest edition of Indian Electricity Rules and latest edition of the Indian Standard Electrifications as applicable to the work, except where such regulations and rules are modified by the specifications. It shall be of high standard and approved construction used in modern electrical works as regards design as well as workmanship. Complete works shall be suitable in all respects for the type of voltage indicated in the relevant items Sch'A'. All electrical works shall be executed by skilled licensed workers. Only licensed supervisors and workers shall be employed, and the contractor on demand by the Engineer-in-charge shall produce such evidence of qualifications for his workers/supervisors either at the commencement of the work or any time there after.

(b) **CABLE :** Cable shall be with multi stranded copper conductors single core PVC insulated unsheathed cable upto 1100 volts grade conforming to IS : 694-1990. For the purpose of colour coding in conduit wiring in particular the samples for make and colour of insulation of wires shall be got approved before the wires/cables are procured and drawn into conduit. Red/yellow/blue wires for phases, black wire for neutral and green wire for earth shall be used wiring in conduit without coding will not be acceptable. Size of cables and voltage grading shall be as per Schedule 'A'/BOQ.

PARTICULAR SPECIFICATION (Contd….)

(c) **POWER CABLES :**Power cables suitable for LT and HT supply shall conform to the following Indian Standards as indicated. The characteristics of the cables viz. the size of the conductors, single core, twin core, three core or multi-core, whether armoured or unarmoured, type of insulation and sheathing and the rated voltage of the cables shall be as indicated :

(i) IS 692-1994, Specification for paper insulted lead-sheathed cables for electric supply.
(ii) IS 1554(Part 1)1988, Specification for PVC insulated (heavy duty) electric cables, Part -I for working voltage upto and including 1100 volts.
(iii) IS 1554-(Part 2)-1988, Specification for PVC insulated (heavy duty) electric cables. Part 2, for working voltages from 3.3 KV upto and including 11 KV.
(iv) IS 7098 (Part 1)-1977 and Part 2-1985 (XLPE) insulated PVC sheathed cables.

(d) **LT CABLES :**All LT cables shall be of 2/3/3 ½ or 4 core aluminium conductors, XLPE insulated heavy duty and suitable for 1100 volts grade and conforming to IS-7098 (Part-1 - 1988). The aluminium conductors used shall be stranded, compacted and circular/shaped and the main insulation shall be of cross linked polyethylene (XLPE) with inner sheathing PVC extruded and each core of the cable shall have colour identification all as specified in IS. Armour over the inner sheath shall be either of strip or wire type and outer sheath shall be of Extruded PVC conforming to ISS. The cables shall be suitable to withstand maximum conductor temperature of 900 C and 2500 C during operation and short circuit respectively.

(e) **CONDUIT**: Specification for fittings for rigid non metallic conduits. Conduit fittings shall be of unplasticiscd PVC as per IS :3419-1988.

(c) MCB DISTRIBUTION BOARDS: Distribution boards shall be factory made and conforming to IS : 8623 suitable for universal mounting copper bus bar, Neutral bar, Earth bar, Standard DIN bar Rail and cable ties for cable management. Top and bottom shall have removable gland plates with knock outs. SPN Distribution boards shall be suitable for provision for DP/SPN MCBs/isolators as incomer and SP outgoings all as specified and shall be flush mounting type. The degree of protection shall be IP -42 protection with acrylic door . TPN Distribution boards shall be suitable for FP/TPN MCBs/Isolator as in comer and TPN/SP outgoings all as specified and shall comply with IP-42 protection with metal door.

(d) MCBs - All MCBs shall be conforming to IS-8828 of 1996 and shall be ISI marked. These shall be suitable for 'C curve and 10 KA Breaking capacity and shall be provided with box terminal on top and bottom both suitable for adopting cable size upto 35 Sqmm. The enclosure of MCBs shall be of' Moulded Self Extinguishing thermo set plastic and these shall be suitable for snap fixing on standard Din Rail.

(e) MCCBs : These shall be of compact and elegant design suitable for reversible load and line terminations without affecting its performance. MCCBs shall be suitable for fixing flush on the panels and shall be provided with handle operating mechanism including Rotary Handle varidepth type. All other specifications shall be asper SSR Part-I Clause19.100.14.

(f) MS BOX HOUSING FOR ACCESSORIES: All electrical accessories shall be mounted on mild steel housing of 16 gauge embedded / flush with the surface of wall. The top cover of housing must be minimum 15 mm larger in all sides. The front of the housing shall be plastic laminated sheet 3 mm thick and smooth finished. Top cover must be removable type fixed to the housing with adequate number of non rusting type polished machine screws (wood screws shall not be used for this purpose). The top surface of housing shall be provided with a groove with chamferred edge for cable entry. The housing should have suitable earth terminal which will be connected to the earth terminal on the main / sub control boards from which the circuit originates by means 1.50 sqmm copper conductor green colour FRLS PVC insulated cable of continuity wire, the cost of which is deemed to be included in the rate per point wiring and it is deemed to include cost of supply and fixing housing and top cover also. Where fan regulators are housed, necessary ventilating holes 2.4 mm dia are to be made on top cover, No. of holes and their arrangements shall be all as directed by the Engineer-in-Charge.

PARTICULAR SPECIFICATION (Contd….)

(g) **CIRCUIT & CONTROL GEARS:** The circuit arrangements is planned for better operation and maintenance and shall not be altered even though it is at variance with clause 19.7.2 of SSR Part-I. Each circuit shall have its separate neutral, originating from the neutral bus / neutral terminal. It shall be ensured that in case of single phase circuits the phase conductor is taken into
the single pole. Switches with single pole switches in 'OFF' position, there should be no conductor at the outlet point controlled by the Switch. Lighting circuits which have three pins outlet, the earth terminal shall be connected by means of 1.50 sqmm copper conductor green colour FRLS PVC insulated with circuit earth wire. Control gear as far as it is related to internal electrical work in building comprises of:-

(i) Distribution Board : Distribution board MCB/TPNL/SPNL, 415/240 volts of ways as described in Sch 'A'/BOQ shall be provided flush with walls. The sheet metal enclosures shall be factory fabricated preferably of the same manufacturer.

(ii) Miniature circuit breaker (MCB's) of ratings as mentioned in Schedule 'A'/BOQ shall be provided in MSB / SMB's. The MCB shall have breaking capacity of 9 KA. The MCB's shall have ISI mark on them.

(iv) The control gear at MCB's shall be housed in metal box made out of 16 gauge MS sheet with hinged front cover and locking arrangement. The box shall be recessed in the wall. Each MSB shall have proper knob as approved by Engineer-in-Charge for the convenience of opening and closing.

(h) **FITTINGS AND ACCESSORIES:** Switch of rating 5 Amp shall be piano type. Switch socket outlet combination 3 pin 5/16 Amp shall be provided as indicated in Sch 'A'/BOQ and as directed by Engineer-in-Charge. Light fittings shall be as per model Nos as described under Sch 'A'/BOQ.

(i) **TESTING OF ELECTRICAL INSTALLATION** : All the electrical installation shall be tested by the contractor in presence of Engineer-in-Charge as specified in MES SSR Part- I(Specifications). Test results shall be recorded duly signed by the contractor and Engineer-in-Charge for Earth test, Continuity test, Insulating resistance test.

(j) **LED LIGHTS & FITTINGS:** LED light and fittings shall be of following specifications:-

(i) **CONSTRUCTION**: Housing, used as a heat sink shall be made of at least 0.8 mm thick sheet Steel conforming to IS: 513 (Grade O)/CRCA polyester powder coated of at least 60 microns) and high U.V. & corrosion resistance. Heat sink used should be aluminum extrusion having high conductivity preferably to grade 6061 alloy or better having thermal conductivity of at least 170-180 W/m.K or aluminium die cast having high conductivity preferably ADC 12 or LM 24. Luminaires shall be provided with toughened glass of min. 0.8 mm thickness of sufficient strength and high efficiency (90%) prismatic diffuser under the LED chamber to protect the LED and luminaries and shall not show yellowness during luminarie life time. Suitable number of LED lamps shall be used in the luminiars. The contractor shall submit the proof of procurement of LEDs from above OEMs at the time of testing. Suitable reflector / lenses may also be provided to increase the illumination uniformity and distribution. Thermal management shall be done in such a way that it shall not affect the properties of the diffuser.

(ii) **FEATURES**

(aa) The efficiency of the LED lamps at 85 Deg C junction temperatures shall be more than 85%. The output of LED shall be more than 100 lumen per watt at minimal operating current.

(bb) The working life of the lamp at junction temperature of 85 Deg C at rated current shall be more than 50,000 working hours of accumulative operation and shall be suitable for continuous operation of 24 hours per day.

(cc) Colour temperature of the proposed white colour LED shall be 5700k (nomonal CCT) and the color variation should be 5665 +/-365K (ANSI binning). Minimum view angle of the LED shall not be less than 1200**.**

(dd) Power factor of complete fitting shall be more than 0.9 at full load 240V. Input frequency range shall be between 50Hz±3%. The LED luminaire shall be free of glare and the color rendering index CRI shall be more than 75. The Diffuser should be used in the luminiar to restrict the glare of LEDs and shall not have yellowness during the entire life of the luminaire

(ee) The fitting shall be so designed that the illumination level shall be evenly distributed and shall be minimum 250 at 1mtr above ground level if fixed at 2.74 Mtr above floor.

PARTICULAR SPECIFICATION (Contd….)

(iii) **STREET LIGHTS:** The street light LED drivers shall have following specifications**:-**

(aa) TheInput voltage Range shall be within 180Vrms to 270Vrms and Operating input voltage shall be 240Vrms. No load power consumption shall be ≤ 500mW.

(bb) Output voltage shall be 105VDC±3% and Output voltage ripple should be within 3%. Output over voltage protection 125VDC. The power factor shall not be less than 0.95.

(cc) Full Load Efficiency shall be ≥ 90%. The THD shall remain ≤8 %.

(dd) Load regulation shall be ±5% and current waveform should meet the EN 61000-3-2.

(ee) Led Driver shall withstand,voltage of 440V for 2 hours and restore normal working when normal voltage is applied. Maximum Temperature rise shall be <= 30ºC @ 45ºC Temp with safety margin of 10oC.

(ff)The driver should comply to CISPR 15 for limits and methods of measurement of Radio Disturbance characteristics. The equipment should comply to all relevant IEC norms.

(iv) **TECHNICAL REQUIREMENTS:** The driver of the luminaires depending on sizes should have following features:-

(aa) 2 x 2 Fixture- Shall have Short Circuit , Over Voltage, String Open protections.
(bb) 1 x 1 Fixture- Shall have Short Circuit , Over Voltage, String Open protections.
(cc) Down Lighters – Shall have Short Circuit , Over Voltage protections
(dd) Tube Light – Shall have Over Circuit , Over Voltage protections.

(v) **ELECTRONIC COMPONENTS:** These components used shall be as follows:-

(aa) IC (Integrated circuit) shall be of industrial grade or above.
(bb) Metallic film / Paper/Polyester Capacitor shall be rated for a temperature of at least 105ºC.
(cc) The resistors shall be preferably made of metal film of adequate rating. The actual loading versus rating shall be 3.
(dd) The junction temperature of the Switching devices such as transistors and MOSFETs etc. shall not exceed 125ºC (allowing thermal margin of 25 ºC).
(ee) The conformal coating used on PCBs must be cleared and transparent and shall not affect colour code of electronic components or the product code of the company.
(ff) LED must be mounted on MCPCB, which should be made up of Metal core (Aluminum metal core) and should have the alumina coating with aerosol spray process of manufacturing to have the better heat conduction or dissipation.
(gg) MCPCB shall be made with OPA dielectric thickness of 0.1mm and Al 5052H34.
MCPCB must comply with IPC-A-600G and IPC-6012A class 2. UL approved solder mask and silkscreen must be used.
(hh) The heat sink compound used should be of silicon with high thermal conductivity with 3W/mk.
(ii) The heavy components shall be properly fixed. The solder connection shall be with good finish.
(jj) The electronics covered for this equipment shall pass all the tests called for in the specification. The tenderer shall indicate the deviation or compliance otherwise the offer may not be considered for evaluation.
(kk) The infrastructure for Quality Assurance facilities as called for in the specification shall be available for the manufacturing of this product. The compliance shall be indicated clearly in the tender itself.

(k) **EARTHING:** Earthing shall be all as per ISS, IEE Regulations. Earthing shall be provided all as described in respective item of Sch 'A'/BOQ and as, shown in electrical plate No. 3 and specified in clause Nos. 19.13.7 and 19.13.7.2 to 19.14.5 of MES Sch Part-I and GI watering of pipe shall be of light grade and GI protection pipe for earth lead conductor shall be of medium grade. PCC (1:3:6) type C-1 shall be used for PCC earth pit. The charcoal dust and return filling shall be done in layers not exceeding 150 mm thick and shall be properly watered and rammed. Surplus soil shall be disposed off and site left clear and tidy on completion to a distance not exceeding 50 m. All earth electrodes shall be tested for earth resistance by means of standard earth meggar. The resistance of earth shall not exceed one ohm. In case the resistance of earth exceeds one ohm due to soil conditions, then the earthing shall be carried out through another method as per IS-3043.

(l) **TESTING OF ELECTRICAL INSTALLATION** : All the electrical installation shall be tested by the contractor in presence of Engineer-in-Charge as specified in MES SSR Part-I(Specifications). Test results shall be recorded duly signed by the contractor and Engineer-in-Charge for Earth test, Continuity test, Insulating resistance test.

(m) **CEILING FANS**: The fan shall have03 blades size of 1200mm, with down rod length 26 cm. The air delivery shall not be less than 230 Cum/Min with rated speed of 400 RPM. The fan shall not draw more than 75 Watt power under rated voltage of 220 to 240 Volts at 50 Hz frequency.

PARTICULAR SPECIFICATION (Contd….)

43. **LT PANEL BOARD INDOOR TYPE**: The LT Panel board indoor type,as mentioned in the BOQ shall be of size not less than 2000x500x1800mm (LxWxH), 415V, 3 phase and Neutral, 4 wire, 50Hz AC supply system having copper bus bars of 300 Amps rating installed on DMC insulators with bus bar chamber. The panel shall be dust and vermin proof with protection class (IP-54) made of 3.00mm thick CRCA mild steel sheet and mild steel angle iron frame work (made out of 40mm X 40mm X 6mm thick). This shall be of front opening type with opening for cable entry with proper ferrule & cable gland with PVC insulated multistranded copper conductor FRLS cable 1.5 sqmm of adequate size for internal wiring complete. All metal surfaces shall be provided with anticorrosive coating of seven tank process and finally epoxy polyester powder coating of Siemens grey shade 7035. The panel shall also comprise with following accessories as tabulated below and the rates quoted shall be inclusive of all these items:-

43. **REPAIR TO PUMPS: -** The tenderers are advised to visit the site and ascertain all at the factors and all minor items of work/repairs required to the execution of terms in schedule 'A'. the unit rates quoted by the contractor shall be deemed to be included for full and entire completion of all works in accordance with good Engineering practice and recognized principles. Details of works which are obviously and fairly interned and are essential for completion of work out which may not have been mentioned in schedule 'A' or specified hereinafter are deemed to have been included in the unit rates quoted Contractor. Derision of the accepting officer on such matters shall be final and binding.

44. **TESTING:-** Each eqipment shall be test before re-fixing the same after repairs/maintenance and certificate to this effect shall be kept in recorded duly signed by the contractor and Engr-in-Charge.

45. **EXCAVATION AND EARTH WORK:-**

(a) Trenches for pipes shall be excavated to gradient as directed by Engineer-in-Charge at site.

(b) The excavation for pipe line in trenches shall be restricted to the authorised width as defined in clause 3.23 on page No 21 of SSR Part-II and shall be measured accordingly. Nothing will, however, be paid extra to the contractor in addition to the authorised width for working space if required for laying water mains and pipes.

(c) Care shall be taken to keep the trenches free from water during the work. All water which may accumulate from rain or other cause shall be bailed or pumped out or otherwise removed. The tendered rate shall be deemed to include for pumping and or bailing out of water as necessary. No claim what so ever shall be entertained on this account at a later date.

(d) Bottom surfaces of the trenches for pipes shall be formed to level all as directed by the Engineer-in-Charge

(e) All excavated earth shall be deposited at least one metre away from the edge of trenches.

(f) Excavated earth from trenches (as approved) shall be used for refilling and surplus earth shall be removed to a distance as specified in Schedule `A', deposited spread and leveled to the satisfaction of Engineer-in-Charge.

(g) No joints shall be covered up with earth till the pipe line has been tested under pressure but certain length of pipes may be covered up if specifically permitted in writing by the Engineer-in-Charge for the purpose of ensuring safety of traffic, persons etc.

(h) The rates quoted by the contractor for excavation in trenches for pipes shall not include use and waste of timbering. Use and waste of timbering considered necessary and ordered by GE in writing shall be paid for extra as a deviation.

46. **MILD STEEL GALVANISED TUBES (PIPES) AND FITTINGS**

46.1 Mild steel galvanised tubes (pipes) and fittings shall bear ISI certification mark with IS-1239 (Part-I) and IS-1239 (Part-II) all as specified in clause 18.4.1 and 18.4.3 on page 352 of MES SSR-2009 (Part-I).

46.2 **LAYING & JOINTING OF GI PIPE**

46.2.1 Laying and jointing of GI pipes shall be carried out as specified in SSR-2009 Part-I.

46.2.2 All steel water tubing shall be galvanised (screwed and socketed) and of medium grade of make as mentioned in Appendix 'D'.

46.2.3 All water supply fittings such as unions, sockets, tee, elbows, bends etc shall be galvanised iron, medium grade, ISI marked.

46.2.4. Stop cocks, bib cocks, shower roses, wall mixers, basin mixtures shall be cast copper alloy chromium plated finish ISI marked. However, pillar cocks for WHBs shall be of fancy type cast copper alloy chromium plated and of make as specified in Appendix 'D'.

46.2.5 All bib cocks shall be 15mm dia, stop cock of the same dia as that of tubing & shower roses of dia as specified in Schedule `A`, swivel type, heavy duty.

PARTICULAR SPECIFICATION (Contd….)

46.3 **WORKMANSHIP OF STEEL TUBING/PIPE**

46.3.1 The contractor shall use proper bends/elbows/tees at turnings/corners. Bending of pipe is not permitted except when the pipe has to follow the contour of brick work or where fittings cannot be inserted. In such cases the bend shall be gradual and formed with the written permission of GE.

46.3.2 Pipes shall not run diagonally.

46.3.3 Unions shall be provided at appropriate places as directed by the Engineer - in-Charge so that in repairs etc. long lengths pipes are not to be taken down.

46.3.4 Contractor shall provide screwed plugs to all open ends of pipes on completion of work without any extra cost to the Department.

46.3.5 The run of pipes shall be straight. All the pipes inside the buildings shall be embedded in floors/fixed to walls. Pipes fixed to walls shall be secured with approved hooks at about 1.50 metre intervals. However, in the case of pipe embedded in the floor etc. the pipe hooks need not be provided.

46.3.6 As far as possible, pipes shall be fixed in shafts (where applicable) in such a way so that they do not give an ugly appearance.

46.3.7 Proper unions sockets shall be provided in rising/delivery main to and from service tank before or after the stop valve without any extra cost to the department.

46.3.8 All concealed water tubing shall be coated with bituminous paint before filling the chases and the rates quoted shall be deemed to have been included for this provision.

47. **VALVE PIT:-**

47.1 Valve pit shall be constructed of size as mentioned in the description of item(s) under Schedule `A' in accordance with the following specifications:-

(a)	Excavation and Earth Work	In any type of soil (soft/lose and hard/dense)
(b)	Foundation concrete.	PCC (1:2:4), type D-2 using 40mm graded stone aggregate.
(c)	PCC Top	PCC (1:2:4), type B-1(nominal mix) using 20mm graded aggregate.
(d)	Brick Work	Sub-class 'B' bricks (old size) in cement and sand mortar (1:4)
(e)	Finishing	(i) 15mm thick cement and sand mortar (1:4) finished even and fair without using extra cement (internally) (ii) External face of brickwork shall be finished flush in the same mortar as the work proceeds.
(g)	White Washing	Three coats of white wash shall be carried out internally to plastered surfaces
(h)	Filling	Filling of earth around chamber shall be carried out with approved earth in layers each of 25cm and well rammed and watered. Surplus soil shall be disposed of within a distance of 50 meters.

48. CAST IRON PRESSURE PIPES AND FITTINGS:-

Work shall be carried out as per clause No 18.2.1 to 18.2.4.6 & 18.46 of SSR Part-I 2009. Joint shall be as per clause No 18.11, 18.9 & 18.48.2 of SSR Part-I 2009.

49. SLUICE VALVE: Shall be fixed as per clause No 18.57.1 to 18.57.6 of SSR Part-I 2009.

50. Non Return Valve. Where specified non return valve (swing check type) shall be provided through which flow can occur in one direction only. It shall be single door swing check type of best quality conforming to IS : 5312.

51. PURCHASE VOUCHERS

In addition to this mandatory submission of purchase vouchers for any material specified elsewhere in tender documents, contractor shall submit the original purchase vouchers to the Engr-in-Charge for his verification for the following items. The said original purchase vouchers shall be released to the contractor after defacing the same as having verified with respect to this contract. No payment shall be released in absence of purchase Vouchers:-

(a)	GI pipe.	(b)	Gate valve.	(c)	Foot valve.
(d)	Ball valve	(e)	Star delta Starter	(f)	CI Pipe.
(g)	PVC Tank.	(h)	Sluice valve. LT panel. Tank nipple. NRB.		

Signature of contractor
Dated:_____

AGE (Contracts)
For Accepting Officer

APPENDIX - 'A'

LIST OF MATERIALS HAVING ISI / BIS CERETIFICATION MARKING

Srl No	Name of Materials	Relevant IS
1.	**CONCRETE**	
(a)	Integral cement water proofing compound	2645
(b)	Ply wood for concrete shuttering work	4990
2.	**JOINERY**	
(a)	Wooden flush door shutters, solid core type	2202(Part-1)
(b)	Particle board and hard board face panel	2202(Part- II)
3	**BUILDERS HARD WARE**	
(a)	Steel Butt hinges	1341
(b)	Non-Ferrous metal butt hinges	205
(c)	Ferrous tower bolts	204 (Part-I)
(d)	Non-Ferrous tower bolts	208
(f)	Parliament hinges	362
(g)	Hydraulically operated door closers	3564
(h)	Continuous piano hinges	3818
(j)	Non-Ferrous metal sliding door bolts	2681
(k)	Tee and strap hinges	206
(l)	Mild steel sliding door bolts	281
4.	**STEEL & IRON WORK**	
(a)	Steel doors, windows, and ventilators.	1038
5.	**ROOF COVERING**	
(a)	Bitumen felts for water proofing and damp proofing	1322
6.	**CELLING AND LINING**	
(a)	Ply wood for general purpose	303
(b)	Block boards	1659
(c)	Veneered particle board	3097
(d)	Marine plywood	710
(e)	Fiber hard board	1658
(f)	Wood particleboard	3087
7.	**FLOORING**	
(a)	White Portland cement	8042
(b)	Cement concrete flooring tiles.	1257
8	**WATER SUPPLY PLUMBING DRAINS AND SANITARY APPLIANCES**	
(a)	Concrete pipes with or without reinforcement	458
(b)	Salt glazed stone ware pipes and fittings	651
(c)	Flushing cisterns for water closets and urinals valveless symphonic type	774
(d)	Cast copper alloy screw down bib tap and stop valves	781
(e)	Galvanised mild steel tubes	1239 (part-I)
(f)	Galvanised mild steel tube fittings	1239 (part-II)
(e)	Sand cast iron, spigot and socket soil, waste and ventilating pipes &fittings	3989
(f)	Ball valves (horizontal plunger type) including floats for water supply purposes	1703
(j)	Cast iron manhole covers and frames	1726(Part I to VII)
(k)	AC pressure pipes	1592
(l)	Automatic flushing cisterns for urinals	2336

APPENDIX 'A' Contd…./-.

Srl No	Name of Materials		Relevant IS
(m)	Vitreous china ware appliances		
	i)	Wash down water closet	2556 (part-II)
	ii)	Squatting pans	2556 (part III)
	iii)	WashBasin	2556 (part IV)
	iv)	Laboratory sinks	2556 (Part V)
	v)	Urinals bowl type	2556 Part VI)
	vi)	Half round channels	2556 (Part VII)
	vii)	Syphonic wash down water closets	2556 (Part VIII)
		Foot rests	2556 (Part X)
(n)	Plastic WC seat covers		2546
(o)	Vertically cast iron pressure pipes for water gas and sewage		1537
(p)	Pillar taps for water supply purposes		1795
(q)	Centrifugally cast pipes for water, gas and sewage		1536
(r)	Centrifugally cast (spun) iron spigot & socket soil, waste and ventilating pipes, fittings and accessories.		3989
(s)	Rubber sealing rings for gas mains, water mains and sewers		5382
(t)	Cast iron fittings for pressure pipe for water, gas and sewage		1538 (Part I to XXIII)

Note: All fittings & fixtures for buildings including hardware and all other items enumerated in this Appendix sanitary fittings & electrical fittings etc shall be ISI marked. In case any fittings/fixtures of ISI marked is not manufactured then the fittings & fixtures conforming to relevant IS may be incorporated as approved by the GE. The contract provision shall be deemed to be amended accordingly & unit rate of buildings under Sch'A' shall be deemed to be inclusive for the same & no extra payment on this account shall be admissible.

Signature of contractor
Dated:_____

AGE (Contracts)
For Accepting Officer

Appendix ‘ B ’

LIST OF PRODUCTS WHICH ARE GENERALLY SUPERIOR TO BIS / ISI CERTIFIED PRODUCTS TO BE INCORPORATED IN WORKS

1. Asbestos Cement Sheets	-	IS-459
2. Asbestos Cement Building Pipes & pipe fittings	-	IS-1626 (Part 1)
3. Distempers washable and dry	-	IS-427 & IS-428
4. Paints	-	IS- 2932
5. Cast copper alloy fancy bib taps and stop valves	-	IS- 8931
6. Cement paint	-	IS- 5410
7. Cast copper alloy fancy pillar taps	-	IS-8934

Signature of contractor
Dated:_____

AGE (Contracts)
For Accepting Officer

Appendix 'C'

S NO	NAME OF PRODUCT	MAKES/ BRANDS/ MANUFACTURER
1	PRESSED STEEL DOOR FRAMES/ CHOWKHATS FOR DOORS/ CUPBOARD AND WINDOW FRAMES	PARSHOTAM STEEL INDUSTRIES (PSI), SHAKTI, MADHU INDUSTRIES, SHIVAM METAL SHAPERS & ASHISH INDUSTRIES
2	STEEL WINDOWS / VENTS	PARSHOTAM STEEL INDUSTRIES (PSI), SHAKTI, MADHU INDUSTRIES, SHIVAM METAL SHAPERS & ASHISH INDUSTRIES
3	FACTORY MADE PANELLED DOOR SHUTTERS	JAIN DOOR PVT LTD, PARSHOTAM STEEL INDUSTRIES (PSI) & MAICA
4	PVC BOARDS /DOOR FRAME / SHUTTERS	ACCUCEL / ECHON, H R ENTERPRISES & RAJSHRI PLASTIWOOD
5	UPVC WINDOWS / DOORS/FRAMES/ VENTILATORS	ACCUCEL / ECHON , YASHPOLY, APARNA VENESTER/OKOTECH, MADHU INDUSTRIES, H R ENTERPRISES, RAJSHRI PLASTIWOOD, FENESTA, ENCRAFT, SUPREME, ASHISH INDUSTRIES & NCL VEKA
6	ANODISED ALUMINIUM DOORS/ WINDOWS/ FRAMES/ PARTITIONS	ASHISH INDUSTRIES, JINDAL, KALCO & ALPURE
7	GALVANISED STEEL SHEETS & COLOUR COATED GALVANIZED STEEL SHEETS	KAMDHENU, TATA & JINDAL
8	ROOF TILES, KERBS STONE	MOHINDRA INDUSTRIES,HARIOM TILES & PAVERS, RN TILES & PAVERS, SWASTIC ALW, GPT & SVRCC
9	GLAZED CERAMIC/ NON SKID WALL/ FLOOR TILES	AGL, ORIENT BELL, SUNHEARRT CERAMIK, RAK CERAMICS, SOMANY, NITCO, JOHNSON & VITERO TILES
10	VITRIFIED TILES (NORMAL / ANTI-SKID)	AGL, ORIENT BELL, SUNHEARRT CERAMIK, VARMORA, RAK CERAMICS, SOMANY & NITCO
11	HDPE & LLDPE STORAGE WATER TANKS	ASTRAL, SINTEX & SUPREME
12	STAINLESS STEEL KITCHEN SINK WITH OR WITHOUT DRAINING BOARD, PLATE RACK	JAINKO, SHAKTI, PRIMA, PRAYAG, DIAMOND, NIRAL, SPC, SHRI NAVKAR METAL LTD, TOPSAN, XEN, JYNAM & SILVERSHINE
13	BRASS VALVES	PRAYAG, L&T VALVES, AMCO, XEN, & VIP VALVES
14	SHOWER ROSES CHOROMIUM PLATED (SWIVEL TYPE)	PRIMA, JAINKO, SHAKTI, SILVERSHINE, VITERO, SOMANY, PRAYAG, PLAYER,JAQUAR, AP ROYALE & BATHSENSE
15	CP BATH ROOM FANCY FITTINGS & ACCESSORIES/ CP BATH FITTINGS	PRIMA, SHAKTI, PLAYER, PRAYAG, VITERO, SOMANY, JAQUAR, ESSCO, AP ROYALE BATH SENSE, TOPSAN, SILVER SHINE, CORONET, XEN & JAYNAM
16	BIB TAPS AND STOP VALVES (CP) FANCY FITTINGS	ESSCO, JAQUAR, PRIMA, PLAYER, AP ROYALE BATH SENSE, SILVER SHINE, JAINKO, TOPSANS, CORONET, SHAKTI & PRAYAG
17	PTMT / PVC FITTINGS	SHAKTI, SILVERSHINE, JAYNAM, PRIMA, XEN, POLYPLAST & PRAYAG
18	HDPE PIPES / SNR PIPES (WATER / SEWAGE)	ASTRAL, SFMC, SUPREME, UTKARSH & DURALINE
19	RCC HUME PIPES	KB SPUN, SVRCC & KALOHA SPUN PIPE
20	CEMENT BASED PAINTS	SNOWCEM, ASIAN, BERGER & DULUX
21	DECORATIVE PAINT (EXR & INT)	SNOWCEM, DULUX, SHALIMAR PAINTS & JK MAX PAINTS
22	HIGH BUILT TEXTURE & WALL FINISHES (EXT & INT)	JK MAX PAINTS, SNOWCEM, ASIAN, BERGER & DULUX
23	PVC WALL PANELLING / PVC CEILING & PARTITION	ECOCELL / ECHON, HR ENTERPRISES, RAJSHRI PLASTIWOOD, DECOINN, BLACK COBRA, SUNBEAM & MAICA
24	HDF / PVC DOOR FRAME SHUTTER MOULDED	HR ENTERPRISES, ECHON, ALCAD,BLACK COBRA ALEX & DECOINN
25	FALSE CEILING TILES / BOARD (GYPSUM BOARD / POP / MINERAL FIBRE)/JOINTING & FINISHING MATERIAL	HI-STEEL, GYPCORE, RAJSHRI PLASTIWOOD & ARMSTRONG
26	FALSE CEILING / METAL FALSE CEILING/ METAL FARMING	HI-STEEL, GYPCORE, RAJSHRI PLASTIWOOD, ROYAL KRAFT & ARMSTRONG
27	WPC DOOR FRAMES/ PANELS/ HYBRID SHEET/ FLOORING/ JALLI	ECOCELL / ECHON, RAJSHRI PLASTIWOOD, H R ENTERPRISES, BLACK COBRA & ASHISH INDUSTRIES
28	CPVC PIPES & FITTINGS	FUSION, SENTINI FLOPIPES, PRAYAG, SUPREME, ASTRAL, FINOLEX & DUTTON
29	PVC/UPVC /SWR PIPES & FITTINGS	FUSION, SENTINI FLOPIPES, PRAYAG, SUPREME, ASTRAL, FINOLEX & DUTTON

Appendix 'C' (CONTD…)

SL NO	NAME OF PRODUCT	MAKES/ BRANDS/ MANUFACTURER
30	SANITARYWARE	PRAYAG, HINDWARE, KOHLER , JAQUAR, AP ROYALE & BATHSENSE, SOMANY & RAK CERAMICS
31	PRE PAINTED GALVALUME STEEL SHEET GALVALUME SHEETS	KAMDHENU, TATA, JINDAL & DYNA ROOFING PVT LDT & ASHISH INDUSTRIES
32	GLASS REINFORCED GYPSUM (GRG) CEILING TILES	DIAMOND CEILING, Hi-STEEL, GYPCORE, MARUTI GYPSUM & SHERA
33	AAC BLOCKS & JOINTING ADHESIVE	MODCRETE, CLAVECON & JK SMART BLOX
34	PVC FLUSHING CISTERNS	PRAYAG, RAK CERAMICS, SILVER SHINE, JAQUAR, HINDWARE, PARRYWARE, AP ROYALALE, JANKO, CORONET, PRIMA, BATHSENSE, POLYPLAST & SUPREME
35	TOILET SEAT COVER	PRAYAG, RAK CERAMICS, PARRYWARE, CERA, SILVERSHINE, JAQUAR, AP ROYALE, JANKO, POLYPLAST & BATHSENSE
36	PCC MANHOLE & DRAIN COVER EARTH PIT, CABLE ROUTE INDICATOR	VENUS, HARIOM TILES & PAVERS, MOHINDRA & SVRCC
37	WOODEN FURNITURE	ZUARI FURNITURE, GODREJ & BOYCE, DURIAN
38	FLY ASH BRICKS	SWASTIK ALWST, INDERBIR INDUTRIES & SHRI RAM FLY ASH BRICKS
39	ELECTRIC CABLE COVER	INDERBIR INDUTRIES & SHRI RAM FLY ASH BRICKS
40	PRECAST RCC COVERS	SWASTIK ALWST, SVRCC & KALOHA SPUN PIPES
41	CONSTRUCTION CHEMICAL AND WATER PROFFING PRODUCTS	M/S CHOKSEY CHEMICALS PVT LTD, M/S TERMAZ LTD, M/S FOSFOC CHEMICALS, MS STP LIMITED, M/S IWL INDIA LIMITED, M/S BERGER PAINT INDIA LTD, M/S ASIAN PAINTS LTD, M/S TIKI TAR DANOSA (INDIA) PVT LTD, M/S STANROSE ENVIROTECH INDIA PVT LTD, M/S BOSTI INDIA PVT LTD, M/S TORCHTAR MEMBRANES & BITUMEN PRODUCT PVT LTD, M/S PIDILITE INDUSTRIES LTD & HINDCON CHEMICAL LTD.
42	TERRAZZO TILES/ PPC TILES/ PPC TILES/ PPC CHEQUERED TILES	SURYA KIRAN, SITCO MOHINDRA, CORAL & SWASTIKA ALW
43	INTERLOCKING TILES/ EXTERIOR TILES / PAVER BLOCKS	SURYA KIRAN, POOJA CONCRETE, MOHINDRA, HARIOM TILES & PAVERS, KB SPUN, RANA TILE CO, SHINO TILES, RN TILES, CORAL & PARVERS
44	HDPE DWC PIPES	ASTRAL, DURTON GEMINI PIPES & FLOW WELL
45	FLY ASH BRICKS	SWASTIKA ALWST, INDERBIR INDUSTRIES & SHRI RAM FLY ASH BRICKS
46	SOLAR WATER HEATING SYSTEM	TATA BP SOLAR & RACOLD
47	LED LIGHTS, PANELS & LUMINARIES INDOOR	SSK, EVEREADY, POLYCAB, HAVELLS, HALONIX, BENLO, HPL,WIPRO, LUKER, ORIENT ELECTRIC, KCSAD, ASCENTECH, JAQUAR, PHILIPS, GOLD MEDAL, CROMPTON, BAJAJ & OSRAM
48	LED LIGHT FITTING & LUMINARIES FOR ROAD AND STREET LIGHTING (OUT DOOR)	SSK, EVEREADY, POLYCAB, HAVELLS, HALONIX, BENLO, HPL,WIPRO, LUKER, ORIENT ELECTRIC, KCSAD, ASCENTECH, JAQUAR, PHILIPS, GOLD MEDAL, CROMPTON, BAJAJ & OSRAM
49	LED LAMPS	EVEREADY, POLYCAB, HALONIX, BENLO,WIPRO, ORIENT ELECTRIC, KCSAD, ASCENTECH, JAQUAR, GOLD MEDAL, PHILIPS, CROMPTON, BAJAJ & OSRAM
50	SMART STREET LIGHTS AND CONTROL SYSTEM	ORIENT ELECTRIC LTD, KCSAD, EVEREADY & ASCENTECH
51	LED FLOOR LIGHT	WIPRO, KCSAD,EVEREADY,ASCENTECH, JAQUAR & GOLD MEDAL
52	TUBE LIGHT/ STREET LIGHT FITTINGS & ACCESSORIES	POLYCAB, HAVELLS, HALONIX, HPL, JAQUAR & GOLD MEDAL
53	FLOOD LIGHT FITTINGS	POLYCAB, HALONIX, KCSAD, EVEREADY & GOLD MEDAL

Appendix 'C' (CONTD…)

54	BOX TYPE DECORATIVE TUBE LIGHT FITTINGS	POLYCAB, HAVELLS & HALONIX
55	POWER TRANSFORMER 132/66/33/22KV	JAY BEE INDUSTRIES PANCHKULA, JAY BEE INDUSTRIES BTI, MUSKAAN, ITE GURGAON, TRANSCON, ABC TRANSFORMERS, BHARAT BIJLEE, SCHEINDER, CROMPTON GREAVES, ANDREW YULE & ALSTOM
56	POWER TRANSFORMER (NON SEALED TYPE) UPTO 12.5 MVA OF 33KV	JAY BEE INDUSTRIES PANCHKULA, JAY BEE INDUSTRIES BTI, MUSKAAN, ITE GURGAON, TRANSCON, ABC TRANSFORMERS, BHARAT BIJLEE, SCHEINDER, CROMPTON GREAVES, ANDREW YULE & ALSTOM
57	OUTDOOR TYPE OIL IMMERSED DISTRIBUTION TRANSFORMERS 11KV UPTO AND INCLUDING 2500 KVA, 33KV CAPACITY	JAY BEE INDUSTRIES PANCHKULA, JAY BEE INDUSTRIES BTI, IECO, MUSKAAN, TRANSCON, BHARAT BIJLEE, SCHEINDER, CROMPTON GREAVES, ANDREW YULE & ALSTOM
58	DISTRIBUTION TRANSFORMERS 11 KV ABOVE 500 KVA CAPACITY	JAY BEE INDUSTRIES PANCHKULA, JAY BEE INDUSTRIES BTI, IECO, MUSKAAN, ITE GURGAON, TRANSCON, POWERWARE, ABC TRANSFORMERS, BHARAT BIJLEE, SCHEINDER, CROMPTON GREAVES, ANDREW YULE & ALSTOM
59	DISTRIBUTION TRANSFORMERS UPTO 500 VA CAPACITY	JAY BEE INDUSTRIES PANCHKULA, JAY BEE INDUSTRIES BTI, IECO, MUSKAAN, ITE GURGAON, TRANSCON, POWERWARE, ABC TRANSFORMERS, BHARAT BIJLEE, SCHEINDER, CROMPTON GREAVES, ANDREW YULE & ALSTOM
60	DRY TYPE TRANSFORMER UPTO 7.5 MVA/33KV CLASS AND OIL TYPE TRANSFORMERS UPTO 20MVA/66KV CLASS	JAY BEE INDUSTRIES PANCHKULA, JAY BEE INDUSTRIES BTI, MUSKAAN & TRANSCON
61	POWER TRANSFORMERS 66/33/11 KV COPPER WIRED	JAY BEE INDUSTRIES PANCHKULA, JAY BEE INDUSTRIES BTI, MUSKAAN TRANSCON & ABC TRANSFORMERS
62	LV SWITCH BOARD	L&T, CROMPTON & HENSEL
63	LT PANELS/APFC/PANELS/AC CONTROL PANELS/PUMP HOUSE PANELS FOR SS HAVING CAPCITY 1000 KVA AND ABOVE/BELOW 1000 KVA	SUPERTECH POWER CONTROL PVT LTD, IECO, BHANDARI, NEPTUNE, CORONET, SPC (SPC ELECTROTECH PVT LTD), DVEPL, PSC, ITE GURGAON, INDO ASIAN, HAVELLS, MEGAWIN, SHALABH, HPL, HARTEK, ABB, L&T & SIEMENS
64	FAN (CEILING, EXHAUST, TABLE, WALL &PEDESTAL)	GOLD MEDAL, POLYCAB, HAVELLS, ANCHOR, LUKER CROMPTON, BAJAJ, KHAITAN, USHA & ORIENT
65	STORAGE WATER HEATER GLS	JAQUAR, GOLD MEDAL, AD SMITH, USHA, CROMPTON, RAMCO & BAJAJ
66	PT SWITCH/SOCKETS/ CEILING ROSES/FAN REGULATORS	GOLD MEDAL, SSK, LEGRAND, CRABTREE, SCHNEIDER, PHILIPS, ANCHOR, BAJAJ & KINJAL
67	MODULAR SWITCH/SOCKETS/ FAN REGULATORS	GOLD MEDAL, BENLO, INDO ASIAN, POLYCAB, HAVELLS, HPL, ANCHOR, LEGRAND, CRABTREE, SCHNEDIER & PHILIPS
68	HT 132/66/33 KV& 11KV XLPE CABLES	LS CABLE, GRANDLAY, V-MARC, GEMSCAB, RALLISON, POLYCAB, HAVELLS, JMW KABEL, CABLE CORPORATION OF INDIA, UNIVERSAL CABLE, ELEKTRON, PARAGON & PARAMOUNT CABLES
69	LT POWER & CONTROL CABLES, 1100 VOLTS, XLPE	LS CABLE, PLAZA CABLES, GRANDLAY, V-MARC, GEMSCAB, SHALABH, RALLISON, POLYCAB, HAVELLS, JMW KABEL, HPL,CABLE CORPORATION OF INDIA, ELEKTRON, PARAGON & PARAMOUNT CABLES
70	PVC WIRES & CABLES 650/1100 VOLTS	LS CABLE, GOLD MEDAL, PLAZA CABLES, V-MARC, GRANDLAY, GEMSCAB, SHALABH, RALLISON, POLYCAB, HAVELLS, BENLO, JMW KABEL, FINLOX, ELEKTRON, PARAGON & PARAMOUNT CABLES
71	HT/LT XLPE AERIAL BUNCHED CABLES	GRANDLAY, PLAZA CABLES, V-MARC, LS CABLE, JMW KABEL & PARAMOUNT CABLES
72	CABLE JUNCTION BOXES, CIRCUIT BREAKER BOXES, DC RATED JUNCTON BOXES	DVEPL, HPL, MEGAWIN, TRANSGUARD & PHOENIX CONTACT

Appendix 'C' (CONTD…)

73	LT AIR CIRCUIT BREAKER	INDO ASIAN, HAVELLS, MEGAWIN, HPL & TRANSGUARD
74	COMPOSITE & RELAY PANEL	SUPERTECH POWER CONTROL PVT LDT & CORONET
75	COMPOSITE POLYMERIC INSULATOR	JAIPURIA, POWER-GRID & MG MATERIALS
76	DG SETS	MAHINDRA, CUMMINS & JAKSONS
77	RIGID PVC CONDUIT PIPES	PLACE CABLES, GOLD MEDAL, MODI'S, KALINGA, PLAZA
78	GI / MS CONDUIT / ERW CONDUIT PIPE	VPL, AKG, BEC, BHARAT STEEL TUBES
79	ACSR CONDUCTOR	ALIND, COYOTE & STERLITE POWER
80	MEDIUM VOLTAGE SFC GAS INSULATED RMU (RING MAIN UNIT) AND OIL INSULATED RMU OF 12KV, 24KV & 36 KV	MEGAWIN, PASCAL, CGPISL & TRANSGUARD
81	HT SWITCH GEAR 33/66 KV, SF6 (OUT DOOR)	MEGAWIN, PASCAL, CGPISL, HARTEK & TRANSGUARD
82	HT SWITCH GEAR 11 KV, VCB (INDOOR/OUT DOOR) HT RING MAIN UNIT, SF6/ACUUM, 11 KV/CT & PT	PASCAL, TRANSGUARD, CGPISL, CORONET, ITE GURGAON, HARTEK & MEGAWIN
83	PACKAGE / UNIFIED SUB-STATION	ITE GURGAON, MEGAWIN, CGPISL & CORONET
84	MCCBs	INDO ASIAN, HAVELLS, BENLO, HPL, GOLD MEDAL, SSK & SHALABH
85	MCB/ ELCB/ RCCB/ RCBO/ ISOLATORS/ MCB DISTRIBUTION BOARD	GOLD MEDAL, POLYCAB, SSK, INDO ASIAN, HAVELLS, BENLO, HPL & SHALABH
86	CHANGE OVER SWITCHES	MEGAWIN, INDO ASIAN, HAVELLS, HPL & SHALABH
87	POWER CONTACTOR	INDO ASIAN, HAVELLS, & HPL
88	SMART/ TAMPER PROOF ENERGY METERS (DIGITAL TYPE)	ANCHOR BY PANASONIC, HPL, ELMEASURE, BENLO, L&T & SECURE
89	STATIC ENERGY METER	BENLO, ELMEASURE, L&T, ABB & SECURE
90	VOLTAGE STABILIZERS UPTO 25 KVA	MUSKAAN, ITE GURGAON, VOLINA, SINETRAC, VINITEC & IECO
91	VOLTAGE STABILIZERS ABOVE 25 KVA (SERVO CONTROLLED)	MUSKAAN, ITE GURUGAON, VOLINA, SINETRAC, VINITEC, IECO & POWERWARE
92	STREET LIGHT TIMERS	SINETRAC, INDO ASIAN, HAVELLS
46	ISOLATION TRANSFORMERS	VOLINA, SINETRAC, VINITEC, IECO, MUSKAAN, ITE GURGAON
47	HT/LT STEEL TUBULAR SWAGED POLES	SHIVAM METEL SHAPERS INDUSTRIES, SURYA & HPL
48	OIL FILLED DISTR TRANSFORMER UPTO 2500KVA (VOLTAGE CLASS UPTO 33KV CLASS)	VOLTMAP, CROMPTON GREAVES & JAYESH ELECTRICAL LTD
49	HT PANEL	PSC, HARTEK & PASCAL
50	BUS DUCTS / RASING MAIN	SPC (SPC ELECTROTECH PVT LTD), PHOENIX CONTRACT & SHALABH
51	GI PIPES / MS PIPES	JINDAL, TATA & OSWAL
52	HDPE/UPVC/PPR PIPES	FUSION, FINOLEX, SFMC, PRINCE & SUPREME
53	BRONZE SLUICE VALVES/NON RETURN VALVES/AIR RELEASE VALVES/PRESSURE REDUCING VALVE/FIRE HYDRANT VALVE/SPRING RELIEF VALVE/BALL VALVE/STEAM VALVE / FLOAT VALVE	PRAYAG , SHAKTI, KIRLOSKAR, ZOLOTO, LEADER & L&T (AUDCO)
54	FLOAT VALVE	PRAYAG, SHAKTI, AIRAEURO, AGROTECH , TELEFLO & TECHNO INTERNATIONAL
55	BUTTTERFLY VALVES/CHECK VALVES	AUDECO, ADVANCE & DEEPAK
56	GUN METAL GATE VALVE/GLOBE VALVE/HORIZONTAL & VRTEICAL CHECK VALVE	KIRLOSKAR , LEADER & L&T (AUDCO)
57	CI PIPES & FITTINGS	JINDAL, TATA METALIKS & TATA KUBOTA
58	GI FITTING	JINDAL, TATA & BST
59	PUMP SETS-MONO BLOCK	KIRLOSKAR, CROMPTON, JOHNSON & JOHNSON, KSB
60	CENTRIFUGAL PUMPS	KIRLOSKAR, KSB, GRANDFOS & ITT
61	SUBMERSIBLE PUMPS	KIRLOSKAR, KSB, GRANDFOS, WASP, ITT & BEACON

Appendix 'C' (CONTD…)

62	NON CLOG SEWAGE PUMPS	KIRLOSKAR, KSB, CROMPTON, JOHNSON & JOHNSON
63	VERTICAL TURBINE PUMP	KIRLOSKAR, KSB, BAJAJ ATLANTA, JOHNSON & JOHNSON
64	DOSING PUMP/ BLEACHING DOZER	ION EXCHANGE, MAIC INDIA & JESCO
65	MOTOR STARTERS	HPL,L&T, BCH, SCHNEIDER, SIEMENS & ABB
66	ELECTRIC MOTOR	BHARAT BIJLEE, KIRLOSKAR ELECTRIC, CROMPTON GREAVES, ABB, SIEMENS & GEC
67	SINGLE PHASE REVENTORS/ PHASE SEQUENCE CORRECTOR	L&T, SIEMENS, LEGRAND, SCHNEIDER & ABB
68	FAN CAPACITORS	HEVELLS, EPICO & TIBCON
69	FAN BEARINGS	SKF, NBC, BOSCH
70	WATER COOLERS	VOLTAS, BLUESTAR, SHRIRAM, USHA & SIDHWAL
71	REFRIGERATOR	WHIRLPOOL, LG, SAMSUNG, VOLTAS, VIDEOCON & KELVINATOR
72	DEEP FREEZER	BLUE STAR, VOLTAS, SIDHWAL, CARRIER & FEDDERS
73	CASSETTE TYPE AC	BLUE STAR, LLYOD, VOLTAS PANASONIC, TOSHIBA & LG
74	PILLAR TYPR AC	LLYDO, LG, BLUE STAR, WHIRLPOOL & PANASONIC
75	DESERT COOLERS	SYMPHONY, KHAITAN, KENSTAR, CROMPTON GREAVES, BAJAJ & OSRAM
76	VARIABLE FREQUENCY DRIVES (VFD)	DANFOSS, HONEYWELL, ABB, AUTOMATION, SIMPHOENIX
77	POT / Y-STRAINER	EMERALD, RAPID COOL, LEADER
78	PRESSURE GAUGE/ DIAL TYPE THERMOMETERS	FIEBIG, H.GURU, EMERALD, AUTOMAT LUB SYSTEM
79	INSULATION (FIBRE GLASS)	LLYOD, SHRI FIRE PACK, IMPEX INSULATION, TRANS INSULATION
80	INSULATION (EXPANDED POLYETHYLENE)	TOSHIBA, STYRENE PACKAGING, CAPRICORN, ECOTHEM INSULATION
81	INSULATION (CROSS POLYTHENE FOAM)	SUPREME, PARAMOUNT, AEROFLEX, TROCELLONE
82	INSULATION (NITRILE RUBBER)	KIFLEX, KFLEX, VIDOFLEX, PARAMOUNT
83	TF THERMOCOL FOR PIPE INSULATION	PK PACKING, BIRDSELL, PROPACK, STYRENE PACKAGING, RYCON SERVICES, KIWI THERMO PACK, SAHARA THERMO PACK
84	CONTROLS (3 WAY VALVES) / THERMOSTATS	ANERGY, SIEMENS, HONEYWELL
85	FLEXIBLE CONNECTOR	CORI ENGINEER, RESISTO FLEX, DUNLOP
86	BUTTERFLY VALVE	AUDECD, ADVANCE, DEEPAK
87	FIRE EXTINGUISHER ALL TYPE	MINIMAX, FIREX, NEWAGE, NITIN, CEASEFIRE
88	FIRE HOSE PIPES & ACCESSORIES	NEWAGE, DUNLOP, COSMOS
89	FIRE HYDRANT / LANDING VALVES	MINIMAX, FIREX, SAFEX
90	FIRE PANEL	MATHER & PLATT, MINIMAX, SAFEX, AGNI, HONEY WELL, SYSTEM SENSOR, FIREX, NOTEFIRE
91	HOOTER / SIREN	MINIMAX, MELTRON, AHUJA, PHILIPS, AGNI
92	MANUAL CALL BELL POINT	AGNI, SURKSHA, MINIMAX, FIREX
93	SMOKE/ HEAT / IONISATION DETECTOR ADDRESSABLE &NON ADDRESSABLE	APPOLO, EDWARDS, HONEY WELL, SYSTEM SENSER
94	WATER SPRINKLER HEAD	APPOLO, TYCO, CEASEFIRE, SAFEX, MINIMAX, STARTECH FIRE SYSTEM, ELECTRONIC CONTROLS, CEASEFIRE

Note:-

(i) In case there are less than 3 make in the approved list for any material/ item / product then only generic description / specification of the material / item / product as specified in tender and relevant codal provision shall be followed.

(ii) Any other make approved by the E-in-C's and or CEWC Chandimandir, upto end date of submission of the bid shall be deemed included in the above list.

Signature of contractor AGE (Contracts)

Dated:_____ For Accepting Officer

APPENDIX 'D'

RECOVERY CHARGES FOR TESTING OF MATERIALS

Legend :- A-Site Lab, B- Zonal Lab, C-National Test House/ SEMT/ Govt approved lab/Engg College

Srl No	Materials	Test	Method of Testing	Frequencies of Test	Level of Test	Rate in Rs	Remarks
1	2	3	4	5	6	7	8
1	Bricks			As per IS 5454 as given under:	Permissible % of defective Bricks		Check for visual and dimensional characteristic shall also be allowed out as per IS-5454
				Lot Size / Sample Size	0		
				1001to10000 / 5	0		
				10001to35000 / 10	1		
				35001to50000 / 15			
	(i) Compressive Strength		IS-3495(Part-II)		A	330	
	(ii) Water Absorption		IS-3495(Part-II)		A	330	
	(iii) Efflorescence		-do-(Part-I)		A	330	
2	Coarse Aggregate	(i) Sieve Analysis	IS-2386(Part-I)	OneTestforevery15cumofAggregate or part thereof.	A	660	
		(ii) Flakiness Indices	-do-	OneTestforevery15cumofAggregateorpart thereof.	A	250	
		(iii) Estimation of deleterious materials	-do-	One Test for every 100 cum of Aggregate or part thereof.	A	600	
		(iv) striping value of aggregate	IS-6241-1971	one test each source of supply or when changes in the quality of aggregates for each source or when source changed	A	330	
		(v) Polished stone value			A		

APPENDIX 'D' (CONTD……..)

1	2	3	4	5	6	7	8
3	Fine Aggregate	(i) Sieve Analysis	(IS-2386) (Part-I)	(I) One Test for every 15m3 of EA or part when brought to site.	A	660	
		(ii) Test for clay silt impurities	-do-	(I) One Test for every 15m3 of EA or part when brought to site.	A	500	
		(iii) Specific Gravity	-do-	One for each source of supply	B	330	
		(iv) Moisture Content	-do- (Part-III)	Regularly as required subject to 2 tests/day when being used.	A	330	
		(v) Test for Organic	-do- (Part-II)	One test for each source of supply.	C	275	
4	Cement	(i) Setting Time	IS-4031 (Latest	Once for each Consignment	B	500	
		(ii) Soundness	IS-4031 (Latest	Once for each Consignment	C	550	
		(iii) Compressive Strength	IS-4031 (Latest	Once for each Consignment	B	550	
		(iv) Fineness	IS-4031 (Latest	Once for each Consignment	C	275	
5	Structure concrete	(i) Slump Test or Compacting Factor Test of Vee Bee Time	IS-1199	The minimum frequency of sampling of concrete of each grade shall be as under:			(i) Random sampling shall be carried out to cover all mixing units
				Qty of Concrete in the work (M3) No of Samples	A	300	
				1-5			
				6-15	A	300	(ii) Refer IS-456-2000, Clause 14 for frequency of sample
				2			
				16-30			
				31-50			
				51 and above 4+1 for each additional 50 cubic metre or part thereof.	A	300	
		(ii) Compressive Strength	IS-516		A	900	

APPENDIX 'D' (CONTD……)

1	2	3	4	5	6	7	8
6	(a) PCC Block for walling(Hollow block)	(i) Compressive Strength	IS-2185(Appx'B')	8 Block out of totalof14	A	900	Sample: 18 Blocksfrom consignment of every 5000 blocks or part thereof.
		(ii) Water Absorption	-do-	3–do-	B	330	
		(iii) Density	-do-(Appx'A')	3–do-	B	330	
	(b) PCC Solid Blocks for walling	(i) Compressive Strength	IS-2185(Appx'A')	12 Block out of totalof18	A	900	Sample: 18 Blocksfrom consignment of every 100000 blocks or part thereof.
		(ii) Water Absorption	-do-	3 –do	A	330	These blocks to be checked for dimension and weight.
		(iii) Density	-do-	3 –do-	B	330	
7	Cement flooring tiles/terrazzotiles	(i) Water Absorption	IS-1237 1980 (Appx 'D')	6 tiles out of total of 18	B	330	
		(ii) Wet transverse strength	-do- (Appx 'E')	6 tiles out of total of 18	B	660	
		(iii) Resistance to wear	do (Appx 'F')	6 tiles out of total of 18	C	1000	Sample 18 tiles from each source of supply selected at random
8	Burnt clay roofing tiles (Hand made) as per IS-2690(Part-II) Length – 150 mm to 250 mm Width – 100 mm to 200 mm Thickness – 35 mm to 50 mm	(i) Water Absorption	IS-3495 (Part-II)	6 tiles out of total of 18	B	660	Sample: 12 tiles from each source of supply selected at random.
		(ii) Compressive Strength	-do-	6 tiles out of total of 18	A	330	Selected at random

Contd..

APPENDIX 'D' (CONTD……)

1	2	3	4	5	6	7	8
9	Mangalore pattern roofingtiles	(i) Water Absorption	IS-654 (Appx 'A')	16 tiles out of total 32	B	660	Sample: 32 tiles from each consignment of every 3000 tiles or part thereof. These tiles shall be checked for dimension& weight.
		(ii) Breaking load	IS-654 (Appx 'C')				
10	Timber	(i) Specific Gravity and weight	IS-1708- 1960	Minimum 3 samples from a lot of 4 cubic meters or 250 pieces of seasoned timber.	B	660	
		(ii) Moisture content	IS-1708- 1960		A	330	
11	Water for constructionpurpose	(i) Test for acidity	IS-456 & 3025	Once the stage of approval	B	660	Also refer Clause 4.8 of IS-456 & its subsequent sub clauses regarding suitability of water
		(ii) Test for alkalinity	IS-456 & 3025		B	660	
		(iii) Test for Solid contents.	IS-456 & 3025		C	1000	
12	Welding of Steelwork	Visual Inspection Test	IS-818-1970, Clause 7.1	100 % by visual inspection	A	330	Specialised tests their method and frequency to be decided on consideration of their importance by the Accepting Officer
13	Timber panelled and glazed door (wooden shutters) (Includingfactory made shutters).			Frequency of sampling from each lot shall be as under:			

Lot Size	Sample Size
20 to 50	05
51 to 100	08
101 to 150	13
151 to 300	20
301 to 500	32
501 to 1000	50
1001 and above	80

APPENDIX 'D' (CONTD…....)

1	2	3	4	5	6	7	8
		(a) Dimensions, sizes, workmanship and finish	IS-1003-1977(Part-I)		A	330	
		(b) Strength test		From each lot 5 % of the factory made shutters shall be tested for strength test.	Manufacturer		
		(i) Slamming	IS-1303-1990				
		(ii) Impact Indication	-do-				
		(iii) Shock resistance	-do-				
		(iv) Edge loading	-do-				
14	PlyWood	(a) Moisture Contents	IS-1734 (Part-VI)	6 Test pieces from each of the board selected.	B	660	
		(b) Water resistance	IS-1734 (Part-I)	As per Table-I shall be subjected to tests.	C	1000	
15	Work particles board(Medium density)IS-3087-1988	(a) Density	IS-2380-1977	Three test specimen each sample (size-150mmx75mm)	A	330	Sampling shall be as per IS-7838 1978 (Table-I)
		(b) Moisture content	-do-	-do-	With Moisture Meter A & B	660	
		(c) Water absorption	-do- (Part 16)	-do- (size-300mmx300mm)	A	330	
		(d) Swelling to surface absorption	-do- (Part 17)	~-do- (size-125mmx100mm)	A	330	
		(e) Swelling in water	-do-	-do- (size 200mmx100mm)	A	330	
		(f) Modulus of rupture	-do- (Part 4)	Three test specimen as per IS-2380-	B	660	
		(g) Screw withdrawal	-do- (Part 14)	~-do- as per IS-2380	C	1000	

APPENDIX 'D' (CONTD)

(e) Mix grading	Stability and Voids analysis	one test for individual constituent and mix aggregate from dyer for each 400 tonne of mix subject 2 minimum 2tests per day per plant	A	330
(f) Binder Content		one test for 400 tonne and subject to minimum 2 tests per day per plant	A	330

Legend : A :Site Lab, B: Zonal Lab, C: Approved Test House / Engineering College

Note :- (i) All 'B' level test shall be conducted in CTL in case there is no functional Zonal Lab in the adjoining area, with testing charges to be recovered as per contract conditions.

(ii) 'C'level test for which facilities are available in CTL shall be conducted in CTL with testing charges to be recovered from contract or as per Contract Conditions Balance 'C' level test shall continue to be conducted in SEMT/Govt Approved Lab/Labof Govt Engg College, as the case be under contractor's own arrangement and actual cost of testing shall be borne by the contractor.

(iii) Various and om/independent testing for material/products required to be conduct in addition to routine tests as stipulated in contracts, be referred to CTL to get verification of quality of material/product.

Signature of Contractor

Dated:_____

AGE (Contracts)

for Accepting Officer

GENERAL CONDITIONS OF CONTRACT (IAFW-2249-1989 PRINT) FOR ITEM RATE CONTRACT WORK (IAFW-1779A)

1. It is hereby agreed by me/us that the followings shall be forming as integral part of the tender documents:-

(a) General conditions of contracts including condition 70 pertaining to the settlement of disputes by Arbitration, IAFW-2249 (1989 print) containing 33 pages (Serial page No 77 to 109) together with amendment 01 to 49 and errata 01 to 20 containing 9 pages (serial page No 110 to 118)

(b) Schedule of Minimum Rates of wages in 8 pages (Serial page No 119 to 126)

2. This tender submitted by me/us is subject to the aforesaid documents and copy of which is in my/our possession and which I/We have read fully and understood before submission of this tender.

3. My/our signature here under is in token of my/our having accepted the aforesaid documents as forming an integral part of this tender.

NOTE: (i) The documents mentioned above can be seen in the office of the GE (North) Delhi Cantt-10 or in any other MES office during working hours.

(ii) In case of difference in interpretation due to wording of English and Hindi versions, the English version will prevail as per article 348 of constitution of India.

(Signature of contractor)
DATED:_____

AGE (Contracts)
For Accepting Officer

SCHEDULE OF CREDIT

1 The credit assessed by the department is only rough indication and the Contractor shall have no claim what-so-ever against the Govt if the actual cost of the salvaged materials turn out to be different from the amount assessed by the department

2 The quantity inserted under Col 4 are approximate and the tenderer will have no claim if the quantities obtained turns out to be different on execution. The assessed rate of credit shown in Col 5 is fixed.

3 The credit amount will be carried over to General Summary to arrive at the contract sum.

4 The materials listed in this Schedule shall be recorded in the Measurement Book and abstracted and this amount shall be deducted from the value of work done under Schedule 'A' items (BOQ). The Contractor shall not be allowed to remove the materials listed in this schedule till such time the value of work done by him is more than the value of credit for such of the materials.

5 If Contractor offers any discount over quoted sum, the same shall not be applicable to this Schedule.

6 Fixed assessed rate of credit has been specified against Serial items as referred above. In each of the item, reference to Serial items of Sch 'A' items (BOQ) is specified from where materials are retrieved. In some of the items materials have not been directly retrieved from said item but through other connected item or may not be specifically indicated there in the said item. This element shall be deemed to have been taken into consideration.

7 Materials obtained from demolition / dismantling/ taking down etc from different items of Sch 'A' items (BOQ) or any other items for which credit is not considered, shall become property of the department and the same shall be deposited by the contractor in MES store yard.

S No	Description of work	A/U	QTY	Rate	Amount
1	2	3	4	5	6
1	Old unserviceable steel scrap	Kg	2000.00	20.00	40000.00
Total					**40000.00**

SIGNATURE OF CONTRACTOR **AGE (CONTRACTS)**

DATED: **FOR ACCEPTING OFFICER**

PROVN OF RCC FOUNDATION WORK FOR INSTALLATION MACHINERY AT COMPONENT LEVEL REPAIR BLDG. (SHOP NO 18) FOR ESTABLISHING COMPONENT LEVEL REPAIR FOR TANK T-90/SK AT 505 ABW AND PROVN OF EXHAUST FAN IN BARREL TYPES SHED OF TECH AREA AT ESD UNDER GE (NORTH) DELHI CANTT -10